Primitive Constructive Logic

Distinction, Report, Set, Index, Number, Operation, Constructive Zero, and the Foundations of Mathematical Meaning

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A dependency-ordered reconstruction from property, distinction, constraint, equivalence, whole closure, count, ratio, and provenance-preserving coordinate compilation.

Abstract

This paper develops Primitive Constructive Logic (PCL) by stripping mathematical assumptions down to realized property sensitivity, realized distinction, retained report, whole occurrence, comparison, constraint equivalence, finite grouping, matching, counting, quotient–remainder, and finite continuation rules. No pre-existing number line, Euclidean space, coordinate frame, completed set-theoretic universe, completed infinity, signed domain, real continuum, complex plane, literal absence, or context-free zero is admitted at the primitive level.

The primitive semantic object is not a geometric point. A meaningful measured report retains three irreducible dependency roles: a property or sensitivity criterion, a realized measurement/distinction under that criterion, and a retained report by which the result remains presently distinguishable. A realized difference between two comparable reports is itself another whole report occurrence. Difference is therefore prior to signed subtraction, distance, line, angle, or coordinate displacement. Difference reports may themselves be compared and grouped into constraint-equivalence classes; repetition of equivalent whole constraints supplies count, while comparison of repeated constraint classes supplies ratio through exact quotient–remainder ledgers.

PCL defines primitive dimension as *constraint rank*: the least number of jointly sufficient and individually irredundant constraint roles required to close a declared report state. Constraint rank is not created by stacking, sweeping, or repeating lower-rank reports. In particular, a finite cycle does not manufacture an interior, a two-index product does not manufacture area, and a three-role semantic closure is not by itself a proof that physical placement has three Euclidean dimensions. A whole closure is the jointly satisfied constraint grouping that constructed the report; no completed continuum of hidden points is inserted between its retained occurrences.

Primitive subtraction is rebuilt as matching plus unmatched residuals. Primitive division of one whole occurrence by a two-occurrence carrier is exactly $1 = 0 \cdot 2 + 1$; it does not manufacture a half-occurrence. A one-to-two relation is retained as a ratio report, while the glyph $1/2$ is only a later coordinate projection. Likewise, negative scalars, exact lengths, right angles, areas, volumes, real coordinates, complex coordinates, and analytic formulas belong to declared provenance-preserving compilers. They may be used conditionally without being mistaken for primitive existence claims.

Zero is reconstructed as a typed comparison/equivalence certificate: a declared comparison was performed and no admitted update remained under the chosen grammar. Finite-resolution analysis, logarithmic refinement, zeta source ledgers, continuation, and reflection remain terminating evaluators whose conventional real or complex notation is admitted only through explicit projection maps that retain their primitive source, residual, refinement, transition, and closure data.

Foundational status of conventional domains

Principle 0.1 (Provenance-preserving domain entry)**.** *A conventional mathematical domain may be used only through a declared compiler from prior PCL reports. The compiler may introduce signed coordinates, real-coordinate evaluators, Euclidean frame relations, complex-pair notation, or analytic scalar formulas, but it must retain the primitive matching, residual, refinement, transition, source, and closure data on which those coordinates depend. A theorem internal to the compiled domain is not, by itself, a proof that the domain exists as a primitive completed carrier.*

Principle 0.2 (Whole-count priority over fractional coordinates)**.** *Primitive quotient–remainder division never subdivides a whole occurrence. In particular,*

$$1 = 0 \cdot 2 + 1$$

is the complete primitive division report of one whole occurrence by a two-occurrence carrier. A one-to-two ratio is constructible as a retained relation of whole carriers, but a glyph such as $1/2$ *is admitted only after a conventional ratio-coordinate compiler has been declared. Every later use of such a glyph must retain the source ratio, denominator carrier, quotient, and remainder or the equivalent whole-count balance relation that licensed the projection.*

Principle 0.3 (No retroactive ontology)**.** *If a later coordinate grammar uses a relation in order to be defined, that grammar may not be cited retroactively as the foundational reason that the relation exists. In particular, signed subtraction cannot justify primitive opposition, a real continuum cannot justify the finite refinement rules used to compile real coordinates, and a complex plane cannot justify the orthogonality or signed axes that its own coordinate representation presupposes.*

Principle 0.4 (Constraint rank precedes geometric dimension)**.** *A report may require several independent constraint roles before any coordinate frame is declared. The minimum number of jointly sufficient and individually irredundant constraint roles is its primitive constraint rank. Repetition, stacking, sweeping, or indexed product of lower-rank reports does not by itself create a new independent constraint role. A later Euclidean dimension count is therefore a coordinate interpretation of an already declared constraint grammar, not the primitive source of that grammar.*

Principle 0.5 (Distinguishable refinement)**.** *A requested refinement becomes a new mathematical stage only when the active grammar realizes and retains a new distinction. A rule saying "refine again" is not evidence that another distinguishable level exists. If the proposed refinement is equivalent to the prior report under every declared criterion, no new primitive occurrence has been constructed.*

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Part I
The existence of mathematical report

1 Primitive foundation

Axiom 1.1 (Realized distinction)**.** *At least one distinction is realized.*

This axiom does not initially assert a number, coordinate, set, duration, history axis, spatial axis, or geometric dimension. It asserts only that a distinguishable update is presently available in some reportable form.

Definition 1.2 (Existence)**.** *An* existence *is anything capable of participating in a realized distinction.*

An existence need not be material. It may be a physical thing, a state, an idea, a relation, a measurement, a record, or another report-capable pattern.

Definition 1.3 (Property or sensing criterion)**.** *A* property criterion P *is a declared respect in which a report system is capable of distinguishing one presently available condition from another. A sensing capacity is the susceptibility by which a distinction under P can produce a report update.*

The property role does not initially contain a numerical value. Color, order, source, placement, duration, transition type, or any other admitted criterion may serve, provided the criterion is itself distinguishable enough to govern a comparison.

Definition 1.4 (Measurement or realized distinction)**.** *A* measurement *under P is the realized outcome of applying the comparison criterion. It may retain a detected difference or a realized no-difference result. Numerical measurement is a later encoding of this already realized relation.*

Definition 1.5 (Record and report)**.** *A* record *is a retained distinguishable consequence of a measurement. A* report *is a presently retained distinguishable state capable of supporting another comparison or causal update.*

Reading, receiving, retransmitting, and remembering are not new primitives. They are further applications of the same causal pattern to an already retained report.

Definition 1.6 (Property–measurement–report closure)**.** *A complete primitive measurement report is written*

$$\mathfrak{T} = (P, M, R; \Gamma),$$

where P is the property or sensing criterion, M is the realized measurement under that criterion, R is the retained report occurrence, and Γ retains that R is a report of M with respect to P.

Theorem 1.7 (Triadic semantic irreducibility)**.** *The roles P, M, and R are jointly irreducible for the meaning of one retained measured distinction.*

Proof. Deleting P leaves no declared respect in which the measurement is interpreted. Deleting M leaves a report with no realized comparison outcome. Deleting R leaves no presently retained occurrence by which the outcome can participate in later mathematics. The three roles may be instantiated by one physical system and need not be three physical objects; the theorem concerns dependency roles, not Euclidean vertices or chronological stages. □

Principle 1.8 (Blank, zero report, and one occurrence are different)**.** *A blank role means that no report occurrence has been retained in that role. A typed zero is a positive report occurrence certifying that an actual comparison closed with no admitted residual distinction. The count* 1 *reports one retained whole occurrence. Literal absence, a zero certificate, and one occurrence are therefore three different report conditions.*

Principle 1.9 (No primitive history axis)**.** *PCL does not assume a reel of past report frames. A prior state may participate in a present comparison only through a presently retained effect or record attributed to that prior state. Thus a comparison is between presently available reports, one of which may encode a propagated record of an earlier condition.*

Definition 1.10 (Measurement grammar)**.** *A* measurement grammar $\mathcal{M}$ *is the finite declaration of which properties, sensitivities, comparison rules, equivalence criteria, and closure conditions are admitted in a report. Number, sign, coordinate, and metric notation are later encodings of already distinguishable relations.*

Principle 1.11 (No retroactive foundation)**.** *If a construction C requires previously admitted operations or relations $A_1, \ldots, A_k$, then C may not be used retroactively to define or justify those same prerequisites unless an independent construction of them has already been supplied. Later completion, projection, or extension cannot become the primitive explanation of an operation required to construct it.*

Principle 1.12 (Rule is not completed totality)**.** *A finite rule capable of further application is not thereby the completed totality of all outputs it could ever produce. "Can continue," "has continued through this finite stage," and "all possible stages exist as one completed object" are distinct predicates.*

2 Object and set

Definition 2.1 (Object occurrence)**.** *An* object occurrence *is any presently distinguishable component admitted to a report under a declared identity criterion. The same projected value may occur more than once when source, role, route, or report position distinguishes the occurrences.*

Definition 2.2 (Report position and index)**.** *A report position is one whole distinguishable role occupied by an admitted occurrence. An* index *is the later label assigned to such a realized role. The index does not exist first and then create the occurrence it names.*

There is no fractional occurrence. If an alleged "part of an occurrence" becomes distinguishable under the active grammar, it is another whole distinguished report role; if it does not become distinguishable, it is not a separately indexed occurrence.

Definition 2.3 (Primitive set-report)**.** *A* primitive set-report *is a finite understandable grouping of already distinguished whole object occurrences under a declared member-identity criterion. Set construction therefore has the dependency order*

distinguished occurrences $\longrightarrow$ *present grouping* $\longrightarrow$ *index/count report* $\longrightarrow$ *optional set notation.*

A primitive set-report is not defined by quantifying over a previously assumed universal object carrier. After a concrete grouping has been constructed, notation such as

$$S = (I_S, v_S)$$

may be used as a compression: I_S is then the index report extracted from the already realized grouping and v_S returns the retained occurrence at each index.

Principle 2.4 (Sethood requires construction)**.** *A predicate $P(x)$ that decides whether a presented occurrence would qualify for a grouping does not by itself construct the completed totality*

$$\{x : P(x)\}.$$

The predicate is an admissibility or selection rule. PCL calls the resulting totality a primitive set only when the relevant member occurrences have actually been grouped into a finite report.

Definition 2.5 (Reindexing equivalence)**.** *Two finite set-reports are reindexing-equivalent when a bijective relabeling pairs the same retained member occurrences while changing only presentation order. The conventional unordered-set notation is a later projection that forgets order; it does not imply that no indexed rendering occurred.*

Principle 2.6 (Infinite carrier notation)**.** *Symbols such as $\mathbb{N}$, $\mathbb{R}$, and $\mathbb{C}$ do not, at the primitive PCL level, denote completed warehouses of already realized members. They first denote finite construction, continuation, refinement, and equivalence grammars. A conventional completed-carrier reading is admissible only as an explicitly declared later information contract.*

3 Absence, empty reports, and zero

Definition 3.1 (Literal absence)**.** Literal absence *means absence of every distinguishable object, sensing capacity, observation, record, and report. PCL does not identify such a notion with a written symbol and does not claim that literal absence has been observationally demonstrated as a state of reality.*

Principle 3.2 (Unreportability of literal absence)**.** *A report cannot certify literal absence by becoming absent itself. Any successful observation, statement, token, or record asserting "nothing was detected" is already a positive report construction containing a criterion, an observation result, and a retained consequence.*

Definition 3.3 (Empty report token)**.** *The token $\varnothing$ is one distinguishable report occurrence whose interpretation is that no member occurrence has been admitted to a referenced grouping under a declared criterion. The token is not literal absence.*

Definition 3.4 (Primitive equivalence report)**.** *Let A^* be a presently retained report attributed to a reference condition and let B be the present comparison report. Under measurement grammar $\mathcal{M}$, a primitive equivalence report is*

$$A^* \equiv_{\mathcal{M}} B$$

when the declared comparison finds no admitted update under $\mathcal{M}$.

The comparison report retains at least the measured properties, the two presently available report roles, the comparison lineage, the full retained result list, the report size, and any unresolved boundary or sensitivity obligations. Equivalence therefore precedes scalar subtraction.

Definition 3.5 (Constructive zero)**.** *A* constructive zero *is the typed scalar or symbolic projection of a completed equivalence or closure report. Schematically,*

$$Z_{\mathcal{M}}(A^*, B) = (\textit{no observed update}, \mathcal{M}, A^*, B, \mathcal{R}, |\mathcal{R}|, \Gamma),$$

followed, when desired, by a scalar projection

$$\pi_{\Delta}(Z_{\mathcal{M}}(A^*, B)) = 0.$$

Thus zero is not primitive nothingness. It is a typed equivalence or closure certificate produced by an actual comparison or operation. Different constructive zero occurrences may share the same scalar projection while retaining different sources, criteria, routes, residuals, and closure meanings.

Principle 3.6 (No zero without a comparison grammar)**.** *"No change," "no admitted member," "zero remainder," "balanced cancellation," and "complete operator closure" are distinct constructive predicates until a theorem proves that a later projection may identify them for the property under study.*

4 First constraint relations

Definition 4.1 (Comparable report occurrences)**.** *Let A and B be presently retained whole occurrences. They are* comparable under property P *when the active measurement grammar supplies a finite comparison rule for P on both reports.*

Definition 4.2 (Primitive comparison)**.** *The primitive comparison report is*

$$\mathrm{Cmp}_P(A, B; \Gamma),$$

retaining the two source occurrences, the property criterion, the comparison rule, and its complete outcome ledger.

A comparison does not initially subtract one report from another. It asks whether the declared property supplies a retained distinction.

Definition 4.3 (Primitive difference or constraint report)**.** *If $\mathrm{Cmp}_P(A, B)$ realizes a distinction, retain that realized distinction as a whole report*

$$D_P(A, B).$$

The report $D_P(A, B)$ is called a primitive difference report *or* constraint report*. It has no primitive sign, length, angle, or coordinate displacement.*

If no distinction is admitted, the comparison instead closes with the typed equivalence report

$$A \equiv_P B.$$

Thus difference and equivalence are alternative report outcomes of one comparison grammar.

Principle 4.4 (Difference is something reported)**.** *Whenever a difference is mathematically available, the difference is itself a retained distinguishable occurrence. PCL does not treat difference as a negative amount of nothingness. Signed differences arise only after a later orientation or coordinate compiler assigns opposing roles to residual or constraint reports.*

Definition 4.5 (Occurrence exclusivity)**.** *If two alleged occurrences are distinguishable under the complete active grammar, they require distinct complete occurrence identities. One complete report index cannot simultaneously denote two mutually distinguishable occurrences without losing the distinction that made them two.*

Principle 4.6 (No primitive line from two reports)**.** *Two distinguishable occurrences supply a comparison relation and, when distinguished, a difference report. They do not thereby construct a Euclidean line, a length, a continuum of intermediate positions, or a spatial axis. Those are later coordinate interpretations requiring additional structure.*

Part II
Distinguishable Multiplicity and the First Mathematical Constructions

5 Distinguishable occurrences

A report cannot contain an object without rendering that object at a distinguishable report position. Therefore an object's reported value and its occurrence within the report must be kept separate.

Definition 5.1 — Report occurrence

Let R be an understandable report. An **occurrence** in R is a pair

$$\omega_i = (i, v_i),$$

where:

- i is the occurrence's report index;
- v_i is the value, identity, pattern, or type reported at that index.

Two occurrences are distinct whenever their indices are distinct:

$$i \neq j \Longrightarrow \omega_i \neq_{\mathrm{C}} \omega_j.$$

This remains true even when their reported values agree:

$$v_i = v_j.$$

Accordingly, value equality does not imply occurrence identity:

$$v_i = v_j \backslash \text{centernot} \Longrightarrow \omega_i = \omega_j.$$

The distinction is between

$$\omega_i =_{report} \omega_j$$

and

$$\omega_i =_{\mathrm{C}} \omega_j.$$

The first means that the two occurrences produce the same value or type report. The second means that they are the same indexed occurrence.

5.1 Repetition requires distinct indices

Consider the primitive set-report

$$R = [a_0, a_1],$$

where

$$\pi(a_0) = \pi(a_1) = a.$$

The two occurrences have the same projected report a, but they occupy different positions:

$$ind(a_0) = 0, ind(a_1) = 1.$$

Therefore,

$$a_0 \neq_{\mathrm{C}} a_1.$$

The report contains two objects:

$$\mid R \mid_C = 2.$$

By contrast,

$$R^{'} = [a_0]$$

contains one occurrence:

$$\mid R^{'} \mid_C = 1.$$

Hence,

$$[a_0, a_1] \neq_{\mathrm{C}} [a_0].$$

If both occurrences are projected only to their common value a, an extensional report may collapse them:

$$\pi_{ext}([a_0, a_1]) = \{a\} = \pi_{ext}([a_0]).$$

That equality belongs to the projection. It is not equality of the original reports.

Theorem 5.1 — Occurrence preservation

Let

$$R = (I_R, v_R)$$

be a report with index carrier I_R and value-reporting map

$$v_R : I_R \to V.$$

The constructive multiplicity of R is

$$N_C(R) = \mid I_R \mid .$$

Its projected extensional value collection is

$$E(R) = im(v_R).$$

Then

$$\mid E(R) \mid \leq N_C(R),$$

with strict inequality whenever distinct indices carry the same projected value.

Proof.

Suppose

$$i \neq j$$

but

$$v_R(i) = v_R(j).$$

The constructive report retains both i and j, so they contribute two occurrences to $N_C(R)$. The image $E(R)$, however, contains only one copy of their common projected value. Therefore projection may reduce multiplicity. □

Corollary 5.1

A value projection cannot be used to count occurrences unless it is injective:

$$v_R(i) = v_R(j) \Longrightarrow i = j.$$

If injectivity has not been proved, extensional value count and constructive occurrence count are different quantities.

6 Primitive set-reports

Definition 6.1 (Finite primitive set-report)**.** *A finite primitive set-report is a presently constructed grouping*

$$S = [s_0, \ldots, s_{m-1}]_{\Gamma_S}$$

of $m > 0$ whole distinguishable occurrences under member criterion Γ_S. The displayed indices are labels extracted from this constructed grouping; they are not a previously assumed index set.

Definition 6.2 (Constructive set equality)**.** *Two primitive set-reports are constructively equal only when a source-preserving pairing matches every member occurrence and every structure declared relevant to the comparison. Count equality or value equality alone is weaker unless the omitted fields are proved irrelevant.*

Definition 6.3 (Reindexing equality)**.** *Two finite reports are reindexing-equivalent when they differ only by a bijective relabeling of the already retained occurrences. Thus*

$$[a_0, a_1, a_2] \sim_{\text{perm}} [a_2, a_0, a_1]$$

may hold without deleting either original order from its source ledger.

Principle 6.4 (No unrestricted set abstraction)**.** *PCL never infers a completed set solely from the statement that a membership predicate can be written. Finite realized groupings are set-reports; indefinitely reusable membership rules remain rules unless a separate finite grouping has actually been constructed.*

7 Literal absence and reported emptiness

Proposition 7.1 (Every denotation of emptiness is a positive report occurrence)**.** *Any mark used to denote emptiness, including ∅ or a typed zero token, is itself a distinguishable report occurrence. Therefore the report containing the mark is not literal absence.*

A report may legitimately state that no member has been admitted or that no update survived a declared comparison. What it may not do is identify that positive report construction with literal nonexistence.

Theorem 7.2 (Constructive zero necessity)**.** *If a zero token occurs in a report, then it is a whole typed report occurrence. Its scalar value may be zero while its report count and source ledger remain positive.*

Proof. Writing, sensing, comparing, retaining, or transmitting the token requires a report occurrence. The zero predicate therefore concerns a declared property of that occurrence rather than erasing the occurrence itself. □

8 Iteration and the first numbers

Number cannot be introduced before distinguishable multiplicity.

Definition 8.1 — Single occurrence

Let

$$A = [a_0].$$

The report A contains one occurrence.

The primitive numerical report associated with this completed occurrence is denoted

$$1.$$

Thus 1 does not create the occurrence. It reports that exactly one distinguishable occurrence has been retained.

Definition 8.2 — Iteration

Given a report A, an iteration of A is a new occurrence produced under the same declared type or construction rule while receiving a new index.

If

$$A = [a_0],$$

then one additional iteration produces

$$Iter(A) = [a_0, a_1],$$

where

$$a_0 \neq_{\mathrm{C}} a_1$$

but perhaps

$$\pi(a_0) = \pi(a_1).$$

The count report is then

$$2.$$

Accordingly,

2

means that two distinguishable occurrences have been completed and retained.

Definition 8.3 — Constructive natural number

A constructive natural number n is the report of a set containing n whole distinguishable occurrences:

$$n \equiv\mid I_S \mid .$$

The numeral is a representation of the multiplicity. It is not the source of that multiplicity.

Thus the primitive order is

$$\text{occurrence} \to \text{iteration} \to \text{count} \to \text{numeral}.$$

9 Joining reports and primitive addition

Definition 9.1 — Source-preserving join

Let

$$A = (I_A, v_A), B = (I_B, v_B).$$

Their source-preserving join is

$$A \sqcup B = (\{A\} \times I_A) \cup (\{B\} \times I_B).$$

The source labels ensure that occurrences from A and B remain distinguishable even when they carry equal projected values.

For example, let

$$A = [a_0], B = [a_1].$$

Even if

$$\pi(a_0) = \pi(a_1),$$

their join contains two occurrences:

$$A \sqcup B = [a_0, a_1].$$

Therefore,

$$\mid A \sqcup B \mid = \mid A \mid + \mid B \mid .$$

Definition 9.2 — Primitive addition

Addition is the count report produced by source-preserving joining:

$$\mid A \sqcup B \mid = \mid A \mid + \mid B \mid .$$

For two singleton reports,

$$\mid A \mid = 1, \mid B \mid = 1,$$

and therefore

$$| A \sqcup B | = 2.$$

The equation

$$1 + 1 = 2$$

is a compressed report of the prior construction

$$[a_0] \sqcup [a_1] = [a_0, a_1].$$

Addition does not create the objects. It reports the multiplicity of their retained join.

Theorem 9.1 — Addition preserves constructive multiplicity

For disjointly indexed reports A and B,

$$N_C(A \sqcup B) = N_C(A) + N_C(B).$$

Projecting equal-valued occurrences to a single extensional value before counting destroys this law.

For example,

$$[a_0] \sqcup [a_1]$$

contains two occurrences, while the projected value set may be only

$$\{a\}.$$

The extensional projection therefore cannot replace the source-preserving join when multiplicity matters.

10 Removal and primitive subtraction

Definition 10.1 — Constructive removal

Let A be a report and let B identify occurrences already contained in A.

Constructive subtraction removes the occurrences indexed by B from A, while retaining the relation describing what was removed and from where.

Write:

$$A \ominus B = R.$$

The complete subtraction state is

$$(A, B, R).$$

The remainder R contains the occurrences of A not removed through B.

Example.

Let

$$A = [a_0, a_1, a_2],$$

and remove

$$B = [a_1].$$

Then

$$A \ominus B = [a_0, a_2].$$

The count report is

$$3 - 1 = 2.$$

The subtraction report is meaningful because the removed occurrence existed previously in A.

Exact removal

If every occurrence of A is removed by an occurrence-identical copy of A, then no positive contribution remains:

$$A \ominus A = 0_A.$$

Here 0_A is the constructive zero report indexed to the removal of A. It certifies complete removal of the declared source occurrences.

It is not literal nonexistence, because the subtraction relation remains understandable:

$$(A, A, 0_A).$$

11 Two distinguishable occurrences and the first constraint

Every retained grouping that contains two distinguishable occurrences admits at least the identity comparison that keeps the two source roles distinct. When a common property grammar is available, the two occurrences may also be compared under that property.

Definition 11.1 (Two-source comparison packet)**.** *For comparable whole occurrences A and B, define*

$$\mathcal{C}_P(A, B) = (A, B, P, \mathrm{Cmp}_P(A, B), \Gamma).$$

If a distinction is realized, the packet retains $D_P(A, B)$; if none is realized, it retains the typed equivalence certificate $A \equiv_P B$.

Theorem 11.2 (Two occurrences do not determine a metric)**.** *A two-source comparison packet does not by itself determine a Euclidean line, signed displacement, numerical distance, angle, or continuum between the sources.*

Proof. The packet contains only the declared sources, property criterion, comparison rule, and outcome. Distinct coordinate frames or geometric interpretations can project the same comparison packet to different numerical displays. Therefore metric structure is not constant on the primitive comparison fiber unless an additional compiler and preservation theorem are supplied. □

12 Constraint equivalence and triadic closure

A difference report is itself a report occurrence and can therefore participate in another comparison.

Definition 12.1 (Constraint-equivalence comparison)**.** *Let*

$$D_1 = D_P(A, B), \qquad D_2 = D_P(C, D)$$

be two realized difference reports admitted to a common difference-comparison grammar $\mathcal{D}$. Write

$$D_1 \equiv_{\mathcal{D}} D_2$$

when the comparison of the two difference reports closes with no admitted distinction under $\mathcal{D}$.

Definition 12.2 (Constraint-equivalence class)**.** *A finite constraint-equivalence class is a presently constructed grouping*

$$[D]_{\mathcal{D}} = (D_1, \ldots, D_n; \Gamma)$$

of whole difference reports that are mutually equivalent under the declared comparison grammar.

The class is finite because only actually retained difference occurrences are grouped. A rule that could classify future differences does not construct the totality of every difference that could ever qualify.

Theorem 12.3 (Count from equivalent constraint recurrence)**.** *If a report contains n source-distinct occurrences of one constraint-equivalence class, then their multiplicity is the whole count n. Number is therefore available from recurrence of distinguishable equivalent constraints without a number line.*

Definition 12.4 (Triadic semantic closure)**.** *The primitive property–measurement–report packet*

$$\mathfrak{T} = (P, M, R; \Gamma)$$

is a three-role closure when all three roles are jointly sufficient for the declared meaning and each role is individually necessary for that same meaning.

Theorem 12.5 (Three-role closure is not primitive area)**.** *Triadic semantic closure does not imply a triangle, planar enclosure, Euclidean area, physical three-dimensionality, or any other geometric filling.*

Proof. The three roles are dependency constraints, not vertices in a pre-existing space. Their closure means that the complete report loses meaning if any role is removed. No metric, incidence plane, orthogonality relation, or coordinate embedding has yet been constructed. □

13 Ordered product reports without a geometric grid

Definition 13.1 (Ordered product report)**.** *Let*

$$X = (I_X, v_X), \qquad Y = (I_Y, v_Y)$$

be finite indexed reports. Their ordered product report is

$$X \times Y = \{(x_i, y_j) : i \in I_X,\ j \in I_Y\},$$

with one whole product occurrence for every ordered source pair.

The ordered product is an indexing construction, not a geometric rectangle. A table may be used to display it, but rows and columns are presentation roles until an additional geometry compiler assigns geometric meaning.

Example. Let

$$X = [0,1], \qquad Y = [0,1].$$

Then

$$X \times Y = \{(0,0),(0,1),(1,0),(1,1)\}$$

has four source-distinct occurrences. Under the later scalar addition projection

$$\pi_+(i,j) = i + j,$$

the two distinct source pairs satisfy

$$\pi_+(0,1) = 1 = \pi_+(1,0),$$

while

$$(0,1) \neq_{\mathrm{C}} (1,0).$$

Thus the same product table already exhibits projection-fiber collapse without invoking a spatial grid.

14 First projection-collapse law

Theorem 14.1 — Equal reports do not imply equal constructions

Let

$$\pi : \mathcal{C} \to V$$

be any report map from constructive objects to projected values.

If

$$\pi(C_1) = \pi(C_2),$$

it does not follow that

$$C_1 = C_2.$$

A proof of constructive equality requires a further theorem showing that π is injective over the relevant domain.

The 2×2 ordered product report gives the first explicit example:

$$\pi_+(0,1) = 1 = \pi_+(1,0),$$

while

$$(0,1) \neq_{\mathrm{C}} (1,0).$$

Thus commutative scalar equality is a report equality, not source identity.

This elementary distinction will later control:

- multiplication;

- fractions;
- quotient–remainder states;
- product-table balance;
- complex conjugation;
- reflected zeta terms;
- and the one-to-two balance report.

Part III
Repetition, Product, Division, and Retained Remainder

15 Repeated equal joining

Addition joins already distinguishable occurrences. Multiplication requires a further construction: the repeated joining of equal-sized occurrence families.

Definition 15.1 — Constructive copy

Let

$$A = (I_A, v_A)$$

be a primitive set-report.

A **constructive copy** of A is a new report

$$A^{(k)} = (I_A^{(k)}, v_A^{(k)})$$

together with an occurrence-preserving correspondence

$$c_k : I_A \to I_A^{(k)}.$$

For each $i \in I_A$,

$$v_A(i) =_\pi v_A^{(k)}(c_k(i)),$$

while the source-indexed occurrences remain distinct:

$$(A, i) \neq_{\mathrm{C}} (A^{(k)}, c_k(i)).$$

Thus a copy repeats the declared structure or type of A; it does not reuse the same occurrence as though one object occupied multiple indices simultaneously.

Definition 15.2 — Repetition index

Let

$$K = [k_0, k_1, \ldots, k_{n-1}]$$

be a report containing n distinguishable repetition indices.

For each $k_j \in K$, construct one indexed copy $A^{(k_j)}$.
The repeated equal join of A over K is

$$Rep_K(A) = \bigsqcup_{k \in K} A^{(k)}.$$

Every occurrence in the result retains both:

- its position inside A;
- the repetition index selecting its copy.

Its complete source address is therefore

$$(k, i) \in K \times I_A.$$

Theorem 15.1 — Repetition count
If A contains m occurrences and K contains n repetition indices, then

$$Rep_K(A)$$

contains one occurrence for every ordered source pair

$$(k, i) \in K \times I_A.$$

Hence its constructive count is reported as

$$n \cdot m.$$

Proof.
Each of the n repetition indices selects one complete copy of the m-occurrence report A. The copies are source-disjoint because their repetition indices differ. Thus there is one retained occurrence for each pair (k, i), giving nm occurrences. □
The scalar product

$$nm$$

is therefore a count report of a prior indexed repetition construction.
It is not the construction itself.

16 Primitive multiplication

Definition 16.1 — Multiplication by repetition
Let m report the count of a primitive occurrence family A, and let n report the count of a repetition-index family K.
The multiplication report

$$n \cdot m$$

is the count of the repeated equal join

$$\mathrm{Rep}_K(A).$$

Thus multiplication is derived from:

$$\begin{aligned}\text{distinguishable occurrence} &\longrightarrow \text{equal constructive copy} \longrightarrow \text{indexed repetition}\\ &\longrightarrow \text{source-preserving join} \longrightarrow \text{count report.}\end{aligned}$$

Multiplication cannot be primitive before these constructions have been established.

Example 16.1

Let

$$A = [a_0, a_1]$$

and

$$K = [k_0, k_1, k_2].$$

Then

$$Rep_K(A)$$

contains the six source-addressed occurrences

$$(k_0, a_0), (k_0, a_1),(k_1, a_0), (k_1, a_1),(k_2, a_0), (k_2, a_1).$$

Its scalar count report is

$$3 \cdot 2 = 6.$$

The number 6 does not retain which of the six source pairs produced it. It is a projection of the six-cell repetition ledger.

17 Multiplication as an ordered source product

Repeated joining and the ordered source product are two views of the same indexed construction.

Let

$$K = [k_0, \ldots, k_{n-1}]$$

and

$$A = [a_0, \ldots, a_{m-1}].$$

The complete multiplication ledger is

$$K \times A = \{(k_i, a_j)\}.$$

The repeated-join view groups the cells by repetition index:

$$K \times A = \bigsqcup_{k \in K} (\{k\} \times A).$$

The product view retains the entire cross-relation at once.

Therefore:

$$\boxed{\text{multiplication ledger} = \text{ordered source product},}$$

while

$$\boxed{\text{scalar multiplication} = \text{count projection of that product.}}$$

17.1 Multiplication is not primitive geometric extent

A table used to display multiplication is not the primitive source of geometric extent.

Multiplication needs only:

- an indexed repetition family;
- an indexed repeated family;
- and their ordered pairings.

Geometric length, area, or volume require an independently declared geometry compiler and constraint interpretation.

Thus:

$$K \times A$$

is a multiplication ledger whether or not its cells are interpreted as a geometric enclosure.

A scalar product does not become area merely because its ordered pairs can be arranged in a rectangular display.

18 Orientation and commutative projection

Let

$$A \times B = \{(a_i, b_j)\}$$

and

$$B \times A = \{(b_j, a_i)\}.$$

These are not literally the same source ledger. Their ordered components have been reversed.

Definition 18.1 — Transposition

Define the transposition map

$$\tau_{A,B} : A \times B \to B \times A$$

by

$$\tau_{A,B}(a_i, b_j) = (b_j, a_i).$$

This map is a bijection.

Therefore the two products have the same count:

$$| A \times B | = | B \times A | .$$

The scalar report is consequently commutative:

$$| A | \cdot | B | = | B | \cdot | A | .$$

But the source ledgers remain orientation-distinct:

$$(a_i, b_j) \neq_{\mathrm{C}} (b_j, a_i)$$

unless a separate equivalence deliberately identifies them.

Theorem 18.1 — Constructive commutativity

Multiplication is commutative at the scalar-count level because transposition pairs every cell of $A \times B$ with exactly one cell of $B \times A$.

Therefore:

$$A \times B \cong B \times A,$$

but generally

$$A \times B \neq_{\mathrm{C}} B \times A.$$

The symbol $\cong$ reports structure-preserving correspondence. It should not be silently replaced by full source identity.

Example 18.1

Let

$$A = B = [0, 1].$$

The cells

$$(0, 1)$$

and

$$(1, 0)$$

are exchanged by transposition. They remain separately indexed product occurrences.

A later scalar map may report them equally, but their product orientations have not thereby ceased to exist.

19 One-copy and zero-copy reports

19.1 Multiplication by one

Let K_1 contain one repetition index:

$$K_1 = [k_0].$$

Then

$$K_1 \times A = \{(k_0, a) : a \in A\}.$$

There is an occurrence-preserving correspondence

$$(k_0, a) \longleftrightarrow a.$$

Therefore the count report satisfies

$$1\cdot \mid A \mid = \mid A \mid .$$

The one-copy product may be canonically normalized back to A, but the full ledger may still retain the fact that a one-copy operation occurred.

19.2 Multiplication by zero

A zero repetition report means that no copy-selection event has occurred.

It cannot be represented by a literally empty understandable report. Instead, the complete multiplication state contains a constructive closure marker

$$0_{Rep(A)}$$

reporting:

No occurrence of A was selected into the repeated join.

Thus:

$$0\cdot \mid A \mid = 0$$

is the scalar projection of a typed no-copy state.

The zero result is not literal nothingness. It is a distinguishable report of zero completed repetitions of the declared object family.

20 Associative regrouping

Consider three indexed reports A, B, and C.

The product

$$(A \times B) \times C$$

contains occurrences of the form

$$((a, b), c).$$

The product

$$A \times (B \times C)$$

contains occurrences of the form

$$(a, (b, c)).$$

These are differently nested source reports.

Define the reassociation map

$$\alpha((a, b), c) = (a, (b, c)).$$

This map is bijective, so the two products have the same count.

Therefore:

$$(\mid A \mid \cdot \mid B \mid)\cdot \mid C \mid = \mid A \mid \cdot(\mid B \mid \cdot \mid C \mid).$$

Again, scalar associativity follows from a structure-preserving correspondence:

$$(A \times B) \times C \cong A \times (B \times C),$$

not from literal identity of their nested source syntax.

21 Constructive distribution

Let $B \sqcup C$ be a source-preserving join.

Then

$$A \times (B \sqcup C)$$

contains two source-distinct classes:

$$\{(a, b) : a \in A, b \in B\}$$

and

$$\{(a, c) : a \in A, c \in C\}.$$

Therefore:

$$A \times (B \sqcup C) \cong (A \times B) \sqcup (A \times C).$$

Taking counts gives:

$$\mid A \mid (\mid B \mid + \mid C \mid) = \mid A \mid\mid B \mid + \mid A \mid\mid C \mid .$$

Distributivity is thus a decomposition theorem for a source-preserving product ledger.

It is not merely a symbolic rearrangement.

Part IV
Comparison, Extraction, Division, and Remainder

22 Primitive comparison by complete containment

Division requires comparison, but comparison cannot yet rely on fractions, decimal magnitudes, or a continuous number line.

Definition 22.1 — Constructive containment comparison

Let A and B be finite occurrence reports.

Write

$$A \preceq_C B$$

when every occurrence of A can be matched injectively to a distinct occurrence of B.
This means that B contains enough occurrence positions to receive one complete copy of A.
Write

$$A \prec_C B$$

when

$$A \preceq_C B$$

but no occurrence-preserving bijection exists between them.
The scalar reports are:

$$| A |\leq| B |$$

and

$$| A |<| B | .$$

Thus numerical order is derived from whole-occurrence containability.

Definition 22.2 — Equal size

Two reports have equal size when each can be injected into the other, equivalently when there exists an occurrence-preserving bijection between their index carriers.
This is size equality, not necessarily full constructive identity.

23 Complete group extraction

Let X be a dividend occurrence report and B a positive divisor template.
A division construction asks:
How many complete source-disjoint copies of B can be identified inside X, and which occurrences remain unassigned?

Definition 23.1 — Extraction copy

An extraction copy of B from X is an injective occurrence map

$$e_k : I_B \to I_X$$

whose image identifies one complete B-sized group inside X.
Distinct extraction copies must have disjoint images:

$$im(e_j) \cap im(e_k) = \text{no shared positive occurrence}$$

for $j \neq k$.

Definition 23.2 — Extraction family

Let

$$Q = [q_0, q_1, \ldots, q_{r-1}]$$

index a family of complete, source-disjoint extractions of B from X.
The extracted occurrence family is represented by

$$Q \times B.$$

The occurrences of X not assigned to any extraction form the positive remainder carrier R.

The complete decomposition is

$$X \cong (Q \times B) \sqcup R.$$

This is the constructive source of the scalar equation

$$| X |=| Q || B | + | R | .$$

24 Primitive division

Definition 24.1 — Constructive quotient

The **constructive quotient** is the indexed report Q counting how many complete copies of B were extracted from X.

The scalar quotient is

$$q =| Q | .$$

The quotient is not initially a fractional scalar. It is a count of completed whole-group extractions.

Definition 24.2 — Constructive remainder

The **constructive remainder** is the source-retained report of dividend occurrences not assigned to a complete extracted copy of B.

Write

$$R = R_{X,B,Q,\gamma},$$

where γ records the declared extraction or grouping rule.

The complete division state is therefore

$$\mathcal{D}(X, B; \gamma) = (X, B, Q, R, \gamma).$$

Its scalar projection is

$$| X |= q | B | + | R | .$$

Remainder condition

Extraction is complete when the remainder cannot contain another full divisor copy:

$$B \not\preceq_C R.$$

At the scalar-count level, this is reported as

$$0 \leq | R |< | B | .$$

The inequality is not primitive. It summarizes the constructive fact that no further complete B-sized extraction is available.

25 Quotient uniqueness and decomposition nonuniqueness

A critical distinction must be retained.

For fixed finite counts

$$| X |= A, | B |= B > 0,$$

the quotient count q and remainder count r satisfying

$$A = qB + r, 0 \leq r < B$$

are unique.

But the source-level grouping need not be unique.

Example 25.1

Let

$$X = [x_0, x_1, x_2, x_3]$$

and let the divisor size be two.

One extraction rule may group:

$$[x_0, x_1], [x_2, x_3].$$

Another may group:

$$[x_0, x_2], [x_1, x_3].$$

Both produce the same scalar quotient:

$$q = 2$$

and the same scalar remainder:

$$r = 0.$$

But the grouping ledgers differ.

Therefore the scalar division state does not uniquely recover the constructive partition unless a grouping rule γ has been declared.

Theorem 25.1 — Scalar uniqueness does not imply ledger uniqueness

The quotient and remainder counts are uniquely determined by the dividend and divisor counts.

The source decomposition is uniquely determined only when the admissible extraction rule is itself deterministic or all admissible decompositions are proved equivalent for the property under study.

Thus:

$$(A, B) \longmapsto (q, r)$$

may be unique while

$$(X, B) \longmapsto (Q, R, \gamma)$$

retains multiple possible construction histories.

26 Exact division and constructive zero

Suppose every dividend occurrence is assigned to a complete divisor copy.

Then there is no positive remainder occurrence.

The division state must nevertheless report that the extraction closed exactly.

Definition 26.1 — Exact-remainder zero

The exact-remainder zero is the typed closure object

$$0_{X:B,\gamma}$$

whose meaning is:

Under extraction rule γ, every occurrence of X has been assigned to one complete copy of B, and no positive occurrence remains unassigned.

The complete division state is

$$\mathcal{D}(X, B; \gamma) = (X, B, Q, 0_{X:B,\gamma}, \gamma).$$

Its scalar projection is

$$A = qB + 0.$$

The zero is not an absent report. It is the report certifying exact closure of a declared extraction obligation.

Theorem 26.1 — Exact division requires an indexed zero

An exact quotient cannot be reported without distinguishing the condition that no positive remainder survives.

Therefore exact division necessarily produces a constructive zero typed to:

- the dividend;
- the divisor;
- the quotient family;
- and the extraction rule.

Two exact divisions may both project to scalar remainder 0 while remaining different zero-closure events:

$$0_{X:B,\gamma} \neq_{\mathrm{C}} 0_{Y:C,\delta}.$$

Their scalar zero reports agree, but their source obligations differ.

27 Division is not multiplication by a primitive reciprocal

At this stage, no reciprocal object has been constructed.

Therefore division cannot be primitively defined by

$$A/B = A \cdot B^{-1}.$$

That formula presupposes:

- a reciprocal carrier;
- multiplication on that carrier;
- an identity object;
- and a proof that the reciprocal closes the declared multiplication obligation.

Primitive division instead begins with:

$$\text{whole dividend occurrences} \longrightarrow \text{complete divisor-copy extraction} \longrightarrow \text{quotient count} \longrightarrow \text{retained remainder}.$$

Only later may a reciprocal be introduced as a report that reproduces the quotient behavior inside a declared extended carrier.

Thus:

$$\boxed{\text{division precedes reciprocal notation constructively.}}$$

A reciprocal is a compiled relation derived from division; it cannot be used to define division without circularity.

28 The complete quotient–remainder identity

The complete constructive division law is:

$$\boxed{X \cong (Q \times B) \sqcup R}$$

with

$$B \not\preceq_C R.$$

Its scalar-count projection is:

$$\boxed{A = qB + r, 0 \leq r < B.}$$

The first expression retains:

- the dividend source X;
- the divisor template B;
- the quotient index family Q;
- every extracted product cell;
- the remainder source R;
- and the extraction rule γ.

The second expression retains only their counts.
Therefore:

$$\boxed{A = qB + r}$$

is not the full division object. It is the scalar report of a prior source-preserving decomposition.

Part V
Ratios, Reciprocal Relations, and the Conventional Fraction Compiler

29 Ratio as a retained two-report relation

Division produces more information than a scalar quotient. For finite reports X and B, the complete division state retains

$$X \cong (Q \times B) \sqcup R,$$

with no further complete copy of B constructible from R.
When

$$\mid X \mid < \mid B \mid,$$

no complete divisor copy can be extracted. The quotient count is therefore zero, but the dividend has not disappeared:

$$Q = 0_{X:B}, R = X.$$

The scalar quotient alone reports

$$0.$$

The complete state reports

$$\mid X \mid = 0 \cdot \mid B \mid + \mid X \mid .$$

This distinction is the primitive source of a proper ratio.

Definition 29.1 — Constructive ratio

Let A and B be finite occurrence reports, with B carrying positive count:

$$\mid B \mid > 0.$$

The **constructive ratio of** A**to** B is the retained relation

$$\mathcal{R}(A, B) = (A, B, \mathcal{D}(A, B)),$$

where $\mathcal{D}(A, B)$ is the quotient–remainder decomposition of A by B.
The ratio retains:

- the numerator source A;

- the denominator source B;
- the complete quotient family Q;
- the remainder R;
- and the grouping rule used in the division.

A ratio is therefore not initially a freestanding magnitude. It is a relation between two declared constructions.

Definition 29.2 — Ratio notation

The notation

$$\frac{\mid A \mid}{\mid B \mid}$$

is the scalar report of the constructive ratio

$$\mathcal{R}(A, B).$$

The fraction bar does not perform an unexplained operation. It abbreviates the prior division relation.

Thus the correct dependency order is

$$(A, B) \to \mathcal{D}(A, B) \to \mathcal{R}(A, B) \to \frac{\mid A \mid}{\mid B \mid}.$$

The fraction notation is last.

30 Proper ratios retain a full residual carrier

Suppose A contains fewer occurrences than B:

$$0 <\mid A \mid <\mid B \mid .$$

No complete B-sized group can be extracted from A. Therefore

$$q = 0, R = A.$$

The complete finite state is

$$\mid A \mid = 0 \cdot \mid B \mid + \mid A \mid .$$

Example 30.1 — The ratio one to two

Let

$$U = [u_0]$$

be a singleton occurrence report, and let

$$D = [d_0, d_1]$$

be a two-occurrence denominator report.

Then

$$| U |= 1, | D |= 2.$$

Dividing U by D permits no complete extraction of D:

$$U \cong (0_{U:D} \times D) \sqcup U.$$

Its scalar quotient–remainder state is

$$1 = 0 \cdot 2 + 1.$$

The exact constructive ratio is therefore represented by

$$(q, R; B) = (0, 1; 2).$$

The primitive construction stops at the retained state $(0, 1; 2)$. If a later ratio-coordinate compiler is declared, it may display that one-to-two relation as

$$\frac{1}{2}.$$

The displayed glyph is not a primitive quotient. Primitive division remains $1 = 0 \cdot 2 + 1$.
The pair

$$(q, R) = (0, 1)$$

is insufficient by itself, because it would also arise from

$$\frac{1}{3}, \frac{1}{4}, \frac{1}{5},$$

and every other ratio in which one occurrence is smaller than the denominator family.
Their complete states are different:

$$(0, 1; 2), (0, 1; 3), (0, 1; 4), (0, 1; 5).$$

Therefore:

$$\boxed{\text{the denominator is part of the ratio object.}}$$

The fraction is not merely a quotient and remainder. It is a typed relation among dividend, divisor, quotient, and residual carrier.

31 A ratio projection does not divide an occurrence

The notation

$$\frac{1}{2}$$

is often spoken of as though one primitive object had been cut into two primitive half-objects.
That interpretation is not required by the constructive ratio.
In the state

$$(0, 1; 2),$$

the residual occurrence remains whole:

$$R = [u_0].$$

Nothing has become half an occurrence. Instead, the whole singleton is being compared with a whole two-occurrence denominator family.

The primitive statement is therefore the retained relation

$$\boxed{\mathcal{R}(U, D_2) \text{ compares one whole occurrence with a two-occurrence carrier.}}$$

If a conventional fraction compiler later displays this relation as $1/2$, that display does not report a primitive object cut into two pieces. It reports a coordinate shadow of the retained one-to-two relation.

$$\boxed{\text{ratio formation never creates a fractional occurrence identity.}}$$

Proposition 31.1 — Whole-index preservation

Let

$$\mathcal{R}(A, B)$$

be a constructive ratio. Every occurrence in A and B retains a whole index in its source report.

Therefore ratio formation does not produce partial indices.

Proof.

Ratio formation performs comparison and grouping between already indexed reports. It introduces no operation that partially admits an occurrence. Every member of A and B remains distinguishable at a whole source position. The fractional scalar is a report of their relative counts, not a subdivision of their indices. □

32 Ratio equality

Two ratio reports may have equal scalar values while retaining different source constructions.

Let

$$\mathcal{R}(A, B)$$

and

$$\mathcal{R}(C, D)$$

be ratios with positive denominator counts.

Definition 32.1 — Scalar ratio equality

The ratios are **scalar-equal** when their cross-products have equal constructive count:

$$| A \times D | = | C \times B | .$$

Equivalently,

$$| A \,||\, D | = | C \,||\, B | .$$

Write

$$\mathcal{R}(A,B) =_{ratio} \mathcal{R}(C,D).$$

This equality is established without first assuming a field of fractions.

Example 32.1

Let

$$\mid A \mid = 1, \mid B \mid = 2,$$

and

$$\mid C \mid = 2, \mid D \mid = 4.$$

Then

$$\mid A \times D \mid = 1 \cdot 4 = 4,$$

while

$$\mid C \times B \mid = 2 \cdot 2 = 4.$$

Therefore:

$$\mathcal{R}(A,B) =_{ratio} \mathcal{R}(C,D).$$

Their coordinate reports are

$$\frac{1}{2} = \frac{2}{4}.$$

However, the underlying source ledgers need not be constructively identical:

$$(A,B) \neq_{\mathrm{C}} (C,D).$$

One ratio compares a singleton with a two-occurrence carrier. The other compares a two-occurrence report with a four-occurrence carrier.

Thus:

equal fractions need not be equal constructions.

They belong to the same scalar ratio class because their cross-product counts balance.

33 Ratio equivalence by common replication

The most direct source of equivalent fractions is common replication.

Let K be a positive repetition-index report. Construct

$$K \times A$$

and

$$K \times B.$$

Their counts are

$$| K || A |, | K || B | .$$

The ratio relation is unchanged at the scalar level:

$$\mathcal{R}(A, B) =_{ratio} \mathcal{R}(K \times A, K \times B).$$

Theorem 33.1 — Common-replication invariance

For every positive repetition carrier K,

$$\frac{| A |}{| B |} = \frac{| K || A |}{| K || B |}.$$

Proof.

The cross-products are

$$A \times (K \times B)$$

and

$$(K \times A) \times B.$$

By reassociation and transposition, both contain one occurrence for each triple

$$(k, a, b) \in K \times A \times B.$$

They therefore have equal constructive count. □

The theorem explains fraction scaling constructively. Multiplying numerator and denominator by the same positive number means replicating both source reports by the same indexed carrier.

The scalar value remains invariant because the same replication structure is added to both sides of the comparison.

34 Common cancellation

The reverse operation is possible only when both numerator and denominator have actually been constructed through a shared repetition carrier.

Suppose

$$A \cong K \times A',$$

and

$$B \cong K \times B'.$$

Then the common carrier K may be removed from the scalar ratio report:

$$\mathcal{R}(A, B) =_{ratio} \mathcal{R}(A', B').$$

Definition 34.1 — Constructive cancellation

Constructive cancellation removes a common declared replication carrier from both sides of a ratio.

It is valid when the decompositions

$$A \cong K \times A^{'}, B \cong K \times B^{'}$$

have been established.

It is not justified merely because two scalar symbols appear to possess a common written factor.

Thus cancellation is a theorem about retained product structure.

Definition 34.2 — Reduced ratio

A ratio

$$\mathcal{R}(A, B)$$

is **constructively reduced** relative to the declared grammar when no non-singleton positive report K has been shown to satisfy

$$A \cong K \times A^{'}, B \cong K \times B^{'}.$$

This definition does not yet require a greatest-common-divisor algorithm. It requires only that no common replication carrier remain in the proven construction.

35 Mixed quotient–ratio reports

Suppose the dividend contains one or more complete divisor copies together with a nonzero remainder:

$$A = qB + R, 0 <\mid R \mid <\mid B \mid .$$

The complete report may be separated into:

- q complete copies of B;
- one retained proper ratio $\mathcal{R}(R, B)$.

Thus:

$$\mathcal{R}(A, B) = q + \mathcal{R}(R, B).$$

Its coordinate report is

$$\frac{A}{B} = q + \frac{R}{B}.$$

This is not a decomposition of A into abstract scalar pieces. It is a report of the actual source decomposition

$$A \cong (Q \times B) \sqcup R.$$

Example 35.1

Let

$$\mid A \mid = 5, \mid B \mid = 2.$$

Then two complete copies of B can be extracted, leaving one occurrence:

$$5 = 2 \cdot 2 + 1.$$

The full ratio state is

$$\mathcal{R}(5,2) = 2 + \mathcal{R}(1,2).$$

Its coordinate notation is

$$\frac{5}{2} = 2 + \frac{1}{2}.$$

The final $\frac{1}{2}$ remains the relation

$$(0,1;2).$$

No half-occurrence has been introduced.

36 Reciprocal reports

A reciprocal cannot be primitive because it presupposes the ratio relation from which it is derived.

Definition 36.1 — Unit report

Let U be a singleton report:

$$\mid U \mid = 1.$$

The unit report supplies the comparison source for reciprocal construction.

Definition 36.2 — Reciprocal of a positive report

Let B be a positive occurrence report:

$$\mid B \mid > 0.$$

The **reciprocal report of** B is the constructive ratio

$$Rec(B) = \mathcal{R}(U, B).$$

Its coordinate notation is

$$\frac{1}{\mid B \mid}.$$

Thus the reciprocal of B is not an independently existing scalar created by inversion. It is the relation of one unit occurrence to the positive denominator carrier B.

For a two-occurrence carrier:

$$Rec(B_2) = \mathcal{R}(U, B_2),$$

whose quotient–remainder state is

$$(0,1;2).$$

A conventional fraction-coordinate compiler may display this retained reciprocal relation as

$$\frac{1}{2}.$$

The primitive reciprocal report itself remains $\mathcal{R}(U, B_2)$ with quotient–remainder state $(0,1;2)$.

Proposition 36.1

The reciprocal report retains its denominator source.

Therefore:

$$Rec(B_2) \neq_{\mathrm{C}} Rec(B_3),$$

even though both produce quotient zero and remainder one before the divisor is restored.

37 Multiplication by a reciprocal

Once reciprocal reports exist, the conventional identity

$$A/B = A \cdot B^{-1}$$

can be recovered as a later theorem.

Let

$$Rec(B) = \mathcal{R}(U, B).$$

Multiplying A by this ratio produces

$$\mathcal{R}(A \times U, B).$$

Because U is a singleton,

$$A \times U \cong A.$$

Therefore:

$$A \cdot Rec(B) =_{ratio} \mathcal{R}(A, B).$$

Its scalar report is:

$$\mid A \mid \cdot \frac{1}{\mid B \mid} = \frac{\mid A \mid}{\mid B \mid}.$$

Theorem 37.1 — Reciprocal multiplication is compiled division

Multiplication by a reciprocal reproduces the scalar ratio only after:

1. a positive denominator report B exists;
2. division has been defined;
3. the unit-to-B ratio has been constructed;
4. ratio multiplication has been defined;
5. the singleton product has been normalized.

Thus:

$$\boxed{A/B = A \cdot B^{-1}}$$

is a compiled identity. It is not the primitive definition of division.

38 Zero ratio and the absence of a reciprocal

Let 0_A be a constructive zero reporting no numerator contribution relative to denominator B.

The ratio

$$\mathcal{R}(0_A, B)$$

has scalar report

$$\frac{0}{|\ B\ |} = 0.$$

This is well formed whenever B is positive.

The source meaning is:

No numerator occurrence contributes relative to the declared positive denominator carrier.

38.1 Why zero has no reciprocal

To construct the reciprocal of a numerical zero, one would require

$$\mathcal{R}(U, 0).$$

But numerical zero supplies no positive divisor carrier. There is no complete occurrence template against which extraction can be performed.

The zero symbol remains an understandable object, but its reported numerical contribution is zero. The symbol's existence does not turn zero contribution into a positive divisor family.

Therefore the positive-divisor compiler cannot be invoked.

Theorem 38.1 — Division by zero is not a ratio state

A constructive ratio

$$\mathcal{R}(A, B)$$

requires

$$|\ B\ | > 0.$$

If the denominator contribution is zero, no positive grouping template exists, no maximal complete extraction family is defined, and no remainder bound of the form

$$0 \leq R < B$$

can be constructed.

Therefore:

$$\boxed{\frac{A}{0} \text{ is not a completed constructive ratio.}}$$

It is not automatically infinity, zero, or an arbitrary value. It is a failure to supply the positive divisor structure required by the division operation.

Corollary 38.1

Zero has no multiplicative reciprocal:

$$0^{-1}$$

is undefined in the ratio grammar.

This is not because the zero symbol is literally absent. It is because its declared numerical contribution cannot serve as a positive divisor carrier.

39 Ratio multiplication

Let

$$\mathcal{R}(A, B)$$

and

$$\mathcal{R}(C, D)$$

have positive denominator reports B and D.

Their product ratio is constructed by pairing numerator sources and denominator sources:

$$\mathcal{R}(A, B) \otimes \mathcal{R}(C, D) = \mathcal{R}(A \times C, B \times D).$$

The scalar report is

$$\frac{\mid A \mid}{\mid B \mid} \cdot \frac{\mid C \mid}{\mid D \mid} = \frac{\mid A \mid\mid C \mid}{\mid B \mid\mid D \mid}.$$

The product retains four source families:

- A;
- B;
- C;
- D.

A reduced scalar fraction may later cancel common replication carriers, but that reduction cannot erase the original multiplication history unless the relevant property factors through the reduced report.

40 Ratio addition

Ratio addition requires a shared comparison carrier.

Let

$$\mathcal{R}(A, B)$$

and

$$\mathcal{R}(C, D)$$

be constructive ratios.

Construct the common denominator carrier

$$B \times D.$$

Replicate A across D:

$$A \times D,$$

and replicate C across B:

$$C \times B.$$

The numerator contributions may then be joined:

$$(A \times D) \sqcup (C \times B).$$

Definition 40.1 — Constructive ratio sum

Define

$$\mathcal{R}(A, B) \oplus \mathcal{R}(C, D) = \mathcal{R}((A \times D) \sqcup (C \times B), B \times D).$$

Its scalar projection is

$$\frac{|\, A \,|}{|\, B \,|} + \frac{|\, C \,|}{|\, D \,|} = \frac{|\, A \,\|\, D \,| + |\, C \,\|\, B \,|}{|\, B \,\|\, D \,|}.$$

This familiar formula is therefore derived from:

1. common carrier construction;
2. numerator replication;
3. source-preserving join;
4. ratio reporting.

The two numerator lanes remain distinguishable:

$$A \times D$$

and

$$C \times B.$$

Their scalar sum does not prove that they arose through the same route.

41 Ratio subtraction

Suppose a constructive matching removes the contribution of

$$C \times B$$

from

$$A \times D.$$

Then the ratio difference is

$$\mathcal{R}(A,B) \ominus \mathcal{R}(C,D) = \mathcal{R}((A \times D) \ominus (C \times B), B \times D).$$

The scalar report is

$$\frac{\mid A \mid}{\mid B \mid} - \frac{\mid C \mid}{\mid D \mid} = \frac{\mid A \parallel D \mid - \mid C \parallel B \mid}{\mid B \parallel D \mid}.$$

If the numerator lanes close exactly, the result is a typed constructive zero:

$$0_{AD:CB}.$$

The zero records which two ratio contributions were matched.

Thus two fraction differences may both project to scalar zero while retaining distinct source cancellation histories.

Part VI
Complement, Balance, and the One-to-Two Coordinate Projection

42 Complement inside a finite carrier

Let B be a positive finite report, and let A be identified as a source-preserving subreport of B.

The complement of A in B is the remainder report C satisfying

$$B \cong A \sqcup C.$$

At the scalar level:

$$\mid B \mid = \mid A \mid + \mid C \mid .$$

The ratio of A to B is

$$\mathcal{R}(A,B),$$

and the ratio of its complement is

$$\mathcal{R}(C,B).$$

Their sum is one complete carrier:

$$\mathcal{R}(A,B) \oplus \mathcal{R}(C,B) = \mathcal{R}(B,B).$$

The ratio $\mathcal{R}(B,B)$ reports one complete extraction:

$$\mid B \mid = 1 \cdot \mid B \mid + 0.$$

Its coordinate report is

$$1.$$

Therefore:

$$\frac{|\,A\,|}{|\,B\,|}+\frac{|\,C\,|}{|\,B\,|}=1.$$

The unit arises from complete carrier closure.

43 Constructive balance

Definition 43.1 — Balanced complement partition

A decomposition

$$B\cong A\sqcup C$$

is **balanced** when A and C have equal size:

$$|\,A\,|=|\,C\,|\,.$$

Because

$$|\,B\,|=|\,A\,|+|\,C\,|,$$

balance gives

$$|\,B\,|=|\,A\,|+|\,A\,|\,.$$

After multiplication has been constructed, this may be reported as

$$|\,B\,|=2\,|\,A\,|\,.$$

The native balance condition requires no subtraction. Construct two copies of A and compare that whole doubled carrier with B by maximal matching.

Definition 43.2 — Whole-count balance comparison

Let D_2 be a two-occurrence replication carrier and form

$$D_2\times A.$$

Maximally match $D_2\times A$ against B. Retain the common matched carrier C_H and the two unmatched residual carriers R_{2A} and R_B:

$$D_2\times A\cong C_H\sqcup R_{2A},\qquad B\cong C_H\sqcup R_B.$$

The *balanced one-to-two relation* holds exactly when

$$\boxed{R_{2A}=0_{2A}\quad\text{and}\quad R_B=0_B.}$$

At the count level this closure is written

$$\boxed{2|A|=|B|.}$$

No negative count, subtraction, or fractional occurrence is required.

Only after a signed coordinate compiler has been independently declared may the same comparison be compressed to the coordinate shadow $2|A|-|B|=0$.

44 Derivation of the balanced one-to-two relation

Assume

$$B \cong A \sqcup C$$

and

$$|A| = |C|.$$

Then

$$|B| = |A| + |A| = 2|A|.$$

Equivalently, the doubled whole carrier $D_2 \times A$ and B match with no residual on either side.

Theorem 44.1 — Balanced one-to-two theorem

A retained ratio $\mathcal{R}(A, B)$ has the same ratio relation as the singleton-to-two-occurrence report $\mathcal{R}(U, D_2)$ exactly when the doubled numerator carrier and denominator carrier balance:

$$\boxed{\mathcal{R}(A, B) =_{ratio} \mathcal{R}(U, D_2) \quad \Longleftrightarrow \quad 2|A| = |B|.}$$

When A is retained as one complementary subreport of B, this is equivalent to a decomposition

$$B \cong A \sqcup C, \qquad |A| = |C|.$$

If a later conventional ratio-coordinate compiler is declared, it may display the ratio class $\mathcal{R}(U, D_2)$ by the glyph $1/2$. That glyph is not the primitive result of dividing one occurrence by two. Primitive division still returns

$$1 = 0 \cdot 2 + 1.$$

Proof.

If $B \cong A \sqcup C$ and $|A| = |C|$, then $|B| = 2|A|$, so the cross-product comparison with the one-to-two report closes:

$$|A| \cdot 2 = |B| \cdot 1.$$

Conversely, suppose $2|A| = |B|$ and $B \cong A \sqcup C$. Compare the two known constructions of B: one is $A \sqcup C$, the other has the count of two copies of A. Match one retained copy of A on both constructions. The unmatched remainder on the first construction is C and on the second is one copy of A, so their counts are equal. □

45 The balanced ratio does not identify either side

A balanced decomposition retains two source-distinct complementary families:

$$A \neq_{\mathrm{C}} C, \qquad |A| = |C|.$$

Both retained ratios are ratio-equal to the same one-to-two reference relation:

$$\mathcal{R}(A, B) =_{ratio} \mathcal{R}(C, B) =_{ratio} \mathcal{R}(U, D_2).$$

A later coordinate compiler may display both by the same glyph $1/2$, but that display does not collapse their source identities.

Thus two complementary constructions may share one ratio-coordinate projection without becoming one construction.

46 Reflection of a balanced carrier

Let

$$B \cong A \sqcup C$$

be balanced. A reflection map

$$\rho_B : A \longrightarrow C$$

pairs whole occurrences of the two equal complementary carriers. Reflection exchanges the retained ratios

$$\rho_B : \mathcal{R}(A, B) \longleftrightarrow \mathcal{R}(C, B).$$

At balance, the whole-count comparison of $D_2 \times A$ with B closes with no residuals; the same is true for $D_2 \times C$ with B.

Away from balance, the primitive report does not contain a negative imbalance. It retains whichever unmatched residual remains after matching the doubled candidate carrier against B. If a later signed frame is declared, exchanging the complementary roles may then be displayed as a sign reversal, but that sign is a coordinate projection rather than a primitive count.

47 No primitive critical half

The glyph 1/2 need not, and under strict PCL must not, be inserted before the one-to-two balance relation has been constructed. The primitive sequence is

$$\begin{gathered}\text{whole carrier } B,\\ \text{whole subreport } A,\\ \text{whole doubled carrier } D_2 \times A,\\ \text{maximal matching of } D_2 \times A \text{ with } B,\\ R_{2A} = 0_{2A}, \quad R_B = 0_B.\end{gathered}$$

Its count shadow is

$$\boxed{2|A| = |B|.}$$

Only after a conventional ratio-coordinate compiler is declared may this balance class be displayed as

$$|A| : |B| = 1 : 2 \quad \rightsquigarrow \quad \frac{|A|}{|B|} = \frac{1}{2}.$$

Therefore

$$\boxed{\text{whole-count balance precedes the half-coordinate glyph.}}$$

The coordinate glyph reports a constructed ratio relation; it does not manufacture a fractional primitive.

48 The primitive ratio hierarchy

The construction developed in this part may be summarized as follows:

$$\begin{gathered}\text{whole occurrences}\\ \Downarrow\\ \text{source-preserving grouping}\\ \Downarrow\\ A \cong (Q \times B) \sqcup R\\ \Downarrow\\ (q, R; B)\\ \Downarrow\\ \mathcal{R}(A, B)\\ \Downarrow\\ \frac{A}{B}.\end{gathered}$$

From this hierarchy follow:

1. the denominator is part of every ratio object;
2. a proper ratio retains the entire dividend as remainder;
3. a fraction does not require a fractional occurrence;
4. equivalent fractions are equal under cross-product balance;
5. scaling is common replication;
6. cancellation removes a proven common carrier;
7. reciprocals are unit-to-denominator ratios;
8. division by zero fails because no positive divisor carrier exists;
9. fraction arithmetic is common-carrier construction;
10. the one-to-two balance relation is the closure of matching $D_2 \times A$ against B with no residuals;
11. the whole-count condition
$$2A = B$$
is prior to any optional coordinate display such as
$$A : B = 1 : 2 \quad \rightsquigarrow \quad A/B = 1/2.$$

Part VII
Finite Positional Reports

49 Finite positional notation and base expansion

Positional notation must be derived after counting, multiplication, division, and remainder. A written digit string is not the number prior to those constructions. It is an indexed report generated by repeated quotient–remainder extraction relative to a declared base carrier.

49.1 Base carrier and digit carrier

Definition 49.1 (Base carrier)**.** *A* base carrier B_b *is a positive finite occurrence report whose count is the base*

$$b = |B_b|, \qquad b \geq 2.$$

The condition $b \geq 2$ ensures that repeated extraction can distinguish at least a null digit report from one positive digit report. The base is not merely a subscript on a numeral. It is part of the positional grammar.

Definition 49.2 (Digit report)**.** *A* digit report in base b *is a remainder-count report d satisfying*

$$0 \leq d < b.$$

Constructively, d reports the remainder carrier left after extracting complete copies of B_b from a current dividend state.

Thus a digit is not primitively chosen from a written alphabet. Its allowed range is derived from the remainder condition.

49.2 Repeated quotient–remainder extraction

Let N_0 be a finite occurrence report with count n_0. Divide by the base carrier:

$$n_0 = q_0 b + d_0, \qquad 0 \leq d_0 < b.$$

The remainder d_0 is the least-position digit report. The quotient carrier becomes the next dividend:

$$q_0 = q_1 b + d_1.$$

Continue recursively:

$$q_k = q_{k+1} b + d_{k+1}, \qquad 0 \leq d_{k+1} < b.$$

Because the initial report is finite and each positive quotient is smaller than its preceding positive dividend when $b \geq 2$, the process terminates at some stage m with

$$q_m = 0.$$

Definition 49.3 (Finite positional ledger)**.** *The* finite positional ledger *of N_0 in base b is the ordered remainder sequence*

$$\Lambda_b(N_0) = [d_m, d_{m-1}, \ldots, d_1, d_0],$$

together with the retained base, digit indices, quotient chain, and termination zero.

The displayed numeral

$$d_m d_{m-1} \cdots d_1 d_{0b}$$

is a compressed projection of this ledger.

Theorem 49.4 (Finite base-expansion theorem)**.** *Every finite natural-number report N_0 and every declared base $b \geq 2$ generate a terminating quotient–remainder ledger. The resulting digit sequence reconstructs the original count uniquely when the base and digit positions are retained.*

Proof. At each step, quotient–remainder uniqueness gives unique q_k and d_k. The quotient chain strictly descends while positive, so it terminates. Substituting the equations backward reconstructs the initial count. Uniqueness follows recursively from uniqueness at each division step. □

49.3 Place weights without primitive powers

Define the first place carrier as the unit report

$$W_0 = U.$$

Define each later place carrier recursively by repeated base product:

$$W_{k+1} = B_b \times W_k.$$

Its count is

$$|W_{k+1}| = b|W_k|.$$

Only after repeated multiplication has been established may this count be abbreviated as

$$|W_k| = b^k.$$

Thus exponent notation is a compressed report of a recursively constructed place carrier.

The complete reconstruction of the numeral ledger is

$$N_0 \cong \bigsqcup_{k=0}^{m} (D_k \times W_k),$$

where D_k is a digit carrier of count d_k. Its scalar projection is

$$n_0 = \sum_{k=0}^{m} d_k b^k.$$

The sum and powers are not primitive ingredients; they summarize the retained place-product ledger.

49.4 A zero digit is an indexed placeholder

A zero digit does not mean that the report position is absent. It means that no positive copy of the corresponding place carrier contributes at that retained index.

For example, the base-two report

$$[1, 0, 1]_2$$

contains three digit positions. The middle position is occupied by a constructive zero digit reporting no 2^1-place contribution. Removing that position would change the positional report:

$$[1, 0, 1]_2 \neq_{\mathrm{C}} [1, 1]_2.$$

Their reconstructed counts are different because the remaining 1 digits are attached to different place carriers.

Theorem 49.5 (Positional-zero theorem)**.** *Whenever a zero digit occurs between retained positional digits, it is a whole indexed report object preserving a place relation. It cannot be replaced by literal absence without changing the positional ledger.*

Leading zero digits require a declared width or boundary to remain significant. Thus

$$[0,1]_2$$

and

$$[1]_2$$

may share a normalized scalar count while remaining different fixed-width reports. Normalization that deletes leading zeros is another projection and must be declared.

49.5 Base dependence and report equality

The same constructive count may admit different positional reports in different bases. For example,

$$[1,0]_2, \qquad [2]_{10},$$

report the same count but use different base carriers, digit alphabets, and place ledgers.

Therefore

$$[1,0]_2 =_{\text{count}} [2]_{10}$$

does not imply full constructive identity. Positional equality is typed by at least:

(a) the base carrier;

(b) the ordered digit positions;

(c) the digit values;

(d) any retained width or leading-zero boundary;

(e) the radix boundary, when present;

(f) and the quotient–remainder construction history if that history is relevant.

49.6 Radix expansion of a proper ratio

Let a proper constructive ratio have numerator count A and positive denominator count D:

$$0 < A < D.$$

Its initial quotient–remainder state is

$$A = 0 \cdot D + A.$$

To generate a base-b positional report after the radix boundary, replicate the current remainder by the base carrier and divide by the retained denominator:

$$bR_k = d_{k+1}D + R_{k+1}, \qquad 0 \le R_{k+1} < D.$$

The quotient d_{k+1} is the next digit report, and the remainder R_{k+1} carries the unresolved ratio obligation into the next position.

Definition 49.6 (Finite radix-state prefix)**.** *After m recurrence steps, the finite radix-state prefix is*

$$0.d_1 d_2 \cdots d_m{}_b$$

together with the retained remainder R_m and denominator D.

The written prefix alone is not the complete ratio. The exact finite state is

$$(d_1, \ldots, d_m; R_m, D, b).$$

49.7 Examples of base-dependent ratio reports

For the retained one-to-two ratio $\mathcal{R}(U, D_2)$, conventionally displayed as $1/2$, a base-two coordinate compiler begins with remainder $R_0 = 1$ and denominator $D = 2$:

$$2R_0 = 2 = 1 \cdot 2 + 0.$$

Thus the first digit is 1 and the remainder closes:

$$\frac{1}{2} = 0.1_2.$$

For the same retained one-to-two relation, the base-ten coordinate compiler gives

$$10R_0 = 10 = 5 \cdot 2 + 0,$$

so

$$\frac{1}{2} = 0.5_{10}.$$

The ratio construction is the same; the positional reports differ because the base carriers differ.

For $1/3$ in base ten,

$$10 \cdot 1 = 3 \cdot 3 + 1.$$

The next remainder is again 1, so the same state recurs and the digit 3 repeats. The exact constructive object remains the ratio $\mathcal{R}(1, 3)$ together with a one-state decimal remainder cycle. The notation

$$0.333\ldots_{10}$$

is a report of that recurring state, not an infinite pile of already completed written digits.

49.8 Termination and recurrence

Theorem 49.7 (Termination criterion)**.** *A radix expansion terminates exactly when some generated remainder is the typed exact-remainder zero:*

$$R_m = 0_{A:D,b}.$$

Theorem 49.8 (Finite remainder recurrence)**.** *For a fixed positive denominator D, every nonzero remainder satisfies*

$$1 \leq R_k < D.$$

There are only finitely many possible positive remainder counts. Therefore a nonterminating radix recursion must eventually revisit a prior remainder state, after which its digit and remainder sequence recurs under the same deterministic division rule.

Proof. The remainder condition restricts each state to one of $D-1$ positive counts. If no zero state is reached, a deterministic sequence with more than $D-1$ steps must repeat a prior state. Equal current remainder, denominator, and base produce equal next quotient and remainder by division uniqueness. □

Thus every rational ratio has either a terminating or recurring positional ledger in a finite base grammar. The recurrence is a finite-state property of retained remainders.

49.9 The radix point is a report boundary

The radix point is not a geometric point inserted into the counted object. It is a report marker separating complete base-place extractions from recursively scaled residual-place reports.

To the left of the radix boundary, positions report complete place carriers. To the right, positions report repeated base scaling of an unresolved remainder relative to the retained denominator. The boundary is part of the notation grammar, not a primitive location in reality.

49.10 Finite prefixes are not completed limits

A finite prefix of a recurring expansion retains a nonzero remainder. Therefore it is not constructively identical to a completed scalar that a later real-number grammar may associate with the full recurrence.

For example, each finite decimal prefix

$$0.9, \quad 0.99, \quad 0.999, \quad \ldots$$

retains a positive unresolved remainder relative to its finite place carrier. No finite prefix is constructively equal to the complete unit report. Any later statement identifying an indefinitely recurring positional rule with 1 must specify a completion or limit operation and prove what information that operation preserves.

This manuscript has not yet introduced such a completion. It therefore retains the distinction among:

(1) a finite digit prefix;

(2) its current remainder state;

(3) a finite recurring-state description;

(4) and a later completed real-number report.

49.11 Positional-report preservation theorem

Theorem 49.9 (Positional-report preservation)**.** *A positional numeral is a lossless report of its finite quotient–remainder construction only when the following are retained:*

(i) the base carrier;

(ii) digit order and whole digit indices;

(iii) the value of every digit, including zero placeholders;

(iv) the radix boundary when present;

(v) the termination zero or current remainder;

(vi) and any fixed-width or normalization rule relevant to identity.

If any of these are discarded, equal displayed values need not imply equal constructive reports.

Before completion and limits can be stated without circularity, the next section constructs matching and unmatched residuals before any signed coordinate is admitted. Conventional negative notation will then be introduced only as a frame-dependent projection of those retained residual roles.

Part VIII
Matching, Residuals, and Signed Coordinate Projection

50 Removal, unmatched obligation, and the origin of signed notation

Primitive removal never constructs a negative multiplicity. Suppose a source report contains A whole occurrences and a request asks for B matching removals. Construct a maximal source-preserving matching with common matched carrier C and residual carriers R_A, R_B satisfying

$$A = C + R_A, \qquad B = C + R_B,$$

with no further admissible match between R_A and R_B. Under universal count-unit matching, at least one residual carrier is the typed zero carrier.

Definition 50.1 (Matched-residual report)**.** *The primitive comparison/removal report is*

$$\mathcal{D}(A, B) = (C, R_A, R_B; \Gamma),$$

where C is the realized common matching, R_A is unmatched source contribution, R_B is unmet request, and Γ retains the declared identity criterion and matching history.

The report is wholly nonnegative as an occurrence construction. It distinguishes which role remains without placing either role on a signed coordinate axis.

Example 50.2 (Five apples and a request for six)**.** *Let five apple occurrences be present and let a request demand six apple removals. Maximal matching gives*

$$C = 5, \qquad R_A = 0, \qquad R_B = 1.$$

Thus five apples are removed, zero apples remain, and one removal request is unfulfilled. No occurrence report contains a negative apple and no primitive count is -1.

Principle 50.3 (No negative existence)**.** *A negative scalar is never interpreted as a negative number of existing occurrences. Primitive construction stops at the typed residual relation (R_A, R_B). A minus sign may be introduced only by a later coordinate frame that chooses one residual role as the opposing direction of a displayed axis.*

50.1 Equivalence and zero before signed difference

Two compared finite reports are equivalent under a declared comparison grammar exactly when both unmatched residuals close:

$$R_A = 0_A, \qquad R_B = 0_B.$$

This typed no-residual report is prior to any scalar equation such as $A - B = 0$.

Definition 50.4 (Primitive comparison zero)**.** *The primitive comparison zero associated with $\mathcal{D}(A, B)$ is the retained report*

$$0_{\mathcal{D}(A,B)} = (C, 0_A, 0_B; \Gamma_0),$$

recording that the declared matching exhausted both roles.

50.2 Optional signed-coordinate compiler

A conventional signed frame may now be declared. Let $\mathcal{F}_\pm$ be a reporting convention that maps residual role R_A to one directed side of a reference axis and residual role R_B to the opposing side. Only after that declaration define the scalar shadow

$$\pi_{\mathcal{F}_\pm}(\mathcal{D}(A, B)) := |R_A| - |R_B|.$$

The displayed subtraction belongs to the declared coordinate grammar. It does not replace the primitive report $(C, R_A, R_B; \Gamma)$.

Definition 50.5 (Integer-coordinate report)**.** *An integer-coordinate report is a pair consisting of a matched-residual construction and a declared signed projection frame:*

$$\mathcal{Z} = (\mathcal{D}(A, B), \mathcal{F}_\pm).$$

The conventional integer glyph is the value of $\pi_{\mathcal{F}_\pm}$ and is not the complete PCL report.

Principle 50.6 (Frame dependence of negativity)**.** *Without a declared opposing coordinate frame, PCL has unmatched role R_A or unmatched role R_B, not positive and negative numbers. Changing the frame may reverse the displayed sign while leaving the primitive matching report unchanged.*

50.3 Coordinate addition and inverse

Two matched-residual reports may be source-preservingly joined before any coordinate normalization. If the same signed frame is then applied to both, conventional integer addition is the scalar shadow of joining like residual roles and rematching opposing residual roles.

The conventional additive inverse operation is therefore a frame operation exchanging the two displayed residual roles. It does not create a primitive negative object. In coordinate notation only,

$$\pi_{\mathcal{F}_\pm}(R_A, R_B) = -\pi_{\mathcal{F}_\pm}(R_B, R_A).$$

Theorem 50.7 (Signed arithmetic is a provenance-preserving projection)**.** *Whenever the conventional signed compiler is applied to finite matched-residual reports, its ordinary addition, subtraction, and inverse identities are valid as coordinate identities, provided the underlying source joins, matchings, and residual roles are retained. Equality of the signed scalar shadows does not imply equality of those construction ledgers.*

50.4 Use of negative notation downstream

From this point onward, a displayed negative integer, signed ratio, negative exponent, or signed error is shorthand inside a declared coordinate grammar. Its complete PCL state retains the underlying role whose scalar shadow was assigned the opposing sign. In particular, a negative exponent abbreviates a reciprocal/inverse obligation after reciprocal construction has independently been earned; it is not a negative repetition count.

Principle 50.8 (No signed coordinate used as its own foundation)**.** *No later theorem about $\mathbb{Z}$, signed real coordinates, vector components, or complex pairs may be used to prove that primitive opposition or negative multiplicity existed before the signed frame was declared.*

Part IX
Powers, Roots, and Operational Depth

51 Powers, repeated composition, and root reconstruction

Multiplication constructs an ordered source product from two declared carriers. A power requires a further relation: the same declared carrier is repeatedly composed with itself under a retained depth index. The scalar notation a^n is therefore not primitive. It is the count projection of a recursively constructed product ledger whose base, depth, grouping, orientation, and zero-depth convention remain part of the complete report.

This section constructs finite powers before any appeal to infinite series, exponential functions, irrational roots, limits, or completion. It also constructs exact root states and finite root-shell residuals. Non-exact roots that require a completed real carrier are deliberately postponed.

51.1 Power depth and repeated self-product

Let $A = (I_A, v_A)$ be a finite occurrence report and let K_n be a finite depth report containing n whole depth indices.

Definition 51.1 (Positive-depth power ledger)**.** *For positive depth, define the power ledger recursively by*

$$\mathrm{Pow}_1(A) = A,$$

and

$$\mathrm{Pow}_{n+1}(A) = \mathrm{Pow}_n(A) \times A.$$

An occurrence of $\mathrm{Pow}_n(A)$ *is an ordered source tuple*

$$(a_{i_1}, a_{i_2}, \ldots, a_{i_n}),$$

with one whole occurrence selected from A at every retained depth position.

The depth positions are not copies of a scalar exponent. They are report indices selecting how many self-product stages are retained. Their order records which factor role each occurrence occupies.

Definition 51.2 (Power source address)**.** *The complete source address of a positive-depth power occurrence is*

$$(k_1, i_1;\, k_2, i_2;\, \ldots;\, k_n, i_n),$$

where k_j is the jth depth role and i_j selects an occurrence from A.

After the recursive construction has been established, the product index carrier may be abbreviated by

$$I_A^n$$

for the n-fold ordered product of I_A with itself. The notation does not create the product; it names the already constructed depth ledger.

Theorem 51.3 (Finite power count)**.** *If A has constructive count a and K_n has positive count n, then*

$$\mathrm{Count}(\mathrm{Pow}_n(A)) = a^n,$$

where a^n abbreviates the count of the n-fold self-product ledger.

Proof. At depth one, $\mathrm{Pow}_1(A) = A$, so the count is a. Suppose the depth-n ledger has count a^n. The next ledger is its placement product with A, so multiplication gives

$$\mathrm{Count}(\mathrm{Pow}_{n+1}(A)) = \mathrm{Count}(\mathrm{Pow}_n(A))\, a = a^n a.$$

The scalar abbreviation for this recursively repeated product is a^{n+1}. □

Thus the dependency is

base carrier ⟶ depth carrier ⟶ ordered self-product ⟶ power ledger ⟶ scalar power report.

51.2 Depth one and depth zero

Depth one retains exactly one factor role, so

$$\mathrm{Pow}_1(A) = A$$

and

$$a^1 = a.$$

Depth zero requires care. It cannot be literal absence, because a zero-depth operation must still be distinguishable as an operation report.

Definition 51.4 (Zero-depth product unit)**.** *The zero-depth power ledger of A is the typed singleton*

$$\mathrm{Pow}_0(A) = U_{\times,A} = [\varepsilon_A],$$

where ε_A reports one completed empty factor selection: no factor occurrence was requested, and the product obligation is closed at depth zero.

The object ε_A is not an occurrence of A. It is a typed product-identity report. Its count is one:

$$\mathrm{Count}(U_{\times,A}) = 1.$$

Therefore the finite fold convention reports

$$a^0 = 1.$$

Theorem 51.5 (Zero-depth identity)**.** *For every declared finite base report A,*

$$\mathrm{Pow}_0(A) \times \mathrm{Pow}_n(A) \cong \mathrm{Pow}_n(A).$$

Proof. The zero-depth carrier contains one identity occurrence. Pairing every depth-n tuple with that single occurrence changes no choice count and admits a canonical normalization that removes the typed identity role. □

Remark 51.6 (The underspecified symbol 0^0)**.** *Inside the finite product-fold grammar, a zero-depth fold has one empty tuple, so a typed expression*

$$\mathrm{Pow}_0(0_A)$$

reports the multiplicative unit. A bare expression 0^0 may also be used in later analytic, limiting, functional, or combinatorial grammars with different boundary requirements. Those grammars are not automatically identical. Therefore the untyped symbol 0^0 has no universal interpretation merely from its typography; the governing operation and domain must be declared.

For every positive depth n,

$$\mathrm{Pow}_n(0_A) = 0_{\mathrm{Pow}_n(A)},$$

because a positive-depth tuple requires a positive selection from the base at every factor role, and the zero base supplies no positive contribution.

51.3 Concatenation, flattening, and the finite exponent laws

The familiar exponent laws arise from explicit transformations of depth tuples.

Theorem 51.7 (Depth concatenation)**.** *For nonnegative depths m and n,*

$$\mathrm{Pow}_{m+n}(A) \cong \mathrm{Pow}_m(A) \times \mathrm{Pow}_n(A).$$

Consequently,

$$a^{m+n} = a^m a^n.$$

Proof. A depth-$(m+n)$ tuple has an initial block of m factor roles and a final block of n factor roles. Splitting at that retained boundary gives one depth-m tuple paired with one depth-n tuple. Concatenation reverses the map. The two ledgers are therefore in bijective, order-preserving correspondence. □

Theorem 51.8 (Depth flattening)**.** *For nonnegative depths m and n,*

$$\mathrm{Pow}_n(\mathrm{Pow}_m(A)) \cong \mathrm{Pow}_{mn}(A).$$

Consequently,

$$(a^m)^n = a^{mn}.$$

Proof. An occurrence on the left is an ordered family of n blocks, each containing m ordered selections from A. Flattening the block addresses gives one ordered tuple with mn factor roles. Restoring the block boundaries reverses the construction. □

Theorem 51.9 (Product power decomposition)**.** *For nonnegative depth n,*

$$\mathrm{Pow}_n(A \times B) \cong \mathrm{Pow}_n(A) \times \mathrm{Pow}_n(B).$$

Consequently,

$$(ab)^n = a^n b^n.$$

Proof. Each depth role on the left selects an ordered pair (a_i, b_j). Unzipping the n selected pairs gives one ordered A-tuple and one ordered B-tuple. Zipping the corresponding depth positions reverses the map. □

These are correspondence theorems among retained ledgers. They do not state that the differently grouped source syntax is literally identical.

51.4 Equal scalar powers and unequal power constructions

Power projection can collapse both base and depth information.

Example 51.10 (Equal value, different depth ledger)**.** *The scalar reports*

$$2^4 = 16 \qquad \textit{and} \qquad 4^2 = 16$$

are equal, but their construction ledgers are different. The first contains four binary factor roles. The second contains two four-choice factor roles. A value-only projection identifies their counts while discarding base carrier, depth carrier, tuple arity, and grouping history.

Even more sharply,

$$1^1 = 1^2 = 1^3 = \cdots = 1$$

at the scalar level, while the corresponding power ledgers retain different numbers of typed factor roles. A normalization may prove those roles irrelevant for a particular property, but scalar equality alone does not do so.

Principle 51.11 (Power projection)**.** *From*

$$a^m = b^n$$

one may infer equal scalar power reports. One may not infer equal bases, equal depths, equal tuple ledgers, or equal operational histories without an additional preservation theorem.

51.5 Signed bases and parity of depth

Signed exponentiation is repeated signed multiplication. Every occurrence remains positively countable as an occurrence; the sign reports the orientation of the resulting product lane.

Definition 51.12 (Parity report)**.** *For a nonnegative depth count n, divide by the two-occurrence carrier:*

$$n = 2q + r, \qquad r \in \{0, 1\}.$$

The depth is even *when $r = 0$ and* odd *when $r = 1$.*

Parity is therefore a quotient–remainder report, not a primitive label attached to the numeral.

Theorem 51.13 (Orientation of a finite signed power)**.** *Let X be a normalized nonzero signed integer report and let $n > 0$.*

(i) If X is positively oriented, then X^n is positively oriented.

(ii) If X is negatively oriented and n is even, then X^n is positively oriented.

(iii) If X is negatively oriented and n is odd, then X^n is negatively oriented.

Proof. Each multiplication of two equally negative orientations contributes to the positive product lane. An even depth partitions all negative factors into pairs. An odd depth leaves one unmatched negative orientation after all possible pairs have been formed. Positive factors never change the orientation. □

At the scalar level this yields the familiar reports

$$(-a)^{2q} = a^{2q}, \qquad (-a)^{2q+1} = -a^{2q+1}.$$

The right-hand signs are reports of the completed orientation ledger, not signs attached independently to each counted occurrence.

51.6 Negative exponents as reciprocal power reports

A negative exponent cannot mean a negative number of multiplication stages. The depth magnitude remains a nonnegative whole count. The negative orientation of the exponent instructs the report to form a reciprocal after the positive-depth power has been constructed.

Definition 51.14 (Signed reciprocal)**.** *Let X be a nonzero signed integer report with magnitude carrier M_X. Its signed reciprocal is the unit-to-magnitude ratio*

$$\mathrm{Rec}_{\mathbb{Z}}(X) = \mathrm{sgn}(X)\,\mathcal{R}(U, M_X),$$

where the ratio receives the same orientation as X.

Definition 51.15 (Negative power)**.** *For positive n and nonzero X, define*

$$X^{-n} = \mathrm{Rec}_{\mathbb{Z}}(X^n).$$

Equivalently,

$$X^{-n} = \frac{1}{X^n}$$

is the scalar report of that signed reciprocal relation.

Thus the minus sign in the exponent does not negate the product count. It changes the final operation from retained product to reciprocal-of-product.

Theorem 51.16 (Positive and negative depth closure)**.** *For nonzero X and positive n,*

$$X^n \cdot X^{-n} =_{\mathrm{ratio}} 1.$$

Proof. By definition, X^{-n} is the unit-to-X^n ratio with the appropriate orientation. Multiplying the magnitude carrier by its reciprocal gives an exact common-carrier cancellation, while the equal orientations multiply to positive orientation. □

Zero has no negative power because the construction would require the reciprocal of a zero-contribution denominator:

$$0^{-n} \quad \text{is undefined for } n > 0.$$

51.7 Powers of retained ratios

Let $\mathcal{R}(A, B)$ be a constructive ratio with positive denominator carrier B.

Definition 51.17 (Finite ratio power)**.** *For nonnegative depth n, define*

$$\mathcal{R}(A, B)^n = \mathcal{R}(\mathrm{Pow}_n(A), \mathrm{Pow}_n(B)).$$

Its scalar projection is

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

At depth zero, both numerator and denominator are typed unit carriers, giving the ratio report one. At positive depth, a zero numerator produces a zero ratio, while the denominator remains positive.

A reduction after powering may remove common replication carriers, but the reduced ratio does not by itself preserve the original base ratio or power depth.

51.8 Roots as inverse power reconstruction

A root is not primitive extraction of a fractional occurrence. It is the inverse reconstruction problem for a previously defined finite power map.

Definition 51.18 (Exact constructive root)**.** *Let N be a nonnegative finite count report and let $n > 0$ be a depth report. A nonnegative report Q is an exact constructive nth root of N when*

$$\mathrm{Pow}_n(Q) =_{\mathrm{count}} N.$$

If the full source structure is also required to match, one must strengthen this to a declared ledger correspondence

$$\mathrm{Pow}_n(Q) \cong N.$$

The distinction matters. Equal counts can certify a scalar root without proving that the radicand was actually constructed by the candidate root ledger.

Definition 51.19 (Perfect power)**.** *A nonnegative count report N is a perfect nth power when an exact finite root report exists at depth n.*

For example, the scalar report 16 is both a perfect square and a perfect fourth power:

$$16 = 4^2 = 2^4.$$

These are different inverse reconstructions of the same projected count.

51.9 Monotonicity and uniqueness of nonnegative roots

Lemma 51.20 (Strict monotonicity of positive-depth natural powers)**.** *Let $0 \le a < b$ and let $n > 0$. Then*

$$a^n < b^n.$$

Proof. A strict containment of finite carriers gives an injection from the a-carrier into the b-carrier together with at least one unmatched occurrence in the latter. Applying the injection at every depth position embeds every a-tuple into a b-tuple. Since $n > 0$, a tuple using an unmatched b-occurrence at one retained position lies outside the image. Hence the product containment is strict. □

Theorem 51.21 (Uniqueness of the nonnegative exact root)**.** *For fixed positive depth n, a nonnegative finite count has at most one nonnegative exact scalar nth root.*

Proof. If $q_1 < q_2$, strict monotonicity gives $q_1^n < q_2^n$, so the two powers cannot both equal the same count. □

This theorem establishes uniqueness only for the nonnegative scalar root report. Different source ledgers can still share the same root count.

51.10 Finite root shells and retained residuals

Most finite counts are not perfect powers at a chosen depth. The constructive inverse state must retain the nearest complete power below the radicand and the unresolved residual above that shell.

Definition 51.22 (Lower root-shell index)**.** *For a nonnegative finite count N and positive depth n, the lower root-shell index is the unique nonnegative count q satisfying*

$$q^n \le N < (q+1)^n.$$

Existence follows because the finite sequence

$$0^n, 1^n, 2^n, \ldots, N^n$$

contains a first power exceeding N unless N is already bounded at the preceding stage; strict monotonicity gives uniqueness.

Definition 51.23 (Root-shell residual)**.** *Given the lower root-shell index q, define*

$$\rho_n(N) = N - q^n.$$

The complete finite root state is

$$\mathrm{RootState}_n(N) = (N, n, q, \rho_n(N)),$$

with

$$0 \le \rho_n(N) < (q+1)^n - q^n.$$

The upper bound is the width of the next power shell. The state is exact precisely when

$$\rho_n(N) = 0_{N:q^n}.$$

Theorem 51.24 (Exact-root closure)**.** *A finite nonnegative count N has exact nth root q if and only if its root-shell residual is the typed zero*

$$\rho_n(N) = 0.$$

The residual is not an error to be silently discarded. It is the positive source difference between the radicand and the largest completed $n$fold self-product below it.

Example 51.25 (Non-exact square state)**.** *For $N = 10$ and depth two,*

$$3^2 = 9 \le 10 < 16 = 4^2.$$

The finite root state is

$$\mathrm{RootState}_2(10) = (10, 2, 3, 1).$$

The report 3 is the lower square-shell index, not an exact square root of 10. Any later decimal or real report for $\sqrt{10}$ must refine this retained shell relation rather than pretending the residual was already zero.

51.11 Signed roots

Signed root reconstruction depends on depth parity.

Theorem 51.26 (Finite signed-root conditions)**.** *Let $n > 0$.*

(i) A positive exact radicand has one nonnegative exact root. When n is even and the root magnitude is nonzero, it also has an oppositely oriented root with the same scalar power.

(ii) A negative exact radicand can have an integer root only when n is odd; that root is negatively oriented.

(iii) A negative radicand has no even-depth integer root.

(iv) The zero radicand has the zero root for every positive depth.

Proof. The parity theorem for signed powers shows that every even-depth power is nonnegative, while odd-depth power preserves the base orientation. Nonnegative magnitude uniqueness then fixes the root magnitude whenever an exact root exists. □

For a positive radicand and even depth, the two signed solutions

$$q \qquad \text{and} \qquad -q$$

are source-distinct orientations with equal scalar power. A *principal root* is obtained only after a convention selects the nonnegative orientation. The radical symbol alone should not be allowed to hide that convention.

51.12 Exact roots of ratios

Let $\mathcal{R}(A, B)$ have positive denominator B.

Definition 51.27 (Exact rational root)**.** *A ratio $\mathcal{R}(C, D)$ is an exact nth root of $\mathcal{R}(A, B)$ when*

$$\mathcal{R}(\mathrm{Pow}_n(C), \mathrm{Pow}_n(D)) =_{\mathrm{ratio}} \mathcal{R}(A, B).$$

A sufficient source-faithful condition is that numerator and denominator admit exact roots:

$$A \cong \mathrm{Pow}_n(C), \qquad B \cong \mathrm{Pow}_n(D).$$

But ratio equality can also permit common-carrier reductions, so exact scalar rational-root status must be checked after declared normalization.

Not every positive rational ratio has a rational exact root. For example, no finite rational carrier has square equal to the ratio 2/1. Constructing such a root requires a later completion grammar. The present section records only the finite shell and exact-ratio predicates; it does not manufacture an irrational object prematurely.

51.13 Power words are not automatically geometric forms

The words *square* and *cube* are often used for both operational depth and geometric enclosure. Those meanings must be separated.

Definition 51.28 (Operational square)**.** *The operational square of A is the two-depth product ledger*

$$\mathrm{Pow}_2(A) = A \times A.$$

It contains ordered pairs of A-occurrences.

Definition 51.29 (Operational cube)**.** *The operational cube of A is the three-depth product ledger*

$$\mathrm{Pow}_3(A) = (A \times A) \times A,$$

up to declared reassociation.

Neither construction alone supplies geometric enclosure. A geometric square additionally requires a declared Euclidean compiler with whatever incidence, closure, equal-length, and angle conditions that compiler uses. A geometric cube likewise requires an independently admitted rank-and-coordinate grammar and cannot be inferred from scalar third power alone.

Therefore

$$A^2$$

means repeated product depth unless a geometric interpretation has been separately constructed. It does not prove that A has been arranged as a Euclidean square.

Likewise, taking a scalar square root of a coordinate-area report does not recover a primitive line object. It returns a side-coordinate magnitude only inside a declared Euclidean square grammar. No primitive area object is generated merely by squaring or rooting a scalar report.

Principle 51.30 (Operational depth before geometric naming)**.** *Powers count repeated operational composition. Geometric squares, cubes, and higher-dimensional objects require an additional coordinate compiler whose constraint roles, incidence rules, metric comparisons, and closure predicates are independently declared.*

51.14 Power units and typed operations

When a base report carries a declared unit or object type, repeated product changes the type of the report.

If A has unit type u, then the depth-n product carries the recursively constructed product type

$$u^{\otimes n}.$$

Only after a unit algebra has been declared may this be abbreviated as u^n.

An exact root operation must therefore invert not only a scalar count but also the declared product type. A numerical equality such as

$$q^n = N$$

does not prove that a typed quantity Q is a valid root of a typed radicand unless the unit and construction grammars also close.

This prevents a scalar coincidence from being mistaken for dimensional or semantic identity.

51.15 Operational-depth preservation theorem

Theorem 51.31 (Power-ledger preservation)**.** *A scalar power report preserves its finite constructive source only when the reporting map retains, or proves irrelevant, all of the following:*

(i) the base carrier and its type;

(ii) the depth carrier and whether its orientation is positive, zero, or negative;

(iii) the ordered factor roles and tuple addresses;

(iv) the grouping and reassociation rule;

(v) the signed orientation and parity route;

(vi) any reciprocal construction used for negative depth;

(vii) the zero-depth identity convention;

(viii) and every residual or typed zero relevant to an inverse root state.

Equal scalar powers do not by themselves imply equal power ledgers.

Theorem 51.32 (Root-state preservation)**.** *A reported root value preserves its finite inverse construction only when it retains, or proves irrelevant:*

(i) the radicand source;

(ii) the power depth;

(iii) the selected orientation or principal-root convention;

(iv) the candidate root carrier;

(v) the exact power comparison;

(vi) the lower and upper power shells when the root is not exact;

(vii) and the root-shell residual.

A decimal approximation without this state is a later projection, not the primitive root construction.

The finite dependency now extends as

$$\begin{aligned}\text{multiplication} &\longrightarrow \text{repeated self-product} \longrightarrow \text{power depth}\\ &\longrightarrow \text{signed and reciprocal powers} \longrightarrow \text{inverse power constraint}\\ &\longrightarrow \text{exact root or retained root-shell residual.}\end{aligned}$$

The next section must distinguish a finite recursive rule from a completed infinite object. Only after that distinction is explicit can recurring expansions, unbounded power families, sequences, limits, and completion be introduced without treating an ellipsis as though it were an already constructed infinity.

Part X
Relations, Mappings, and Indexed Construction Families

52 Relations, mappings, and indexed construction families

The preceding sections have already used source-preserving correspondences, projections, transpositions, reassociations, extraction rules, and recurrence steps. Those constructions now permit a general account of relation and function. A function must not be inserted as a primitive black box. It is a declared rule relating already distinguishable source occurrences to already distinguishable target occurrences.

The present section remains finite. It constructs relations, mappings, composition, inverse relations, fibers, and indexed families without yet asserting a completed infinite domain or an infinite sequence.

52.1 A relation begins with retained source and target roles

Let A and B be constructed reports. Their placement product

$$A \times B$$

contains ordered pairs (a, b) with a retained source role from A and a retained target role from B.

Definition 52.1 (Binary relation report)**.** *A binary relation from A to B is a source-preserving subreport*

$$\mathcal{R} \preceq_{\mathrm{C}} A \times B.$$

An occurrence $(a, b) \in \mathcal{R}$ reports that the source occurrence a is related to the target occurrence b under the declared relation.

The order of the pair is part of the relation ledger. In general,

$$(a, b) \neq_{\mathrm{C}} (b, a).$$

A symmetric scalar statement about the pair does not erase its source and target roles.

Definition 52.2 (Relation membership)**.** *For a relation report $\mathcal{R} \preceq_{\mathrm{C}} A \times B$, write*

$$a \,\mathcal{R}\, b$$

when the ordered occurrence (a, b) is admitted to $\mathcal{R}$.

The notation $a \,\mathcal{R}\, b$ is therefore a projection of the retained pair occurrence. It does not create the relation.

52.2 Converse, diagonal, and relational identity

Definition 52.3 (Converse relation)**.** *The converse of $\mathcal{R} \preceq_{\mathrm{C}} A \times B$ is*

$$\mathcal{R}^{-1} = \{(b, a) : (a, b) \in \mathcal{R}\} \preceq_{\mathrm{C}} B \times A.$$

At this stage the superscript -1 means reversal of relation orientation. It does not yet assert that the converse is a function.

Definition 52.4 (Diagonal relation)**.** *The diagonal relation on A is*

$$\Delta_A = \{(a, a) : a \in A\} \preceq_{\mathrm{C}} A \times A.$$

The diagonal retains two roles for the same admitted occurrence: source role and target role. Its scalar values may agree, but the relation still records an input occurrence and an output occurrence.

Definition 52.5 (Relational composition)**.** *Let*

$$\mathcal{R} \preceq_{\mathrm{C}} A \times B, \qquad \mathcal{S} \preceq_{\mathrm{C}} B \times C.$$

Their composition is the relation

$$\mathcal{S} \circ \mathcal{R} = \{(a, c) : \textit{there is a retained } b \in B \textit{ with } (a, b) \in \mathcal{R} \textit{ and } (b, c) \in \mathcal{S}\}.$$

The intermediate b is part of the full composition ledger. Projecting only (a, c) discards the route through B.

Theorem 52.6 (Relational identity)**.** *For every relation $\mathcal{R} \preceq_{\mathrm{C}} A \times B$,*

$$\mathcal{R} \circ \Delta_A = \mathcal{R}, \qquad \Delta_B \circ \mathcal{R} = \mathcal{R}$$

at the admitted-pair level.

Proof. The diagonal adds no new target choice. Composing on the source side retains exactly the pairs (a, b) already in $\mathcal{R}$, because the only diagonal intermediate paired with a is a itself. The target-side argument is analogous. □

52.3 Input support and output support

A relation need not use every occurrence of either carrier.

Definition 52.7 (Source support)**.** *The source support of $\mathcal{R} \preceq_{\mathrm{C}} A \times B$ is*

$$\mathrm{src}(\mathcal{R}) = \{a \in A : \textit{some } b \in B \textit{ satisfies } (a, b) \in \mathcal{R}\}.$$

Definition 52.8 (Target support)**.** *The target support of $\mathcal{R}$ is*

$$\mathrm{tgt}(\mathcal{R}) = \{b \in B : \textit{some } a \in A \textit{ satisfies } (a, b) \in \mathcal{R}\}.$$

These supports are later reported as domain and image when the relation satisfies the function condition. The ambient target carrier B remains distinct from the target occurrences actually reached.

52.4 Single-output construction

A general relation permits one source to relate to no target, one target, or several targets. A deterministic mapping requires a declared single-output condition.

Definition 52.9 (Single-valued relation)**.** *A relation $\mathcal{R} \preceq_{\mathrm{C}} A \times B$ is single-valued on a source report $D \preceq_{\mathrm{C}} A$ when, for every $a \in D$, any two retained pairs*

$$(a, b_1) \in \mathcal{R}, \qquad (a, b_2) \in \mathcal{R}$$

have the same target occurrence under the declared target identity:

$$b_1 =_{\mathrm{C}} b_2.$$

Equal projected target values are insufficient when target occurrence identity matters. If

$$\pi(b_1) = \pi(b_2)$$

but $b_1 \neq_{\mathrm{C}} b_2$, then the relation has two source-distinct target routes unless a later equivalence deliberately identifies them.

Definition 52.10 (Totality on a declared source)**.** *A relation $\mathcal{R} \preceq_{\mathrm{C}} A \times B$ is total on $D \preceq_{\mathrm{C}} A$ when every $a \in D$ occurs as the source of at least one admitted pair in $\mathcal{R}$.*

Definition 52.11 (Constructive mapping)**.** *A constructive mapping*

$$f : D \longrightarrow B$$

is a relation report $\Gamma_f \preceq_{\mathrm{C}} D \times B$ that is total and single-valued on D, together with any rule, trace, or source structure declared relevant to producing its target occurrences.

The pair report Γ_f is the graph projection of the mapping. The complete constructive mapping may contain more than the graph.

52.5 Rule, graph, and evaluation ledger

Three objects must be kept distinct.

Definition 52.12 (Mapping rule)**.** *A mapping rule ρ_f is a declared construction that, when supplied an admissible input occurrence, selects or constructs a target occurrence.*

Definition 52.13 (Extensional graph)**.** *The extensional graph of f is the pair report*

$$\Gamma_f = \{(a, f(a)) : a \in D\} \preceq_{\mathrm{C}} D \times B.$$

Definition 52.14 (Evaluation ledger)**.** *An evaluation ledger for input a is a retained construction*

$$\Lambda_f(a) = (a, \rho_f, \tau_a, r_a, f(a)),$$

where τ_a denotes any admitted transition route and r_a denotes any admitted residual, quotient, boundary, or closure state produced during evaluation.

The graph reports input-output pairs. The evaluation ledger reports how the output was constructed. Two mapping rules may have the same graph while retaining different evaluation ledgers.

Principle 52.15 (Graph equality is a projection)**.** *The implication*

$$\Gamma_f = \Gamma_g \Longrightarrow f =_{\mathrm{C}} g$$

is valid only if the declared identity criterion for mappings discards every rule, route, residual, and construction difference between f and g.

This does not reject extensional function equality. It identifies it as a declared quotient of richer constructive mappings.

52.6 Domain, codomain, and image

Definition 52.16 (Declared domain)**.** *The declared domain of*

$$f : D \to B$$

is the source report D on which the graph is total and single-valued.

Definition 52.17 (Declared codomain)**.** *The declared codomain is the target carrier B into which outputs are admitted.*

Definition 52.18 (Image)**.** *The image of f is the reached target support*

$$\mathrm{im}(f) = \{b \in B : \textit{some } a \in D \textit{ satisfies } f(a) =_{\mathrm{C}} b\}.$$

The codomain and image need not be equal. A codomain is part of the mapping declaration; the image is constructed by actual output occurrences.

Definition 52.19 (Partial mapping)**.** *A partial mapping from A to B is a single-valued relation whose source support is a proper subreport*

$$D \preceq_{\mathrm{C}} A.$$

It may be written

$$f : A \rightharpoonup B,$$

with constructed domain $D = \mathrm{src}(\Gamma_f)$.

Undefinedness is not automatically a target value. If the mapping has no admitted output for a, then either a lies outside the constructed domain or a separate typed failure report must be constructed in an enlarged codomain.

52.7 Injective, surjective, and bijective mappings

Definition 52.20 (Injective mapping)**.** *A mapping $f : D \to B$ is injective when*

$$f(a_1) =_{\mathrm{C}} f(a_2) \Longrightarrow a_1 =_{\mathrm{C}} a_2.$$

Equivalently, every reached target occurrence has at most one source occurrence in the graph.

Definition 52.21 (Surjective mapping)**.** *A mapping $f : D \to B$ is surjective onto its declared codomain when*

$$\mathrm{im}(f) =_{\mathrm{C}} B.$$

Definition 52.22 (Bijective mapping)**.** *A mapping is bijective when it is both injective and surjective.*

These properties depend on the chosen identity criteria for source and target occurrences. A projection may collapse several target occurrences to one scalar value and thereby destroy injectivity even when the pre-projection mapping was injective.

Example 52.23 (Projection destroys injectivity)**.** *Let*

$$D = \{(0,1),(1,0)\}$$

and let f be the identity mapping on these ordered cells. Then f is injective. Under the scalar projection

$$\pi_+(i,j) = i + j,$$

both outputs report 1. The composed report $\pi_+ \circ f$ is not injective on D. The failure belongs to the projection fiber, not to the original identity mapping.

52.8 Composition of mappings

Let

$$f : A \to B, \qquad g : B \to C.$$

Definition 52.24 (Constructive composition)**.** *The composite mapping*

$$g \circ f : A \to C$$

has graph

$$\Gamma_{g\circ f} = \{(a,c) : \text{the retained intermediate } b = f(a) \text{ satisfies } c = g(b)\}.$$

Its complete evaluation ledger retains

$$\Lambda_{g\circ f}(a) = (a, \Lambda_f(a), f(a), \Lambda_g(f(a)), g(f(a))).$$

Theorem 52.25 (Mapping composition)**.** *If $f : A \to B$ and $g : B \to C$ are total single-valued mappings, then $g \circ f : A \to C$ is total and single-valued.*

Proof. Each $a \in A$ has one target $b = f(a)$, and that b has one target $c = g(b)$. Therefore each a has one composite target c. □

At graph level, mapping composition is associative. At ledger level, the two parenthesizations

$$h \circ (g \circ f) \qquad \text{and} \qquad (h \circ g) \circ f$$

retain canonically corresponding nested evaluation structures. They need not be literally identical syntactic ledgers unless reassociation is included in the declared equality.

52.9 Identity mapping

Definition 52.26 (Identity mapping)**.** *The identity mapping on A is the mapping*

$$\mathrm{id}_A : A \to A$$

with graph

$$\Gamma_{\mathrm{id}_A} = \Delta_A.$$

For every $a \in A$,

$$\mathrm{id}_A(a) = a.$$

The identity output is the same admitted occurrence under the declared identity criterion. The evaluation ledger may still record that an identity operation was invoked.

Theorem 52.27 (Mapping identity laws)**.** *For mappings $f : A \to B$,*

$$\mathrm{id}_B \circ f = f, \qquad f \circ \mathrm{id}_A = f$$

at graph level. Ledger equality additionally requires the declared normalization of inserted identity steps.

52.10 Inverse relation and inverse mapping

The converse relation Γ_f^{-1} always exists as a reversed pair report. It is not always a mapping.

Theorem 52.28 (Inverse-function criterion)**.** *Let $f : A \to B$ be a mapping. Its converse graph defines a mapping*

$$f^{-1} : B \to A$$

exactly when f is bijective onto B.

Proof. The converse is single-valued precisely when no two source occurrences of f reach the same target, which is injectivity. It is total on B precisely when every target in B is reached, which is surjectivity. □

When f is bijective,

$$f^{-1} \circ f = \mathrm{id}_A, \qquad f \circ f^{-1} = \mathrm{id}_B$$

at graph level.

A noninjective mapping has inverse fibers rather than a single-valued inverse. A nonsurjective mapping has target occurrences with no converse source. These are structural conditions, not failures of notation.

52.11 Projections and fibers

Definition 52.29 (Reporting projection)**.** *A reporting projection is a mapping*

$$\pi : C \to V$$

from a constructive carrier C to a selected report carrier V.

A projection may preserve all relevant structure, or it may deliberately discard source, order, orientation, residual, route, width, base, or closure information.

Definition 52.30 (Fiber)**.** *For $v \in V$, the fiber of π over v is*

$$\pi^{-1}(v) = \{c \in C : \pi(c) = v\}.$$

Two constructive objects are projection-equal under π when they lie in the same fiber:

$$c_1 =_\pi c_2 \quad \Longleftrightarrow \quad \pi(c_1) = \pi(c_2).$$

Theorem 52.31 (Injective projection criterion)**.** *Projection equality implies constructive identity on a declared subreport $D \preceq_{\mathrm{C}} C$ exactly when the restricted projection*

$$\pi|_D : D \to V$$

is injective under the declared identities.

The theorem does not say that noninjective projections are invalid. It says that an additional proof is required before an output report may stand in for the full source object.

52.12 Higher-arity relations and mappings

An n-input relation is constructed on a product carrier

$$A_1 \times A_2 \times \cdots \times A_n \times B.$$

A mapping of n inputs is a total single-valued relation from the input product

$$A_1 \times \cdots \times A_n$$

to B.

Thus a binary operation on A is a mapping

$$\star : A \times A \to A.$$

Closure is not supplied by the symbol $\star$. It is the property that every admitted input pair has an admitted output in A.

The arithmetic operations constructed earlier can now be retrospectively encoded as mappings on their declared carriers. This encoding does not replace their source constructions.

52.13 Indexed construction families

A family is not initially an infinite set. It is a mapping from a constructed index report into a target carrier.

Definition 52.32 (Finite indexed family)**.** *Let I be a finite index report and let X be a target carrier. A finite indexed family in X is a mapping*

$$x : I \to X.$$

The occurrence at index i is written

$$x_i = x(i).$$

The full family report retains at least

$$(I, X, \Gamma_x),$$

and may additionally retain the rule and evaluation ledger producing each x_i.

Definition 52.33 (Finite sequence report)**.** *A finite sequence is an indexed family whose index carrier carries the constructed successor order*

$$i_0 \to i_1 \to \cdots \to i_m.$$

Order is therefore supplied by the index carrier and its successor relation. The target values need not themselves be ordered.

Definition 52.34 (Reindexing of a family)**.** *Given a bijection*

$$\sigma : J \to I,$$

the reindexed family is

$$x \circ \sigma : J \to X.$$

A reindexing may preserve the multiset of output reports while changing adjacency, order, recurrence, or causal route. Therefore a sequence property is reindexing-invariant only after that invariance is proved.

52.14 Transition maps and finite recurrences

Definition 52.35 (Transition map)**.** *A transition map on a state carrier X is a mapping*

$$T : X \to X.$$

Given an initial state x_0, finite iteration constructs

$$x_1 = T(x_0), \qquad x_2 = T(x_1), \qquad \ldots, \qquad x_m = T(x_{m-1}).$$

For a constructed finite index carrier

$$I_m = [0, 1, \ldots, m],$$

the recurrence ledger is the finite indexed family

$$x^{(m)} : I_m \to X, \qquad x^{(m)}(k) = T^k(x_0),$$

where T^k abbreviates the already constructed finite composition depth.

Definition 52.36 (Recurrence state)**.** *A recurrence state at index k is the complete ledger*

$$\Lambda_T(k) = (x_0, k, T, x_k, \tau_k),$$

where τ_k retains the sequence of intermediate transitions when that route is relevant.

Equal endpoint states need not imply equal recurrence histories. A cycle may return to a prior state after a positive transition depth:

$$x_j =_{\mathrm{C}} x_k \qquad \text{with} \qquad j < k.$$

The endpoint equality does not erase the completed transition count $k - j$.

52.15 Finite family equality

Several equality criteria are possible.

Definition 52.37 (Pointwise report equality)**.** *Families $x : I \to X$ and $y : I \to X$ are pointwise equal under a report projection π when*

$$\pi(x(i)) = \pi(y(i))$$

for every $i \in I$.

Definition 52.38 (Constructive family equality)**.** *The families are constructively equal when the index carrier, index roles, target occurrences, mapping rule, and every other declared relevant ledger component are preserved by the chosen correspondence.*

Pointwise scalar equality may leave different source routes at every index. Conversely, two differently indexed families may be isomorphic under a declared reindexing while not being literally the same report.

52.16 Mapping-preservation theorem

Theorem 52.39 (Constructive mapping preservation)**.** *A report of a mapping preserves the complete finite mapping only when it retains, or proves irrelevant, all of the following:*

(i) the source carrier and its identity criterion;

(ii) the target carrier and its identity criterion;

(iii) the admitted domain;

(iv) the input-output graph;

(v) the rule selecting or constructing each output;

(vi) every intermediate state, route, quotient, residual, boundary, and closure component relevant to the property under study;

(vii) the equality relation imposed on rules and evaluation ledgers;

(viii) and any normalization that deletes identity steps, source tags, ordering, or repeated equal outputs.

Equality of output graphs alone proves complete mapping identity only after all omitted components have been shown irrelevant under the declared identity criterion.

Theorem 52.40 (Indexed-family preservation)**.** *A finite indexed-family report preserves its construction only when it retains, or proves irrelevant:*

(i) the index carrier;

(ii) the successor, order, or adjacency relation on indices;

(iii) the target carrier;

(iv) the mapping from indices to target occurrences;

(v) the generating rule, when one is used;

(vi) the initial and terminal conditions;

(vii) and the evaluation route or recurrence ledger relevant to the stated property.

A displayed list of values is therefore not automatically the complete family object.

The dependency now extends as

$$\begin{aligned}\text{ordered product} &\longrightarrow \text{relation report} \longrightarrow \text{single-output condition} \longrightarrow \text{mapping}\\ &\longrightarrow \text{composition and inverse criteria} \longrightarrow \text{projection fibers}\\ &\longrightarrow \text{finite indexed families and recurrence ledgers.}\end{aligned}$$

The next section must distinguish a finite rule that admits another iteration from a completed infinite carrier. A finite recurrence description, an ellipsis, and an asserted infinity are not yet the same constructive object.

Part XI
Finite Rules and Indefinite Continuation

53 Finite rules, extensible prefixes, and continuation

The constructions developed so far are finite. A numeral has finitely many retained digit positions, a quotient–remainder computation has finitely many completed extraction stages, a power has finite operational depth, and a sequence report has a finite index carrier. Mathematics nevertheless uses notations such as

$$x_0, x_1, x_2, \dots$$

and phrases such as "continue the construction indefinitely." These expressions must not silently replace a finite construction by an already completed infinite carrier.

This section distinguishes four objects that conventional notation often compresses into one:

(i) a completed finite prefix;

(ii) a finite rule that constructs a further stage;

(iii) a proof that the rule remains applicable after every completed finite stage;

(iv) and a completed infinite carrier, should such a carrier later be admitted.

Only the first three are constructed here. The fourth requires a separate completion or admission rule and an identity criterion not yet introduced.

53.1 Finite prefixes

Definition 53.1 (Finite prefix report)**.** *Let X be a state carrier and let*

$$I_m = [0, 1, \dots, m]$$

be a constructed finite successor-index carrier. A finite prefix report in X is a finite indexed family

$$x^{[m]} : I_m \to X.$$

Its displayed form is

$$[x_0, x_1, \ldots, x_m], \qquad x_k = x^{[m]}(k).$$

The superscript $[m]$ records the terminal index of the completed prefix. It is not an exponent. The prefix contains exactly the occurrences attached to the indices already admitted into I_m.

Definition 53.2 (Prefix restriction)**.** *For $j \le m$, the restriction of $x^{[m]}$ to I_j is*

$$x^{[m]} \restriction_{I_j} : I_j \to X.$$

A prefix $x^{[j]}$ is compatible with $x^{[m]}$ when

$$x^{[m]} \restriction_{I_j} =_{\mathrm{C}} x^{[j]}$$

under the declared family identity.

Compatibility says that the longer report extends the earlier report without revising its retained occurrences, indices, or relevant construction ledger.

Definition 53.3 (One-stage extension)**.** *A one-stage extension of $x^{[m]}$ is a prefix*

$$x^{[m+1]} : I_{m+1} \to X$$

such that

$$x^{[m+1]} \restriction_{I_m} =_{\mathrm{C}} x^{[m]}.$$

The newly admitted occurrence is

$$x_{m+1} = x^{[m+1]}(m+1).$$

An extension therefore preserves the old prefix and adds one whole indexed occurrence. It does not retroactively turn the earlier finite prefix into an incomplete fragment of an object that had already existed in completed form.

53.2 Continuation rules

Definition 53.4 (Continuation rule)**.** *A continuation rule on a class of admissible finite prefixes is a mapping or relation*

$$E : x^{[m]} \longmapsto x^{[m+1]}$$

that constructs at least one compatible one-stage extension whenever its input prefix satisfies the rule's declared admissibility conditions.

A continuation rule may use only the terminal state, the entire retained prefix, an auxiliary state ledger, or some combination of them. These are different rule types and must not be identified without proof.

Definition 53.5 (State-local continuation)**.** *A continuation rule is state-local when there exists a transition map*

$$T : X \to X$$

such that

$$x_{m+1} = T(x_m).$$

Definition 53.6 (History-dependent continuation)**.** *A continuation rule is history-dependent when the next occurrence is selected from the complete prefix ledger:*

$$x_{m+1} = F(x^{[m]}).$$

Equal terminal states guarantee equal next states only for a state-local deterministic rule. A history-dependent rule may assign different continuations to prefixes sharing the same terminal value.

Definition 53.7 (Continuation certificate)**.** *A continuation certificate for a prefix $x^{[m]}$ is a retained report*

$$\kappa_m$$

showing that the admissibility conditions of E are satisfied at stage m and recording the construction of at least one compatible extension.

The complete one-stage continuation state is therefore not merely x_{m+1}. It may be represented as

$$\Lambda_E(m) = (x^{[m]}, E, \kappa_m, x^{[m+1]}).$$

53.3 Deterministic and branching continuation

Definition 53.8 (Deterministic continuation)**.** *A continuation rule is deterministic on its admissible prefixes when every admissible prefix has exactly one compatible next prefix under the declared identity:*

$$E(x^{[m]}) =_{\mathrm{C}} x^{[m+1]}.$$

Definition 53.9 (Branching continuation)**.** *A continuation relation is branching when some admissible prefix has more than one constructively distinct compatible extension:*

$$x_a^{[m+1]} \neq_{\mathrm{C}} x_b^{[m+1]}.$$

A displayed common prefix does not identify a unique indefinite continuation in the branching case. The branch-selection rule or branch address is part of the report.

Example 53.10 (Ambiguous continuation)**.** *The finite list*

$$[1, 2, 3]$$

can be extended by many rules, including

$$[1, 2, 3, 4], \qquad [1, 2, 3, 5], \qquad [1, 2, 3, 6],$$

or a rule that returns to 1*. The finite prefix alone does not determine which rule was intended.*

Thus the notation

$$1, 2, 3, \ldots$$

is not self-interpreting. Its meaning depends on a separately declared continuation grammar.

53.4 The ellipsis is a report token

Definition 53.11 (Ellipsis token)**.** *An ellipsis token is a distinguishable report object indicating that a declared continuation rule is intended to remain available beyond the displayed finite prefix.*

The mark ... is not the unrendered collection of every omitted occurrence. It occupies one report position and refers to a rule or continuation obligation.

Accordingly,

$$[x_0, x_1, x_2, \ldots]$$

contains, as written, finitely many displayed occurrences and one continuation token. Its interpretation requires at least:

(i) the displayed prefix;

(ii) the continuation rule;

(iii) the stage at which the rule begins;

(iv) any state or history needed by the rule;

(v) and the identity criterion for later extensions.

Theorem 53.12 (Ellipsis noncompletion)**.** *An ellipsis token does not by itself construct a completed infinite carrier.*

Proof. The token is one finite report occurrence. It may refer to an extension rule, but reference to a rule does not supply all of the rule's possible future output occurrences as already admitted indexed members. A completed carrier would additionally require an admission rule for all its indices and an identity criterion for the resulting total report. Neither is supplied by the mark alone. □

53.5 Indefinite extendability

Definition 53.13 (Indefinitely extendable construction)**.** *A prefix construction governed by E is indefinitely extendable when, for every completed finite admissible prefix $x^{[m]}$, a continuation certificate constructs at least one compatible extension $x^{[m+1]}$.*

In retrospective numerical notation, the condition is

$$\text{for every constructed finite } m, \quad x^{[m]} \text{ admits an } E\text{-extension } x^{[m+1]}.$$

This is a statement about the availability of a next finite construction at every finite stage. It is not yet the assertion that one completed object containing all stages has been constructed.

Theorem 53.14 (No terminal prefix)**.** *If a continuation rule is indefinitely extendable, no admissible finite prefix is terminal under that rule.*

Proof. Let $x^{[m]}$ be any admissible completed finite prefix. Indefinite extendability supplies a continuation certificate and a compatible extension $x^{[m+1]}$. Hence $x^{[m]}$ is not terminal. Since the argument applies to every completed finite admissible prefix, none is terminal. □

Corollary 53.15 (No last constructed natural stage)**.** *For every constructed finite stage index m, the successor stage $m+1$ can be constructed. Hence there is no greatest finite natural-number report under the successor construction.*

This corollary establishes unbounded finite succession. It does not yet identify the unbounded succession rule with a completed set of all natural numbers.

53.6 Can continue, is continued, and has been completed

Three claims must be distinguished:

availability: $x^{[m]}$ admits another stage;
execution: $x^{[m+1]}$ has actually been constructed;
completion: a total carrier containing every stage has been admitted.

The first does not imply that the next stage has been executed. The second produces only a longer finite prefix. Repeating the first two claims at every finite stage still does not, without another construction principle, supply the third claim.

Principle 53.16 (Nonretroactive extension)**.** *The construction of a later prefix does not alter the finite identity of an earlier prefix. It adds a new compatible report containing the earlier one as a retained restriction.*

Thus

$$x^{[m]} \neq_{\mathrm{C}} x^{[m+1]}$$

even though

$$x^{[m+1]} \restriction_{I_m} =_{\mathrm{C}} x^{[m]}.$$

53.7 Universal finite-stage statements

A theorem may apply uniformly to every finite stage without first treating all stages as members of a completed carrier.

Definition 53.17 (Uniform finite-stage rule)**.** *A finite-stage property P is uniform under E when:*

(i) $P(x^{[0]})$ *is established for the initial prefix;*

(ii) whenever $P(x^{[m]})$ *holds and* E *constructs* $x^{[m+1]}$*, the property* $P(x^{[m+1]})$ *follows by one retained extension argument.*

This is the constructive content later compressed into induction. The induction report records an initial construction and a successor-preservation rule. It does not require that every stage have already been separately rendered.

Theorem 53.18 (Finite-stage induction)**.** *If a property has a constructed initial witness and is preserved by every admissible one-stage extension, then the property holds for each finite prefix obtained by finitely many applications of the extension rule.*

Proof. The initial prefix carries the property. Each completed extension carries the property from its predecessor to its successor. Therefore any prefix reached after a specified finite number of extension steps inherits the property through a finite chain of preserved reports. □

The theorem concerns every finitely reached stage. It does not yet define a property of a completed infinite totality.

53.8 Coherent prefix systems

Definition 53.19 (Coherent prefix system)**.** *A coherent prefix system is a continuation rule together with a family of finite prefix reports satisfying*

$$x^{[m+1]} \restriction_{I_m} =_{\mathrm{C}} x^{[m]}$$

at every constructed stage.

A coherent system preserves all prior finite information under extension. It is the finite data needed for a later completion construction, but it is not itself automatically identical to the completed object that such a construction may introduce.

Definition 53.20 (Prefix projection)**.** *For $j \leq m$, define the prefix projection*

$$\rho_{m,j}(x^{[m]}) = x^{[m]} \restriction_{I_j} .$$

Coherence requires

$$\rho_{m,j}(x^{[m]}) =_{\mathrm{C}} x^{[j]}.$$

For $i \leq j \leq m$, restriction composes:

$$\rho_{m,i} = \rho_{j,i} \circ \rho_{m,j}.$$

This equation reports compatibility among finite observations. A later completed carrier, if admitted, must reproduce every finite prefix under its own finite-report projections.

53.9 Potential continuation and completed infinity

Definition 53.21 (Potential continuation report)**.** *A potential continuation report is the finite package*

$$\mathcal{P}_E = (x^{[0]}, E, \mathcal{K}),$$

where $x^{[0]}$ is an initial prefix, E is a continuation rule, and $\mathcal{K}$ is a uniform certificate scheme establishing extension after every completed finite admissible stage.

The word *potential* here does not mean imaginary or uncertain. It means that each further stage is constructible when requested, while no terminal totality has yet been admitted.

Definition 53.22 (Completed infinite carrier – deferred)**.** *A completed infinite carrier would be a report object X_∞ equipped with an index-admission rule and finite projections such that every constructed finite prefix is recovered compatibly:*

$$\rho_m(X_\infty) =_{\mathrm{C}} x^{[m]}.$$

This definition states the obligation such an object would have to satisfy, but this section does not assert its existence. Existence requires a later completion principle and a declared equality relation among candidate completions.

Theorem 53.23 (Quantifier separation)**.** *The statement*

for every finite m, an admissible prefix $x^{[m]}$ can be constructed

does not by itself construct an object X_∞ containing every stage as an already admitted member.

Proof. The first statement supplies a uniform finite construction scheme and, for any selected finite m, a finite witness. The second statement introduces a new total report object, a total index carrier, and compatibility maps from that object to every finite prefix. Those additional objects and relations are not components of any one finite witness and do not follow merely by renaming the extension scheme. □

The theorem does not deny that a completed carrier may be consistently constructed. It requires the construction to be stated rather than smuggled in through notation.

53.10 The infinity symbol is not an infinite carrier

The mark

$$\infty$$

is a distinguishable finite report token. Depending on the declared grammar, it may report unboundedness, a direction of continuation, a projective boundary, divergence, or a completed extension object. These meanings are not interchangeable.

Principle 53.24 (Typed infinity)**.** *Every use of ∞ must declare the carrier and relation in which it is interpreted. The existence of the symbol does not supply infinitely many indexed occurrences, and two uses of the symbol need not denote the same construction.*

For example,

$$m \longmapsto m + 1$$

reports unbounded finite succession, whereas

$$\lim_{m \to \infty} x_m$$

will later report a limiting relation over that succession. Neither expression has yet constructed a scalar object that may be manipulated as an ordinary finite count.

53.11 Finite recurrence and cyclic compression

A nonterminating recurrence need not require an infinite amount of descriptive information.

Suppose a deterministic transition map

$$T : S \to S$$

acts on a finite state carrier S. If no terminal state is reached, the recurrence eventually revisits a prior state. Once

$$x_j =_{\mathrm{C}} x_k, \qquad j < k,$$

the subsequent state sequence repeats with cycle length $k - j$ under the same deterministic rule.

Definition 53.25 (Finite recurrence descriptor)**.** *A finite recurrence descriptor is the retained package*

$$\mathcal{C} = (x_0, T, j, k, [x_0, \ldots, x_k]),$$

where $x_j =_{\mathrm{C}} x_k$ certifies the entrance and closure of a repeated state cycle.

The descriptor finitely determines each later finite stage. It does not consist of infinitely many already written state occurrences.

Theorem 53.26 (Cycle compression)**.** *For a deterministic recurrence on a finite state carrier, a finite transient prefix together with one retained cycle determines the state at every requested finite stage.*

Proof. Before the cycle entrance, the retained transient prefix supplies the state directly. After the entrance, divide the additional transition count by the positive cycle length. The remainder selects the corresponding retained position in the cycle. This uses the quotient–remainder construction already established and produces the requested finite state without constructing every intervening occurrence as a permanent total carrier. □

53.12 Recurring radix reports

The decimal report

$$0.333\ldots_{10}$$

should now be read as a finite continuation report. For the ratio $1/3$, the radix recurrence retains the state

$$10 \cdot 1 = 3 \cdot 3 + 1.$$

The remainder returns to 1, so the same digit and remainder state recur.

The exact finite descriptor may be written

$$\mathcal{C}_{1/3,10} = (R_0 = 1,\ d_1 = 3,\ R_1 = 1,\ E_{10,3}).$$

This descriptor constructs any requested finite decimal prefix

$$0.3, \quad 0.33, \quad 0.333, \quad \ldots$$

while retaining the unresolved remainder at each finite stage.

Theorem 53.27 (Recurring report exactness)**.** *A recurring radix descriptor may exactly specify a rational ratio even though no finite digit prefix is itself equal to the completed positional value later associated with the recurrence.*

Proof. The descriptor retains the original numerator, denominator, base, digit output, and repeated remainder state. These finite data determine the quotient–remainder relation exactly. Each finite written prefix, by contrast, terminates at a finite place and retains a nonzero residual obligation. Exactness belongs to the recurrence relation, not to any one truncated digit list. □

This distinction will be essential for statements such as

$$0.999\ldots = 1.$$

At present, the left side is a finite recurrence descriptor and each of its finite prefixes differs from the unit. Equality with a completed real unit cannot be stated until a completion relation is constructed.

53.13 Indefinite sums and products are deferred

The notation

$$\sum_{k=0}^{\infty} a_k$$

or

$$\prod_{k=0}^{\infty} a_k$$

does not yet denote a completed sum or product. What is currently constructed is the finite family of partial operations

$$S_m = \sum_{k=0}^{m} a_k, \qquad P_m = \prod_{k=0}^{m} a_k,$$

for each requested finite m.

The continuation rules

$$S_{m+1} = S_m + a_{m+1}, \qquad P_{m+1} = P_m \cdot a_{m+1}$$

construct longer finite ledgers. A completed infinite aggregate requires a later convergence or completion criterion.

Principle 53.28 (No aggregation by punctuation)**.** *An infinity mark or ellipsis does not perform infinitely many additions, multiplications, unions, intersections, or evaluations. It reports a continuation grammar whose completed aggregate, if any, must be separately constructed.*

53.14 Stage-local and completion-dependent properties

Definition 53.29 (Stage-local property)**.** *A property is stage-local when its truth for a continuation report can be decided from a specified finite prefix and its retained ledger.*

Examples include the value at a selected finite index, whether a selected transition occurred, and whether a finite remainder closed at a specified stage.

Definition 53.30 (Completion-dependent property)**.** *A property is completion-dependent when no specified finite prefix alone supplies its declared witness and the property instead refers to the behavior of all finite extensions or to a completed carrier.*

Convergence, equality of completed recurring expansions, boundedness over every stage, and the value of an infinite series are completion-dependent. They cannot be inferred from a few observed prefixes unless a uniform theorem supplies the missing universal relation.

Theorem 53.31 (Finite observation insufficiency)**.** *No finite observed prefix, by itself, proves that an unspecified continuation rule is indefinitely extendable, terminating, recurring, convergent, or unique.*

Proof. Any finite prefix is compatible with multiple later rules unless the continuation grammar has been separately fixed. Some compatible rules terminate, some continue, some branch, and some repeat. Therefore the observed values alone do not determine a global continuation property. □

53.15 Continuation-preservation theorem

Theorem 53.32 (Continuation-report preservation)**.** *A finite report of an indefinitely extendable construction preserves the construction only when it retains, or proves irrelevant, all of the following:*

(i) the initial prefix and its complete identity;

(ii) the finite index and successor grammar;

(iii) the continuation rule;

(iv) the admissibility conditions for each extension;

(v) the continuation certificates or a uniform certificate scheme;

(vi) whether continuation is deterministic or branching;

(vii) every branch-selection address relevant to the displayed route;

(viii) the compatibility relation among successive prefixes;

(ix) any terminal, recurrence, or closure conditions;

(x) and the distinction between an extension scheme and a completed carrier.

An ellipsis or infinity symbol without these declarations is not a lossless report of the construction.

Theorem 53.33 (Finite-rule noncollapse)**.** *Let r map a continuation ledger to a displayed prefix with an ellipsis. If two distinct continuation rules produce the same displayed prefix, then r is noninjective. Any property depending on which continuation occurs cannot factor through the displayed ellipsis report without an additional fiber-constancy theorem.*

Proof. Let C_1 and C_2 share the same displayed prefix but use different continuation rules. Then

$$r(C_1) = r(C_2)$$

while

$$C_1 \neq_{\mathrm{C}} C_2.$$

A property that distinguishes their later stages is not constant on the fiber of r. Hence the common displayed report cannot determine that property. □

53.16 Dependency summary

The construction developed in this section is

$$\begin{aligned}\text{finite indexed prefix} &\longrightarrow \text{compatible one-stage extension} \longrightarrow \text{continuation rule}\\ &\longrightarrow \text{uniform extension certificate} \longrightarrow \text{indefinite extendability}\\ &\longrightarrow \text{coherent finite-prefix system.}\end{aligned}$$

It does not yet make the further step

$$\text{coherent finite-prefix system} \longrightarrow \text{completed infinite carrier.}$$

That step requires a declared completion construction, a criterion for equality among completed reports, and a proof that every finite projection is preserved.

The next section should unify the equality relations used throughout the manuscript and establish the exact fiber condition under which a later projection may replace a prior constructive ledger. Only after that audit should convergence and completion identify distinct finite approximation routes with one completed value.

Part XII
Equality Hierarchy and Projection-Fiber Preservation

54 Equality hierarchy and projection-fiber preservation

The manuscript has now constructed several ways in which two reports may agree. These agreements must not be compressed into one untyped equality symbol. An equality statement is always relative to the structure that the comparison is required to preserve.

A report may agree with another in count while differing in source, agree in projected value while differing in orientation, agree pointwise while differing in evaluation history, or agree under a completion that has not yet been constructed. Every substitution therefore requires a declared equality relation and a theorem showing that the property being substituted is preserved by that relation.

54.1 Equality is a typed comparison relation

Definition 54.1 (Typed equality)**.** *Let $\mathcal{C}$ be a carrier of constructive reports and let Σ be a declared family of structures to be preserved. The relation*

$$C_1 =_\Sigma C_2$$

means that C_1 and C_2 agree under the identity criterion determined by Σ.

The declaration Σ may include any combination of:

(i) indexed occurrence identity;

(ii) source labels;

(iii) report order;

(iv) orientation;

(v) grouping or bracketing;

(vi) transition history;

(vii) boundary and closure obligations;

(viii) quotient–remainder state;

(ix) base and positional width;

(x) evaluation trace;

(xi) or only a selected scalar or coordinate report.

The bare symbol $=$ is therefore safe only after the relevant carrier and preservation criterion have already been fixed. Where those conditions are not evident, the equality must be typed explicitly.

Principle 54.2 (No untyped substitution)**.** *From $C_1 =_\Sigma C_2$, one may substitute C_1 for C_2 only in a context proved invariant under $=_\Sigma$.*

This principle is the equality analogue of the earlier warning that equal projected outputs do not imply equal constructions.

54.2 Constructive identity

Definition 54.3 (Constructive identity)**.** *Two reports are constructively identical, written*

$$C_1 =_{\mathrm{C}} C_2,$$

when every structure declared constitutive of the reports is preserved, including every relevant indexed occurrence, source address, relation, orientation, boundary, and construction obligation.

Constructive identity is the strongest equality used in this manuscript. It does not mean that two marks printed at different locations are literally the same physical ink event. It means that, within the declared mathematical report, no retained constitutive distinction separates them.

Thus

$$C_1 =_{\mathrm{C}} C_2 \quad \Longrightarrow \quad \pi(C_1) = \pi(C_2)$$

for every projection π defined on the retained structure, but the converse generally fails.

Theorem 54.4 (Constructive identity preserves every declared property)**.** *Let P be a property defined entirely from the structures included in constructive identity. If*

$$C_1 =_{\mathrm{C}} C_2,$$

then

$$P(C_1) = P(C_2).$$

Proof. By definition, constructive identity preserves every structure from which P is evaluated. Replacing one report by the other therefore changes no input to the evaluation of P. □

This theorem is strong because the equality relation is strong. Most later report equalities deliberately forget some structure and therefore preserve fewer properties.

54.3 Occurrence, source, and order equality

Definition 54.5 (Occurrence equality)**.** *Two report occurrences*

$$\omega_i = (i, v_i), \qquad \omega_j = (j, v_j)$$

are occurrence-identical only when their complete occurrence addresses agree under the declared report identity.

Value equality

$$v_i = v_j$$

does not imply occurrence equality when $i \neq j$.

Definition 54.6 (Source equality)**.** *Two reports are source-equal when their retained source addresses correspond and every source-dependent relation is preserved.*

Definition 54.7 (Order equality)**.** *Two finite reports are order-equal when corresponding occurrences appear under the same declared successor and precedence relations.*

A permutation may preserve multiplicity and projected values while failing order equality. A source-erasing projection may preserve order while failing source equality. These are independent comparison obligations.

54.4 Structural equivalence and isomorphism

Definition 54.8 (Structure-preserving equivalence)**.** *Let $\mathcal{S}(C)$ denote a selected structural signature of a report C. Write*

$$C_1 \cong_{\mathcal{S}} C_2$$

when there exists a bijective correspondence between their admitted components that preserves every relation in $\mathcal{S}$.

For example,

$$(A \times B) \times C \cong A \times (B \times C)$$

under reassociation, although their nested source syntax differs. Likewise,

$$A \times B \cong B \times A$$

under transposition at the product-structure level, although ordered product orientations differ.

Theorem 54.9 (Isomorphism preserves structural invariants)**.** *If*

$$C_1 \cong_{\mathcal{S}} C_2,$$

then every property defined solely from the preserved signature $\mathcal{S}$ has equal reports on C_1 and C_2.

An isomorphism therefore licenses substitution only for $\mathcal{S}$-invariant properties. It does not prove source identity, construction-history identity, or equality of structures omitted from $\mathcal{S}$.

54.5 Reindexing and extensional equality

Definition 54.10 (Reindexing equality)**.** *Two indexed reports are reindexing-equal when a bijection between their index carriers preserves the admitted occurrences while changing only the names or presentation order of the indices declared irrelevant.*

Definition 54.11 (Extensional value equality)**.** *Two reports are extensionally value-equal when they produce the same collection of projected values after occurrence multiplicity, order, source, and any other omitted structure have been quotiented away.*

Thus

$$[a_0, a_1] \quad \text{with} \quad \pi(a_0) = \pi(a_1) = a$$

may project extensionally to

$$\{a\},$$

as does the one-occurrence report $[a_0]$. The extensional reports agree while constructive multiplicity does not.

Principle 54.12 (Extensional equality is domain-limited)**.** *Extensional equality is sufficient only for properties that depend solely on the retained extensional value collection.*

Counting source occurrences, reconstructing routes, distinguishing repeated evaluations, and auditing cancellation cannot in general be performed from extensional value equality alone.

54.6 Count, scalar, coordinate, and ratio equality

Definition 54.13 (Count equality)**.** *Two finite reports are count-equal when their occurrence carriers admit a bijection:*

$$C_1 =_{\text{count}} C_2 \quad \Longleftrightarrow \quad |I_{C_1}| = |I_{C_2}|.$$

Count equality ignores the values, sources, order, and relations occupying those indices unless separately retained.

Definition 54.14 (Scalar-report equality)**.** *Let*

$$\pi_{\text{sc}} : \mathcal{C} \to S$$

be a scalar reporting map. Then

$$C_1 =_{\text{sc}} C_2 \quad \Longleftrightarrow \quad \pi_{\text{sc}}(C_1) = \pi_{\text{sc}}(C_2).$$

Definition 54.15 (Coordinate equality)**.** *Let*

$$\pi_{\text{coord}} : \mathcal{C} \to K$$

be a declared coordinate projection. Coordinate equality means equality in K after that projection.

Coordinate equality does not imply equality of the pre-coordinate placements or relations unless the coordinate projection is injective over the relevant domain.

Definition 54.16 (Ratio equality)**.** *For positive denominator reports, constructive ratios $\mathcal{R}(A, B)$ and $\mathcal{R}(C, D)$ are ratio-equal when their cross-product counts balance:*

$$|A||D| = |C||B|.$$

Ratio equality preserves the normalized comparison while generally discarding numerator and denominator source identities. Primitively one may write

$$\mathcal{R}(U, D_2) =_{\text{ratio}} \mathcal{R}(D_2, D_4)$$

without requiring the two ratio ledgers to be constructively identical. A later fraction-coordinate compiler may display these two ratio classes by the conventional glyphs 1/2 and 2/4.

54.7 Orientation, balance, and closure equality

Definition 54.17 (Orientation equality)**.** *Two signed reports are orientation-equal when their positive and negative lanes agree after the declared cancellation and normalization rules.*

A positive lane and a negative lane may have equal magnitude while differing in orientation. Magnitude equality alone does not imply orientation equality.

Definition 54.18 (Balance equality)**.** *Two balance reports are balance-equal when they have the same retained carrier and the same signed imbalance state.*

In particular,

$$H(A; B) = 0$$

reports equal complementary counts relative to the retained carrier B. Another zero balance produced by a different carrier or decomposition may be scalar-equal to zero while remaining a different balance ledger.

Definition 54.19 (Closure equality)**.** *Two operation states are closure-equal when the same declared obligations have been discharged and no relevant unmatched source, inverse, boundary, or remainder survives.*

Closure equality is stronger than equality of a selected net scalar. A scalar zero can result from opposing nonzero contributions, while a closure zero additionally certifies that every declared obligation has closed.

54.8 Pointwise, rule, and family equality

Definition 54.20 (Pointwise equality)**.** *Mappings $f, g : X \to Y$ are pointwise equal on X when*

$$f(x) = g(x)$$

for every admitted $x \in X$.

Pointwise equality compares output reports. It need not preserve implementation, evaluation trace, resource ledger, or source route.

Definition 54.21 (Rule equality)**.** *Two mapping or continuation rules are rule-equal when their admissible inputs, transition conditions, branch conditions, outputs, and retained evaluation obligations agree under the declared rule grammar.*

Definition 54.22 (Indexed-family equality)**.** *Two indexed families are equal under a declared family criterion only when both their index correspondence and their member comparison satisfy that criterion.*

Two families may be pointwise scalar-equal while differing in source, recurrence rule, or proof certificate. Therefore

$$f(x) = g(x) \text{ for all } x$$

does not by itself imply

$$f =_{\mathrm{C}} g$$

as constructive mappings.

54.9 Report-induced equivalence relations

Let

$$r : \mathcal{C} \to V$$

be any reporting map.

Definition 54.23 (Report-induced equivalence)**.** *Define*

$$C_1 \sim_r C_2 \quad \Longleftrightarrow \quad r(C_1) = r(C_2).$$

The relation $\sim_r$ is an equivalence relation on $\mathcal{C}$ whenever equality in V is an equivalence relation.

Proof. Reflexivity follows from $r(C) = r(C)$. Symmetry follows from symmetry of equality in V. Transitivity follows because $r(C_1) = r(C_2)$ and $r(C_2) = r(C_3)$ imply $r(C_1) = r(C_3)$. □

Definition 54.24 (Projection fiber)**.** *For $v \in \mathrm{im}(r)$, the fiber of r over v is*

$$r^{-1}(v) = \{C \in \mathcal{C} : r(C) = v\}.$$

Equivalently, the fiber of C is its equivalence class

$$[C]_r = \{D \in \mathcal{C} : D \sim_r C\}.$$

A projection fiber is exactly the family of constructive reports that the selected report map treats as equal.

Example 54.25 (Addition fiber)**.** *For the ordered product report*

$$\pi_+(i, j) = i + j,$$

the scalar fiber over 1 *contains at least*

$$\pi_+^{-1}(1) = \{(0, 1), (1, 0)\}.$$

The scalar value 1 *identifies the fiber but does not identify which oriented source pair produced it.*

54.10 Quotients are deliberate information contracts

The quotient carrier

$$\mathcal{C}/\!\sim_r$$

contains one equivalence class for each report value in $\mathrm{im}(r)$. Passing from $\mathcal{C}$ to this quotient deliberately declares every distinction within each fiber irrelevant for the quotient-level task.

Principle 54.26 (Quotient declaration)**.** *A quotient does not prove that the forgotten distinctions never existed. It declares that they will not be distinguished by the quotient identity criterion.*

The quotient may be mathematically useful and exactly suited to a chosen task. The remaining question is whether a later property is well defined on that quotient. That requires a preservation theorem.

54.11 Projection-fiber factorization theorem

Let

$$P : \mathcal{C} \to Z$$

be any property, measurement, predicate, or further report of the full constructive ledger.

Theorem 54.27 (Projection-fiber factorization)**.** *Let $r : \mathcal{C} \to V$ and $P : \mathcal{C} \to Z$. The following statements are equivalent:*

(i) There exists a map

$$\widehat{P} : \mathrm{im}(r) \to Z$$

such that

$$P = \widehat{P} \circ r.$$

(ii) P is constant on every fiber of r; that is, for all $C_1, C_2 \in \mathcal{C}$,

$$r(C_1) = r(C_2) \quad \Longrightarrow \quad P(C_1) = P(C_2).$$

Proof. Assume first that $P = \widehat{P} \circ r$. If $r(C_1) = r(C_2)$, then

$$P(C_1) = \widehat{P}(r(C_1)) = \widehat{P}(r(C_2)) = P(C_2).$$

Hence P is constant on every fiber.

Conversely, suppose that P is constant on every fiber of r. For each $v \in \mathrm{im}(r)$, select any $C \in \mathcal{C}$ satisfying $r(C) = v$ and define

$$\widehat{P}(v) = P(C).$$

If D is another representative with $r(D) = v$, then C and D lie in the same fiber, so $P(C) = P(D)$. Therefore $\widehat{P}$ is well defined. For every $C \in \mathcal{C}$,

$$\widehat{P}(r(C)) = P(C),$$

and hence $P = \widehat{P} \circ r$. □

Corollary 54.28 (Exact condition for ledger erasure)**.** *A constructive ledger may be discarded with respect to a property P exactly when P is constant on every fiber of the retained report map.*

This condition is necessary and sufficient. It is not a convention, intuition, or appeal to the usefulness of scalar notation.

54.12 Predicate preservation

A predicate is a map

$$Z : \mathcal{C} \to \{\mathrm{true}, \mathrm{false}\}.$$

Corollary 54.29 (Predicate descent)**.** *A predicate Z is well defined on the quotient induced by r if and only if*

$$r(C_1) = r(C_2) \quad \Longrightarrow \quad Z(C_1) = Z(C_2).$$

Thus a projected equality licenses a projected predicate only when every constructive report in the same fiber receives the same predicate value.

Example 54.30 (Scalar-zero fiber)**.** *Let*

$$\pi_{\mathrm{sc}} : \mathcal{C} \to S$$

be a scalar projection, and let Z_{cl} report complete closure. Scalar zero determines complete closure only if

$$\pi_{\mathrm{sc}}(C_1) = \pi_{\mathrm{sc}}(C_2) = 0 \quad \Longrightarrow \quad Z_{\mathrm{cl}}(C_1) = Z_{\mathrm{cl}}(C_2)$$

for every admissible pair in the scalar-zero fiber.

If one zero-output ledger is completely closed while another retains an unmatched source or boundary obligation, then complete closure is not constant on the zero fiber. In that case scalar zero cannot represent complete operator zero.

54.13 Injectivity is sufficient but not necessary

Theorem 54.31 (Injective-report sufficiency)**.** *If $r : \mathcal{C} \to V$ is injective, then every property $P : \mathcal{C} \to Z$ factors through r on* $\mathrm{im}(r)$*.*

Proof. Every fiber of an injective map contains exactly one constructive report. Every property is therefore constant on each fiber, and the factorization theorem applies. □

Injectivity is stronger than necessary. A noninjective projection may still preserve a particular property if that property is constant on the larger fibers. For example, a count projection may discard object labels while still preserving whether the count is even. It cannot thereby preserve every property of the original report.

Principle 54.32 (Property-by-property preservation)**.** *Noninjective report maps require a separate fiber-constancy proof for every later property that is to be inferred from the report alone.*

54.14 Counterexample to universal projection preservation

Theorem 54.33 (Universal history erasure is false)**.** *There is no general theorem stating that equal reported outputs preserve every property of their constructive ledgers.*

Proof. Let $C_1 \neq_{\mathrm{C}} C_2$ satisfy

$$r(C_1) = r(C_2).$$

Define the property

$$P(C) = \begin{cases} 0, & C = C_1, \\ 1, & C \neq C_1. \end{cases}$$

Then

$$P(C_1) \neq P(C_2)$$

while their reports agree. Hence P is not constant on the common fiber and cannot factor through r. □

The theorem does not say that no history may ever be erased. It says that erasure is legitimate only relative to properties proved insensitive to the erased distinctions.

54.15 Congruence and lawful substitution

Let $\sim$ be an equivalence relation on a carrier $\mathcal{C}$ and let

$$F : \mathcal{C}^n \to \mathcal{D}$$

be an n-input operation.

Definition 54.34 (Congruence for an operation)**.** *The relation $\sim$ is a congruence for F when*

$$C_k \sim D_k \quad \textit{for every } k$$

implies

$$F(C_1, \ldots, C_n) \sim' F(D_1, \ldots, D_n)$$

under the declared output equivalence $\sim'$.

Theorem 54.35 (Quotient operation criterion)**.** *An operation F descends to a well-defined operation on equivalence classes if and only if its output class is constant under replacement of each input by any equivalent representative.*

Proof. If the quotient operation is well defined, equivalent representatives must produce the same output class. Conversely, if the output class is invariant under representative replacement, define the quotient operation by selecting any representatives; the result is independent of the selection. □

This is the multi-input version of fiber factorization. It supplies the exact requirement behind symbolic substitution rules.

Example 54.36 (Ratio representatives)**.** *The pairs $(1, 2)$ and $(2, 4)$ represent the same ratio class. Addition and multiplication of ratio classes are well defined because the cross-product equivalence is a congruence for the constructed ratio operations. Source-specific properties of the original numerator and denominator reports do not descend merely because the ratio operations do.*

54.16 Canonical representatives do not restore erased history

A quotient system may select one canonical representative from each equivalence class. For example, a ratio may be reduced, a signed ledger may be normalized, or a positional numeral may drop leading zero digits.

Definition 54.37 (Canonicalization)**.** *A canonicalization map is a function*

$$c : \mathcal{C} \to \mathcal{C}$$

that assigns a chosen representative to each report equivalence class and satisfies

$$r(c(C)) = r(C).$$

Principle 54.38 (Canonical representative limitation)**.** *A canonical representative records the chosen normal form of a fiber. It does not reconstruct which original member of the fiber produced the report.*

If

$$C_1 \sim_r C_2$$

and

$$c(C_1) = c(C_2),$$

then canonicalization has intentionally merged the source distinction. Applying c twice may stabilize the normal form,

$$c(c(C)) = c(C),$$

but this idempotence is not recovery of the original ledger.

54.17 Sections and representative choice

A right inverse or section of r is a map

$$s : \mathrm{im}(r) \to \mathcal{C}$$

satisfying

$$r(s(v)) = v.$$

Such a section chooses one constructive representative for each report value.

Theorem 54.39 (Representative selection is not inversion)**.** *If r is noninjective, no section s can satisfy*

$$s(r(C)) =_{\mathrm{C}} C$$

for every $C \in \mathcal{C}$.

Proof. Let $C_1 \neq_{\mathrm{C}} C_2$ lie in the same fiber, so $r(C_1) = r(C_2) = v$. The section returns the single report $s(v)$. It cannot be constructively identical to both distinct reports. Therefore at least one original ledger is not recovered. □

A decoding rule can invert a projection only after the relevant domain has been restricted so that the projection is injective, or after sufficient side information has been retained to separate the fibers.

54.18 Minimal sufficient reports

Definition 54.40 (Sufficient report for a property family)**.** *Let $\mathcal{P}$ be a family of properties on $\mathcal{C}$. A report map $r : \mathcal{C} \to V$ is sufficient for $\mathcal{P}$ when every $P \in \mathcal{P}$ factors through r.*

Definition 54.41 (Property indistinguishability)**.** *For a property family $\mathcal{P}$, define*

$$C_1 \sim_{\mathcal{P}} C_2 \quad \Longleftrightarrow \quad P(C_1) = P(C_2) \text{ for every } P \in \mathcal{P}.$$

A report is sufficient for $\mathcal{P}$ precisely when every report fiber lies inside one $\mathcal{P}$-indistinguishability class.

Theorem 54.42 (Sufficiency criterion)**.** *The report map r is sufficient for $\mathcal{P}$ if and only if*

$$r(C_1) = r(C_2) \quad \Longrightarrow \quad C_1 \sim_{\mathcal{P}} C_2.$$

This theorem converts the vague question "Does the report contain enough information?" into an exact comparison between report fibers and the distinctions made by the intended property family.

54.19 Composition of projections

Suppose

$$\mathcal{C} \xrightarrow{r} V \xrightarrow{q} W$$

are successive report maps. The composite report is

$$q \circ r : \mathcal{C} \to W.$$

Every fiber of r lies inside a fiber of $q \circ r$, but the composite may merge several r-fibers.

Theorem 54.43 (Successive-erasure criterion)**.** *A property $P : \mathcal{C} \to Z$ factors through $q \circ r$ if and only if P is constant on every fiber of the composite report. Factoring through r alone does not guarantee factoring through the coarser report $q \circ r$.*

Thus preservation must be re-audited after every additional projection. A count may preserve parity; reducing parity further to a constant symbol does not.

54.20 Equality ladders

The equality relations developed so far form a preservation ladder rather than a single chain valid in every case. A typical sequence is

$$\text{constructive identity} \Longrightarrow \text{source-preserving structural equivalence} \Longrightarrow \text{count or scalar equality},$$

but reverse implications require additional injectivity or fiber-constancy theorems.

For ratio reports one may have

$$\mathcal{R}(A,B) =_{\mathrm{C}} \mathcal{R}(C,D) \Longrightarrow \mathcal{R}(A,B) =_{\mathrm{ratio}} \mathcal{R}(C,D),$$

while

$$\mathcal{R}(A,B) =_{\mathrm{ratio}} \mathcal{R}(C,D) \not\Longrightarrow \mathcal{R}(A,B) =_{\mathrm{C}} \mathcal{R}(C,D).$$

For signed reports one may have

$$C_1 =_{\mathrm{C}} C_2 \Longrightarrow \mathrm{net}(C_1) = \mathrm{net}(C_2),$$

while equal net values need not preserve the positive and negative source lanes.

For mapping evaluations,

$$E_1 =_{\mathrm{C}} E_2 \Longrightarrow \mathrm{out}(E_1) = \mathrm{out}(E_2),$$

but equal outputs do not imply equal evaluation traces.

Principle 54.44 (Direction of implication)**.** *Every weakening of equality may be used in the forward direction established by its projection. No reverse implication is available without a separate recovery or preservation theorem.*

54.21 Zero-predicate hierarchy

The manuscript now distinguishes several zero reports:

(i) a zero occurrence count;

(ii) a typed no-contribution report;

(iii) an exact subtraction closure;

(iv) an exact division remainder zero;

(v) a signed balance zero;

(vi) a scalar or coordinate zero;

(vii) and a complete closure zero in which every declared obligation has closed.

These zero reports may share the scalar symbol 0 while answering different questions.

Definition 54.45 (Zero-predicate projection)**.** *Let Z_{full} be a complete zero predicate on a ledger space $\mathcal{C}$, and let*

$$\pi_0 : \mathcal{C} \to S$$

be a selected scalar report. Scalar zero represents complete zero exactly when there exists a scalar predicate $\widehat{Z}$ such that

$$Z_{\mathrm{full}} = \widehat{Z} \circ \pi_0.$$

By the factorization theorem, this is equivalent to complete zero being constant on every scalar fiber, especially the scalar-zero fiber.

Theorem 54.46 (No automatic promotion of zero)**.** *The equation*

$$\pi_0(C) = 0$$

may be promoted to

$$Z_{\text{full}}(C) = \text{true}$$

only after proving that every admissible ledger with scalar report zero satisfies the complete zero predicate.

This theorem will later govern the relation between complex scalar zero and complete operator closure.

54.22 One-to-two balance preservation condition

The primitive balance predicate is not a signed equation. Let $\mathcal{B}_{2:1}(A; B)$ denote the report obtained by maximally matching the doubled whole carrier $D_2 \times A$ against B. Constructive balance means that both unmatched residual carriers are empty.

A conventional ratio-coordinate compiler may map the retained ratio class $\mathcal{R}(A, B)$ to a displayed scalar coordinate $\nu(A; B)$. On the admitted complement domain,

$$\mathcal{B}_{2:1}(A; B) = \text{closed} \quad \Longrightarrow \quad \nu(A; B) \text{ is the coordinate image of the } 1 : 2 \text{ ratio class.}$$

If that compiler chooses the traditional fraction glyph, the image may be written $1/2$. The glyph itself does not reconstruct the balance report.

Theorem 54.47 (One-to-two coordinate-fiber condition)**.** *Let ν be a normalized coordinate projection and let $Z_{2:1}$ be the predicate that the whole-count matching of $D_2 \times A$ with B closes with no residuals. A displayed coordinate value representing the $1 : 2$ ratio class determines constructive balance exactly when $Z_{2:1}$ is constant on that coordinate fiber over the admitted domain.*

This is not an RH claim. It is the preservation condition that any later one-to-two coordinate application must satisfy.

54.23 Finite-rule equality and future completion

Section 53 distinguished a finite continuation descriptor from a completed infinite carrier. The present equality hierarchy sharpens that distinction.

Two continuation descriptors may:

(i) share every displayed finite prefix encountered so far;

(ii) produce equal values at every requested finite index;

(iii) be pointwise equal on all finite indices;

(iv) use different source rules or proof certificates;

(v) or later be identified by a completion relation not yet constructed.

Pointwise equality of all finite values is stronger than agreement on a sampled prefix, but it still need not imply constructive rule identity. A later completion may deliberately quotient multiple approximation families into one completed value. That quotient will be legitimate only for properties constant on its completion fibers.

Principle 54.48 (Completion must declare its equality)**.** *A completion is not merely the addition of new objects. It must declare which finite approximation families are identified, prove that the relation is an equivalence relation, and verify that every operation and property used on the completion is well defined on those equivalence classes.*

This is the equality work that must precede any assertion such as

$$0.999\ldots = 1$$

or any identification of distinct convergent sequences with one real report.

54.24 Equality-audit protocol

For every equality used later in the manuscript, the following audit should be possible:

(i) What are the two compared report carriers?

(ii) Which structures are required to be preserved?

(iii) Which structures are intentionally forgotten?

(iv) What map or correspondence induces the equality?

(v) What are the fibers or equivalence classes?

(vi) Is the later property constant on those fibers?

(vii) Does the relevant operation respect representative replacement?

(viii) Is a canonical representative being confused with an original source ledger?

(ix) Is a scalar zero being promoted to a stronger closure zero?

(x) Is a later completion equality being retrojected into a finite stage?

An equality claim that cannot answer these questions is not yet sufficiently typed for Primitive Constructive Logic.

54.25 Equality-preservation theorem

Theorem 54.49 (General equality-preservation theorem)**.** *Let $\sim$ be an equivalence relation on a constructive carrier $\mathcal{C}$, and let*

$$q : \mathcal{C} \to \mathcal{C}/\!\sim$$

be the quotient report. A property, predicate, or operation is well defined using only the quotient report if and only if it is invariant under replacement by $\sim$-equivalent representatives.

Proof. For a one-input property this is the projection-fiber factorization theorem applied to q. For a multi-input operation, apply the same argument to the product quotient and require output equivalence to be constant on every product fiber. □

Corollary 54.50 (No free reverse equality)**.** *From a weaker report equality one may infer a stronger constructive equality only when the quotient fibers have been proved singleton on the relevant domain or when additional retained data select the original member uniquely.*

54.26 Dependency summary

The hierarchy developed in this section is

$$\begin{aligned}\text{constructive reports} &\longrightarrow \text{declared comparison structure} \longrightarrow \text{typed equality}\\ &\longrightarrow \text{report-induced equivalence classes} \longrightarrow \text{projection fibers}\\ &\longrightarrow \text{fiber-constancy test} \longrightarrow \text{lawful quotient property or operation.}\end{aligned}$$

The central theorem is

$$\boxed{P = \widehat{P} \circ r \quad \Longleftrightarrow \quad r(C_1) = r(C_2) \Rightarrow P(C_1) = P(C_2).}$$

Consequently:

(1) equal reports need not be equal constructions;

(2) a quotient is a declared information contract, not proof that erased distinctions are unreal;

(3) injectivity preserves every property, but noninjective projections preserve only fiber-constant properties;

(4) symbolic substitution requires congruence;

(5) canonicalization does not reconstruct source history;

(6) scalar zero does not automatically imply complete closure;

(7) the scalar half does not automatically reconstruct its carrier and balance lane;

(8) and every future completion must prove that its equality and operations are well defined on completion fibers.

The next section should construct convergence from signed residuals, nested whole bounds, and finite-resolution agreement. It must distinguish exact finite equality from limit equality and must use the present fiber theorem to state precisely what a completed value preserves and what it discards.

Part XIII
Finite Resolution Rules and Conventional Real-Coordinate Grammars

55 Constructive convergence and finite resolution

Convergence is formulated before any claim that a completed real continuum exists. It compares finite stages through finite common-carrier matching and asks whether a finite witness can certify every requested positive resolution. A later real-coordinate notation may summarize the resulting evaluator, but the evaluator and its witnesses remain the primitive PCL content.

55.1 Finite ratio comparison without signed numbers

Let X and Y be finite constructive ratio reports. Resolve them over a common positive denominator carrier D so that their numerator counts are A_X and A_Y. Perform maximal matching:

$$A_X = C + R_X, \qquad A_Y = C + R_Y,$$

with no further admissible match between R_X and R_Y.

Definition 55.1 (Residual magnitude)**.** *The residual magnitude between X and Y is the nonnegative finite ratio*

$$d_{\mathrm{C}}(X, Y) := \frac{|R_X| + |R_Y|}{D}.$$

The full comparison report retains $(C, R_X, R_Y, D; \Gamma)$. The magnitude projection forgets only which residual role survived.

Lemma 55.2 (Finite residual laws)**.** *For finite ratio reports X, Y, Z:*

(i) $d_{\mathrm{C}}(X, Y)$ *is a nonnegative ratio report;*

(ii) $d_{\mathrm{C}}(X, Y) = 0$ *exactly when both unmatched residual roles are empty under the declared comparison;*

(iii) $d_{\mathrm{C}}(X, Y) = d_{\mathrm{C}}(Y, X)$*;*

(iv) the direct residual magnitude from X to Z is no larger than the joined residual magnitudes obtained through Y.

Definition 55.3 (Positive resolution report)**.** *A positive resolution report is a finite ratio ρ containing a positive numerator contribution and a positive denominator carrier. It specifies a requested residual bound and is never identified with literal zero.*

A finer requested resolution is another finite report, not an infinitesimal object. Thus resolution can be refined indefinitely by rule while every actual evaluation remains finite.

55.2 Approximation families and residual ledgers

Definition 55.4 (Finite-stage approximation family)**.** *An* approximation family *in $\mathbb{Q}_{\mathrm{C}}$ is an indexed construction family*

$$X = (X_n)_{n \in \mathbb{N}}$$

generated by a retained finite rule, recurrence, explicit lookup report, or other admitted mapping

$$f_X : \mathbb{N} \longrightarrow \mathbb{Q}_{\mathrm{C}}.$$

Each X_n is a completed finite ratio report.

The family descriptor is finite when its rule and required source data are finite, even though the rule admits another finite stage for every natural index. Section 53 already established that this indefinite extendability does not itself construct a completed infinite carrier.

Definition 55.5 (Target residual ledger)**.** *Let $L \in \mathbb{Q}_{\mathrm{C}}$ be a candidate target. The* target residual ledger *of X relative to L is the full finite matching report between X_n and L, together with magnitude*

$$e_n(X; L) := d_{\mathrm{C}}(X_n, L).$$

The full ledger retains both unmatched roles, the common carrier, source route, and finite index. If a later ordered coordinate frame is declared, the surviving residual role may be projected as a side-of-approach token; that token is not primitive negativity.

Definition 55.6 (Tail)**.** *For a natural index N, the* tail beginning at N *is the restricted family*

$$X_{\geq N} := (X_n)_{n \geq N}.$$

This is a continuation-rule restriction, not an already completed infinite set.

55.3 Finite-resolution containment

Definition 55.7 (Resolution neighborhood)**.** *For $L \in \mathbb{Q}_{\mathrm{C}}$ and positive resolution ρ, define the finite-resolution neighborhood report*

$$B_\rho(L) := \{X \in \mathbb{Q}_{\mathrm{C}} : d_{\mathrm{C}}(X, L) < \rho\}.$$

Membership means that the finite signed residual between X and L is constructively smaller than the declared positive resolution.

The set notation is retrospective compression of the membership rule. It does not posit a spatial ball or a completed continuum.

Definition 55.8 (Eventual resolution containment)**.** *The family X is* eventually contained at resolution ρ around L *when there exists a finite index N_ρ such that*

$$n \geq N_\rho \quad \Longrightarrow \quad X_n \in B_\rho(L).$$

Equivalently,

$$n \geq N_\rho \quad \Longrightarrow \quad e_n(X; L) < \rho.$$

The witness N_ρ is a whole finite index. Nothing in the definition requires a final stage after which no further stages exist.

Lemma 55.9 (Resolution monotonicity)**.** *If X is eventually contained at resolution ρ around L and $\rho \leq \sigma$, then X is eventually contained at resolution σ around L.*

Proof. The same witness index works because

$$e_n(X; L) < \rho \leq \sigma.$$

□

The reverse implication is not generally valid. Satisfying a coarse resolution does not certify a finer one.

55.4 Constructive convergence

Definition 55.10 (Resolution-certified convergence to a retained target)**.** *Let $X = (X_n)$ be a finite-stage approximation rule and let Λ be the declared observable signature. The rule converges constructively to a retained target L under Λ when, for every presented positive finite resolution report ρ, one can construct a finite witness N_ρ and a finite tail proof such that every lawful finite stage $n \geq N_\rho$ satisfies*

$$d_\Lambda(X_n, L) < \rho.$$

The conventional notation

$$\lim_{n\to\infty} X_n = L$$

is a compressed statement about this finite witness grammar. The infinity mark is not a final natural index and does not assert that an infinite list has been completed.

Definition 55.11 (Target-free Cauchy certificate)**.** *A finite-stage rule is resolution-coherent under Λ when, for every positive finite ρ, one can construct N_ρ such that all finite $m, n \geq N_\rho$ satisfy*

$$d_\Lambda(X_m, X_n) < \rho.$$

This certificate compares actual finite reports and does not, by itself, manufacture a target object.

Definition 55.12 (Convergence certificate)**.** *A convergence certificate records the generation rule, observable signature, requested resolution, finite witness, retained target when one has independently been constructed, and the finite proof bounding every later lawful finite stage.*

Principle 55.13 (No retrojected completion)**.** *Convergence is a rule-level relation among finite reports. A later abstraction, completion label, or coordinate projection may summarize the rule, but it does not retroactively turn the finite stages into members of an already-existing completed continuum.*

Principle 55.14 (Typed convergence)**.** *A family may converge under one observable signature while failing to stabilize under another. In particular, aggregate effect may become arbitrarily small while the complete source ledger continues to gain new distinguishable occurrences. Scalar convergence therefore does not imply source identity or source stabilization.*

55.5 Exact equality, stabilization, and convergence

Three different closure conditions must be separated.

Definition 55.15 (Finite-stage equality)**.** *The family has* finite-stage equality *with L at index n when*

$$X_n =_{\mathrm{C}} L.$$

Definition 55.16 (Eventual stabilization)**.** *The family* eventually stabilizes *at L when there exists a finite N such that*

$$n \geq N \quad \Longrightarrow \quad X_n =_{\mathrm{C}} L.$$

Theorem 55.17 (Stabilization implies convergence)**.** *If X eventually stabilizes at L, then $X_n \to_{\mathrm{C}} L$.*

Proof. Beyond the stabilization index, every residual is the typed exact-zero report, whose magnitude is smaller than every positive resolution. □

The converse fails.

Example 55.18 (Positive finite residuals converging to zero)**.** *For each positive natural index n, let*

$$X_n = \frac{1}{n}.$$

Every finite stage is a positive ratio report, so

$$X_n \neq_{\mathrm{C}} 0$$

for every finite n.

Let the requested positive resolution be

$$\rho = \frac{a}{b}, \qquad a, b \in \mathbb{N}, \quad a > 0, \quad b > 0.$$

Choose

$$N_\rho = b + 1.$$

For every $n \geq b + 1$,

$$0 < \frac{1}{n} \leq \frac{1}{b+1} < \frac{a}{b},$$

because $a \geq 1$. Hence

$$\frac{1}{n} \longrightarrow_{\mathrm{C}} 0.$$

No numerator occurrence became literal absence. The family satisfies every requested positive resolution while retaining a positive numerator at each finite stage.

Theorem 55.19 (Fixed-positive nonvanishing)**.** *Let $X = (X_n)$ and L be given. Suppose there exists a fixed positive ratio δ such that, beyond every proposed finite index N, some later stage satisfies*

$$d_{\mathrm{C}}(X_n, L) \geq \delta.$$

Then X_n does not converge constructively to L.

Proof. Request the positive resolution $\rho = \delta$. No tail can remain strictly inside that resolution, because every tail contains a stage whose residual magnitude is at least δ. □

Corollary 55.20 (A fixed positive report is not zero by refinement)**.** *If $c > 0$ is fixed and $X_n = c$ for every n, then X_n does not converge to zero.*

This blocks the informal substitution

$$\text{“arbitrarily refined”} \quad \rightsquigarrow \quad \text{“a fixed positive primitive became zero.”}$$

55.6 Orientation of approach

Convergence controls residual magnitude. It does not, by itself, erase residual orientation.

Definition 55.21 (Eventual approach from below)**.** *A family approaches L* eventually from below *when it converges to L and there exists N such that*

$$n \geq N \quad \Longrightarrow \quad X_n \preceq_{\mathbb{Q}_{\mathrm{C}}} L.$$

Definition 55.22 (Eventual approach from above)**.** *A family approaches L* eventually from above *when it converges to L and there exists N such that*

$$n \geq N \quad \Longrightarrow \quad L \preceq_{\mathbb{Q}_C} X_n.$$

A convergent family need satisfy neither one-sided condition.

Example 55.23 (Alternating orientation with contracting magnitude)**.** *Let*

$$X_n = \frac{(-1)^n}{n}$$

for positive n. The residual orientation alternates at every stage, but

$$d_C(X_n, 0) = \frac{1}{n} \longrightarrow_C 0.$$

Therefore $X_n \to_C 0$ while the approach route does not stabilize to one orientation.

The scalar limit report forgets this alternating orientation unless the approach ledger is retained separately.

55.7 Uniqueness of a retained limit

Theorem 55.24 (Uniqueness of constructive limit in $\mathbb{Q}_C$)**.** *If*

$$X_n \longrightarrow_C L \qquad \text{and} \qquad X_n \longrightarrow_C M,$$

then

$$L =_{\mathbb{Q}_C} M.$$

Proof. Assume instead that

$$d := d_C(L, M) > 0.$$

Choose the positive resolution

$$\rho = \frac{d}{3}.$$

There are finite witnesses N_L and N_M such that beyond their maximum,

$$d_C(X_n, L) < \frac{d}{3} \qquad \text{and} \qquad d_C(X_n, M) < \frac{d}{3}.$$

The finite residual triangle law gives

$$d = d_C(L, M) \leq d_C(L, X_n) + d_C(X_n, M) < \frac{2d}{3},$$

a contradiction. Therefore $d = 0$, so the two targets are equal as normalized signed-ratio reports. □

Limit uniqueness is thus a theorem of the separated finite residual grammar. It is not supplied by the notation lim itself.

55.8 Algebraic preservation of limits

Limit arithmetic must be proved from finite residual bounds. It cannot be licensed merely by symbolic substitution.

Theorem 55.25 (Sum, inverse, and difference laws)**.** *Suppose*

$$X_n \to_{\mathrm{C}} L \qquad \text{and} \qquad Y_n \to_{\mathrm{C}} M.$$

Then

$$X_n + Y_n \to_{\mathrm{C}} L + M,$$
$$-X_n \to_{\mathrm{C}} -L,$$

and

$$X_n - Y_n \to_{\mathrm{C}} L - M.$$

Proof. Given positive resolution ρ, choose tails satisfying

$$d_{\mathrm{C}}(X_n, L) < \frac{\rho}{2}, \qquad d_{\mathrm{C}}(Y_n, M) < \frac{\rho}{2}.$$

The residual triangle law gives

$$d_{\mathrm{C}}(X_n + Y_n, L + M) \le d_{\mathrm{C}}(X_n, L) + d_{\mathrm{C}}(Y_n, M) < \rho.$$

Orientation reversal preserves magnitude, proving the inverse law. The difference law follows by combining the first two. □

Theorem 55.26 (Fixed scaling law)**.** *Let $C \in \mathbb{Q}_{\mathrm{C}}$ be fixed. If $X_n \to_{\mathrm{C}} L$, then*

$$CX_n \to_{\mathrm{C}} CL.$$

Proof. If $C = 0$, every product is an exact typed zero. If $C \neq 0$, request resolution $\rho/\|C\|_{\mathrm{C}}$ for the residual of X_n from L. Finite ratio multiplication then gives

$$d_{\mathrm{C}}(CX_n, CL) = \|C\|_{\mathrm{C}}\, d_{\mathrm{C}}(X_n, L) < \rho.$$

□

Lemma 55.27 (Convergent families are eventually bounded)**.** *If $X_n \to_{\mathrm{C}} L$, then there exist a positive finite bound B and a finite index N such that*

$$n \ge N \quad \Longrightarrow \quad \|X_n\|_{\mathrm{C}} \le B.$$

Proof. Use resolution 1. Beyond its witness index,

$$d_{\mathrm{C}}(X_n, L) < 1.$$

The residual triangle law yields

$$\|X_n\|_{\mathrm{C}} \le \|L\|_{\mathrm{C}} + 1.$$

Take $B = \|L\|_{\mathrm{C}} + 1$. □

Theorem 55.28 (Product law)**.** *If $X_n \to_{\mathrm{C}} L$ and $Y_n \to_{\mathrm{C}} M$, then*

$$X_n Y_n \to_{\mathrm{C}} LM.$$

Proof. Choose an eventual bound B for $\|X_n\|_{\mathrm{C}}$ and set

$$K = B + \|M\|_{\mathrm{C}} + 1.$$

Given positive resolution ρ, choose a common tail on which

$$d_{\mathrm{C}}(Y_n, M) < \frac{\rho}{K} \qquad \text{and} \qquad d_{\mathrm{C}}(X_n, L) < \frac{\rho}{K}.$$

Using the finite decomposition

$$X_nY_n - LM = X_n(Y_n - M) + M(X_n - L),$$

finite magnitude and triangle laws give

$$\begin{aligned} d_{\mathrm{C}}(X_nY_n, LM) &\le \|X_n\|_{\mathrm{C}} d_{\mathrm{C}}(Y_n, M) + \|M\|_{\mathrm{C}} d_{\mathrm{C}}(X_n, L) \\ &< \frac{B\rho}{K} + \frac{\|M\|_{\mathrm{C}}\rho}{K} < \rho. \end{aligned}$$

□

Theorem 55.29 (Reciprocal law away from zero)**.** *If $X_n \to_{\mathrm{C}} L$ and $L \neq 0$, then there exists a finite index after which every X_n is nonzero and*

$$\frac{1}{X_n} \longrightarrow_{\mathrm{C}} \frac{1}{L}.$$

Proof. Choose resolution $\|L\|_{\mathrm{C}}/2$. Beyond its witness index,

$$d_{\mathrm{C}}(X_n, L) < \frac{\|L\|_{\mathrm{C}}}{2},$$

so

$$\|X_n\|_{\mathrm{C}} > \frac{\|L\|_{\mathrm{C}}}{2} > 0.$$

Thus the reciprocal compiler is defined on the tail. The finite identity

$$\frac{1}{X_n} - \frac{1}{L} = \frac{L - X_n}{X_nL}$$

gives

$$d_{\mathrm{C}}\left(\frac{1}{X_n}, \frac{1}{L}\right) \le \frac{2\, d_{\mathrm{C}}(X_n, L)}{\|L\|_{\mathrm{C}}^2}.$$

For any requested output resolution ρ, require

$$d_{\mathrm{C}}(X_n, L) < \frac{\rho\|L\|_{\mathrm{C}}^2}{2}$$

on a sufficiently late common tail. The displayed bound then yields the desired reciprocal resolution.

□

Every law above is a fiber-preservation theorem for the limit report. None licenses replacement of a finite approximation ledger by its target when stagewise source information matters.

55.9 Nested whole bounds

Convergence may be certified by whole finite upper and lower reports rather than by a single displayed error formula.

Definition 55.30 (Nested bound family)**.** *A* nested bound family *is a pair of finite-ratio families*

$$(A_m, B_m)_{m\in\mathbb{N}}$$

satisfying

$$A_m \preceq A_{m+1} \preceq B_{m+1} \preceq B_m.$$

Its width report is

$$W_m := B_m - A_m \geq 0.$$

Every stage is a whole finite interval report with two finite endpoints. No endpoint is a partial object, and no interval width is presumed to vanish at a finite stage.

Definition 55.31 (Target retained by nested bounds)**.** *A target $L \in \mathbb{Q}_{\mathrm{C}}$ is retained by the nested bound family when*

$$A_m \preceq L \preceq B_m$$

for every finite m.

Theorem 55.32 (Nested-bound convergence)**.** *Suppose L is retained by (A_m, B_m) and*

$$W_m \longrightarrow_{\mathrm{C}} 0.$$

If a family X_m satisfies

$$A_m \preceq X_m \preceq B_m$$

for every m, then

$$X_m \longrightarrow_{\mathrm{C}} L.$$

Proof. Both X_m and L lie inside the same finite bound carrier, so

$$d_{\mathrm{C}}(X_m, L) \leq W_m.$$

Every requested positive resolution eventually dominates W_m, and hence also the residual magnitude. □

Principle 55.33 (Bounds without a retained target)**.** *A nested rational bound family whose widths converge to zero need not yet possess a target inside $\mathbb{Q}_{\mathrm{C}}$. Without such a retained target, it is a completion specification, not a proved rational limit.*

This distinction prevents the completeness of the real carrier from being smuggled into the definition of rational convergence.

55.10 The constructive squeeze theorem

Theorem 55.34 (Squeeze by retained bounds)**.** *Suppose*

$$A_n \preceq X_n \preceq B_n$$

for every sufficiently late stage and

$$A_n \to_{\mathrm{C}} L, \qquad B_n \to_{\mathrm{C}} L.$$

Then

$$X_n \to_{\mathrm{C}} L.$$

Proof. For a requested positive resolution ρ, choose a tail on which both bounding residuals from L are smaller than ρ. Since X_n lies between the two bounds, its signed residual from L cannot exceed the larger bounding residual magnitude. Hence it is also smaller than ρ on that tail. □

The theorem is a finite order argument repeated at every requested resolution. It does not require a geometric picture of continuously shrinking intervals.

55.11 Cauchy tail coherence

A family may exhibit internally contracting tails even when no target has yet been constructed in the present carrier.

Definition 55.35 (Constructive Cauchy family)**.** *An approximation family $X = (X_n)$ in $\mathbb{Q}_{\mathrm{C}}$ is* Cauchy *when, for every positive resolution ρ, there exists a finite index N_ρ such that*

$$m, n \geq N_\rho \quad \Longrightarrow \quad d_{\mathrm{C}}(X_m, X_n) < \rho.$$

The Cauchy condition compares finite tail stages with one another. It does not assert that a target already exists in the carrier.

Theorem 55.36 (Convergence implies Cauchy coherence)**.** *Every constructively convergent family in $\mathbb{Q}_{\mathrm{C}}$ is Cauchy.*

Proof. Suppose $X_n \to_{\mathrm{C}} L$. For requested resolution ρ, choose a tail on which

$$d_{\mathrm{C}}(X_n, L) < \frac{\rho}{2}.$$

For m, n in that tail, the residual triangle law gives

$$d_{\mathrm{C}}(X_m, X_n) \leq d_{\mathrm{C}}(X_m, L) + d_{\mathrm{C}}(L, X_n) < \rho.$$

□

Principle 55.37 (No converse before completion)**.** *The assertion that every Cauchy family has a target is not a theorem of the finite rational carrier constructed so far. It is the defining obligation of a later completion.*

Definition 55.38 (Cauchy zero-distance equivalence)**.** *For Cauchy families $X = (X_n)$ and $Y = (Y_n)$, write*

$$X \sim_{\mathrm{Cau}} Y$$

when

$$d_{\mathrm{C}}(X_n, Y_n) \longrightarrow_{\mathrm{C}} 0.$$

Equivalently, for every positive resolution ρ, there is a finite N_ρ such that

$$n \geq N_\rho \quad \Longrightarrow \quad d_{\mathrm{C}}(X_n, Y_n) < \rho.$$

Theorem 55.39 (Cauchy zero-distance is an equivalence relation)**.** *The relation $\sim_{\mathrm{Cau}}$ is reflexive, symmetric, and transitive on the family of Cauchy approximation rules.*

Proof. Reflexivity follows from exact zero residual at every synchronized stage. Symmetry follows from residual symmetry. For transitivity, if $X \sim_{\mathrm{Cau}} Y$ and $Y \sim_{\mathrm{Cau}} Z$, construct a finer positive resolution report σ whose two-copy join is bounded by the requested resolution ρ, use σ for each relation, and apply the residual triangle law. □

This equivalence relation is the natural candidate for the completion quotient in the next section. At this stage it is only an explicitly constructed relation among finite approximation families.

55.12 Limit fibers and preservation obligations

Let $\mathcal{D}_{\mathbb{Q}}$ denote the family of approximation rules that converge to retained targets in $\mathbb{Q}_{\mathrm{C}}$. Uniqueness permits the limit-value map

$$\mathrm{Lim} : \mathcal{D}_{\mathbb{Q}} \longrightarrow \mathbb{Q}_{\mathrm{C}}, \qquad X \longmapsto L.$$

The fiber

$$\mathrm{Lim}^{-1}(L)$$

contains every admitted approximation family whose limit report is L. These families may differ in source, formula, recurrence, rate, side of approach, oscillation, finite-stage zeros, and proof certificate.

Example 55.40 (One limit fiber, different stagewise constructions)**.** *The constant-zero family*

$$Z_n = 0$$

and the positive family

$$X_n = \frac{1}{n}$$

both lie in the zero-limit fiber:

$$\mathrm{Lim}(Z) = 0 = \mathrm{Lim}(X).$$

Yet:

(i) every stage of Z is an exact typed zero;

(ii) no stage of X is zero;

(iii) Z stabilizes immediately;

(iv) X never stabilizes;

(v) their source and residual ledgers are different.

Theorem 55.41 (Limit-fiber preservation condition)**.** *A property P of approximation families can be computed from the limit value alone exactly when P is constant on every fiber of* Lim*.*

Proof. This is the projection-fiber factorization theorem of Section 54 applied to the limit-value map. □

Therefore properties such as "has limit zero" descend to the limit report, while properties such as "contains an exact zero stage," "approaches from above," "stabilizes," or "uses a particular recurrence" generally do not.

Principle 55.42 (Limit equality is typed equality)**.** *From*

$$\lim X_n = \lim Y_n$$

one may infer equality of limit values. One may not infer constructive equality of the approximation families unless an additional theorem proves that the relevant source property is constant on the common limit fiber.

55.13 Finite prefixes do not determine convergence

Theorem 55.43 (Finite-prefix underdetermination)**.** *No finite prefix by itself determines a unique limit or even determines whether an unrestricted continuation converges.*

Proof. Let

$$[X_0, X_1, \ldots, X_N]$$

be any completed finite prefix. It admits, among others:

(i) a continuation that is constantly 0 after N, which converges to 0;

(ii) a continuation that is constantly 1 after N, which converges to 1;

(iii) a continuation alternating between 0 and 1, which does not converge.

All three preserve the given finite prefix. Therefore convergence belongs to the retained continuation rule and its proof, not to the prefix alone. □

A finite descriptor may nevertheless certify convergence when its rule implies the required tail bounds. The theorem blocks only unsupported extrapolation from a finite sample.

55.14 Radix limits and the report $0.999\ldots$

For each positive natural m, define the finite decimal prefix

$$p_m := \sum_{k=1}^{m} 9 \cdot 10^{-k}.$$

Finite induction using the already constructed power and ratio laws gives

$$p_m = 1 - 10^{-m}.$$

Hence the exact finite residual is

$$1 - p_m = 10^{-m} > 0.$$

Therefore

$$p_m \neq_{\mathrm{C}} 1$$

for every finite m.

Theorem 55.44 (Decimal-nine prefix convergence)**.** *The finite prefix family satisfies*

$$p_m \longrightarrow_{\mathrm{C}} 1.$$

Proof. Let the requested resolution be $\rho = a/b > 0$. Choose $m = b$. Since $10^b > b$ for every positive natural b,

$$10^{-b} < \frac{1}{b} \leq \frac{a}{b}.$$

For every $n \geq b$,

$$d_{\mathrm{C}}(p_n, 1) = 10^{-n} \leq 10^{-b} < \rho.$$

□

There are now three distinct report types:

(i) a finite prefix p_m, which is never equal to 1;

(ii) the finite recurrence descriptor "append another 9 digit," which is a rule object;

(iii) the limit-value report of the generated prefix family, which equals 1.

Principle 55.45 (Typed interpretation of recurring notation)**.** *If the notation*

$$0.999\ldots_{10}$$

is declared to denote the finite recurrence descriptor, then it is not constructively identical to the unit report. If it is declared to denote the limit value of its finite-prefix family, then

$$0.999\ldots_{10} =_{\mathrm{lim}} 1.$$

The equality is a limit-value equality, not equality of any finite prefix with 1 *and not equality of the source descriptor with the unit construction.*

This resolves the apparent conflict without erasing either finite residuals or the proved limit value.

55.15 Divergence and nonconvergence witnesses

Failure to observe stabilization is not itself a proof of divergence. Nonconvergence requires a witness showing that at least one required resolution condition fails for every proposed target.

Definition 55.46 (Resolution-separated tail behavior)**.** *A family has* resolution-separated tail behavior *when there exists a positive δ such that, beyond every finite index, two later stages m, n satisfy*

$$d_{\mathrm{C}}(X_m, X_n) \geq \delta.$$

Theorem 55.47 (Separated tails cannot converge)**.** *A family with resolution-separated tail behavior is not constructively convergent.*

Proof. Every convergent family is Cauchy. Request resolution δ in the Cauchy condition. The separated-tail witness contradicts the existence of any tail whose pairwise residuals are all strictly smaller than δ. □

Example 55.48 (Persistent alternation)**.** *The family*

$$X_n = \begin{cases} 0, & n \text{ even}, \\ 1, & n \text{ odd} \end{cases}$$

has later stages separated by residual magnitude 1 *beyond every finite index. It therefore does not converge.*

The conclusion follows from a retained positive separation, not from a picture of oscillation on a pre-existing coordinate line.

55.16 Cofinal resolution families and base independence

Testing every positive ratio is conceptually exact but can be compressed when a declared resolution family refines beneath every positive ratio.

Definition 55.49 (Cofinal resolution family toward zero)**.** *A family $\mathcal{E}$ of positive ratio reports is* cofinal toward zero *when, for every positive ratio ρ, some $\varepsilon \in \mathcal{E}$ satisfies*

$$0 < \varepsilon < \rho.$$

Theorem 55.50 (Cofinal resolution test)**.** *If $\mathcal{E}$ is cofinal toward zero, then $X_n \to_{\mathrm{C}} L$ exactly when eventual resolution containment holds for every $\varepsilon \in \mathcal{E}$.*

Proof. The forward direction is immediate because every ε is a positive resolution. Conversely, given arbitrary positive ρ, choose $\varepsilon \in \mathcal{E}$ with $\varepsilon < \rho$. A tail lying within ε also lies within ρ by resolution monotonicity. □

Theorem 55.51 (Radix-power resolutions are cofinal)**.** *For every finite base $b \geq 2$, the family*

$$\mathcal{E}_b = \{b^{-m} : m \in \mathbb{N},\ m > 0\}$$

is cofinal toward zero in the positive constructive ratios.

Proof. Let $\rho = a/c > 0$. Choose $m = c$. Since $b \geq 2$ and $c > 0$,

$$b^c > c.$$

Therefore

$$b^{-c} < \frac{1}{c} \leq \frac{a}{c} = \rho.$$

□

Thus binary, decimal, or any other finite-base resolution ladder tests the same convergence relation when the base and comparison theorem are retained. No radix is ontologically privileged by the definition of convergence.

55.17 No positive finite infinitesimal in the rational carrier

Theorem 55.52 (No fixed positive rational lies below every positive resolution)**.** *For every fixed $\delta \in \mathbb{Q}_C$ with $\delta > 0$, there exists a positive resolution ρ such that*

$$0 < \rho < \delta.$$

Proof. Take $\rho = \delta/2$. □

Corollary 55.53 (No finite positive rational is an undetectable residual)**.** *A fixed positive rational residual cannot satisfy all positive-resolution tests against zero.*

Convergence to zero therefore requires a family of changing finite residuals whose refinement responds to every requested positive bound. It cannot be represented by one fixed positive rational relabeled as zero.

55.18 Limit preservation theorem

Theorem 55.54 (Constructive limit preservation)**.** *A limit-value report preserves exactly the properties of an approximation family that are constant on the corresponding limit fiber. In particular, the limit value alone generally does not preserve:*

(i) the finite rule or recurrence generating the family;

(ii) the source object at each index;

(iii) the sign or side of approach;

(iv) the rate at which requested resolutions are met;

(v) the witness-selection rule $\rho \mapsto N_\rho$;

(vi) whether any finite stage equals the target;

(vii) whether the family eventually stabilizes;

(viii) the base or positional grammar of the approximations;

(ix) or the proof route establishing convergence.

These data may be discarded only for a later property proved constant on the limit fiber.

The limit map is therefore another quotient report. It is useful precisely because it suppresses distinctions irrelevant to limit-value arithmetic, but it cannot be used to prove that those distinctions never existed.

55.19 Dependency summary and handoff to completion

The construction of convergence is now

$$\begin{aligned}\text{finite signed ratios} &\longrightarrow \text{finite residual and magnitude} \longrightarrow \text{positive resolution report}\\ &\longrightarrow \text{eventual tail containment} \longrightarrow \text{constructive convergence}\\ &\longrightarrow \text{limit-value fibers and preservation audit.}\end{aligned}$$

The essential distinctions are:

(1) finite-stage equality is not convergence;

(2) eventual stabilization implies convergence but is not required by it;

(3) a positive residual may survive at every finite stage while its family converges to zero;

(4) no fixed positive finite ratio becomes zero through refinement;

(5) convergence to a retained target implies Cauchy tail coherence;

(6) Cauchy coherence alone does not yet construct a target in $\mathbb{Q}_{\mathrm{C}}$;

(7) equal limit values do not imply equal approximation families;

(8) a finite prefix does not determine convergence without a continuation rule;

(9) recurring notation must distinguish its finite descriptor from its limit-value projection;

(10) and every operation on future completed values must be proved invariant under Cauchy zero-distance equivalence.

The next section constructs a real-report evaluator grammar from presented Cauchy refinement rules under $\sim_{\mathrm{Cau}}$. It must prove that every admitted operation respects the declared equivalence fibers and must state exactly which finite approximation data each later projection forgets.

56 Rational refinement and real-report grammar

Principle 56.1 (Real-domain status)**.** *The term "real" in this part names a conventional coordinate/refinement grammar, not a primitively demonstrated completed continuum. Every displayed signed real coordinate must retain the finite refinement evaluator, matching residuals, and resolution certificates from which the coordinate was compiled. Claims internal to this grammar are conditional algebraic or analytic claims about the compiler; they are not ontological proofs that a totality of real numbers exists.*

The finite rational grammar already constructed contains finite ratio reports and their exact arithmetic. A rule may support arbitrarily fine rational refinement without any one execution becoming infinite. PCL therefore treats a real report first as an evaluator grammar rather than as a member drawn from a completed set of all Cauchy sequences.

56.1 The finite rational carrier and the refinement obligation

Let $\mathbb{Q}_{\mathrm{C}}$ denote the grammar of finite signed constructive ratios under the ratio equality already proved. A *Cauchy approximation rule* is any particular finite rule X whose stages $X_n \in \mathbb{Q}_{\mathrm{C}}$ admit a target-free Cauchy certificate at every requested finite resolution.

Principle 56.2 (No set of all Cauchy families)**.** *PCL does not require, and at the primitive level does not admit, a completed set containing every possible Cauchy rule. Quantification over admissible rules means: for any presented rule satisfying the declared finite obligations, the following construction applies.*

For two presented Cauchy rules X and Y, write

$$X \sim_{\mathrm{Cau}} Y$$

when their synchronized finite-stage residual is resolution-certified to zero:

$$\forall \rho > 0 \; \exists N_\rho \; \forall n \geq N_\rho : \quad d_{\mathrm{C}}(X_n, Y_n) < \rho.$$

Every quantifier here ranges over finitely presentable reports and finitely checked stages or rules for all later finite stages.

56.2 Real reports as finite evaluators

Definition 56.3 (Constructive real report)**.** *A constructive real report is a finite evaluator specification*

$$\mathcal{R}_X = (X, \mathcal{C}_X, \Gamma_X)$$

that accepts a requested positive rational resolution ρ and returns a finite rational enclosure or representative together with a certificate that every lawful further refinement remains inside the requested tolerance.

The notation

$$[X]$$

is henceforth a compiled label for the real-report rule generated by X under the equivalence relation $\sim_{\mathrm{Cau}}$. It is not a set-builder expression collecting all equivalent infinite sequences.

Definition 56.4 (Constructive real grammar)**.** *The symbol*

$$\mathbb{R}_{\mathrm{C}}$$

is conventional shorthand for the declared real-coordinate evaluator grammar and its proved operations. It is not a primitively completed set of all real numbers, and membership notation involving it is always shorthand for presentation to that compiler.

Definition 56.5 (Real-report equality)**.** *For two presented real evaluators $[X]$ and $[Y]$, define*

$$[X] =_{\mathbb{R}_{\mathrm{C}}} [Y] \quad \Longleftrightarrow \quad X \sim_{\mathrm{Cau}} Y.$$

This compares their finite refinement behavior. It does not erase source-rule identity.

Theorem 56.6 (Every admitted Cauchy rule yields a real evaluator)**.** *Given any particular Cauchy approximation rule X with a supplied Cauchy witness selector, the pair consisting of that rule and its witness grammar defines a constructive real evaluator $[X]$.*

Proof. For every requested positive finite resolution, the Cauchy certificate supplies a finite stage after which every later finite stage lies in the declared tolerance. Returning such a finite representative or nested rational enclosure therefore produces the requested evaluation. No completed totality of all possible rules is required. □

56.3 Embedding finite rationals

Definition 56.7 (Constant-family embedding)**.** *For $q \in \mathbb{Q}_{\mathrm{C}}$, let*

$$\underline{q} = (q, q, q, \ldots)$$

denote the constant Cauchy family. Define

$$\iota : \mathbb{Q}_{\mathrm{C}} \longrightarrow \mathbb{R}_{\mathrm{C}}, \qquad \iota(q) := [\underline{q}].$$

Theorem 56.8 (The rational embedding is injective)**.** *If*

$$\iota(p) =_{\mathbb{R}_{\mathrm{C}}} \iota(q),$$

then

$$p = q$$

in $\mathbb{Q}_{\mathrm{C}}$.

Proof. Equality of the completed reports means

$$d_{\mathrm{C}}(p, q) \longrightarrow_{\mathrm{C}} 0$$

for the constant residual family. If $p \neq q$, then the residual magnitude is one fixed positive rational. The fixed-positive nonvanishing theorem from Section 55 says that such a family cannot converge to zero. Hence $p = q$. □

The embedding therefore preserves finite rational distinction. From this point onward, a rational may be written inside $\mathbb{R}_{\mathrm{C}}$ without displaying ι when the carrier is clear, but the embedding remains part of the dependency ledger.

Theorem 56.9 (Rational limits agree with the embedding)**.** *If a rational Cauchy family X converges in $\mathbb{Q}_{\mathrm{C}}$ to a retained rational q, then*

$$[X] =_{\mathbb{R}_{\mathrm{C}}} \iota(q).$$

Proof. Rational convergence to q is exactly the statement

$$d_{\mathrm{C}}(X_n, q) \longrightarrow_{\mathrm{C}} 0,$$

which says $X \sim_{\mathrm{Cau}} \underline{q}$. □

56.4 Nested rational intervals as completion reports

Cauchy families are not the only finite description of a completed real. Nested rational bounds provide an equivalent route when their widths converge to zero.

Definition 56.10 (Cauchy interval specification)**.** *A* Cauchy interval specification *is a family of rational endpoint pairs*

$$I_n = [L_n, U_n]$$

satisfying

$$L_n \leq L_{n+1} \leq U_{n+1} \leq U_n$$

and

$$U_n - L_n \longrightarrow_{\mathrm{C}} 0.$$

Every stage is a finite rational interval report. The interval need not collapse at any finite stage.

Theorem 56.11 (Nested endpoints are equivalent Cauchy families)**.** *For a Cauchy interval specification (I_n), both endpoint families (L_n) and (U_n) are Cauchy and*

$$(L_n) \sim_{\mathrm{Cau}} (U_n).$$

Proof. For $m \geq n$, nesting gives

$$L_n \leq L_m \leq U_m \leq U_n.$$

Therefore

$$0 \leq L_m - L_n \leq U_n - L_n$$

and similarly

$$0 \leq U_n - U_m \leq U_n - L_n.$$

Since the width tends to zero, each endpoint family is Cauchy. Also

$$0 \leq U_n - L_n \longrightarrow_{\mathrm{C}} 0,$$

so the two endpoint families are Cauchy-equivalent. □

Definition 56.12 (Real report retained by nested intervals)**.** *The completed real retained by (I_n) is*

$$\mathrm{Real}(I) := [(L_n)] = [(U_n)].$$

Theorem 56.13 (Interior choice invariance)**.** *If $X_n \in \mathbb{Q}_{\mathrm{C}}$ satisfies*

$$L_n \leq X_n \leq U_n$$

for every n, then (X_n) is Cauchy and

$$[(X_n)] = \mathrm{Real}(I).$$

Proof. Every interior choice lies within the interval width of either endpoint:

$$0 \leq X_n - L_n \leq U_n - L_n.$$

The width tends to zero, so $X \sim_{\mathrm{Cau}} L$. Cauchy coherence follows from equivalence with the Cauchy endpoint family. □

The completed value therefore depends on the shrinking interval fiber, not on which finite point is selected from each interval.

56.5 Every Cauchy family admits nested rational bounds

Theorem 56.14 (Nested-bound representative theorem)**.** *Every rational Cauchy family $X = (X_n)$ admits a Cauchy interval specification whose retained completed real is $[X]$.*

Proof. Use the cofinal dyadic resolution ladder. Choose strictly increasing indices N_k such that

$$m, n \geq N_k \Longrightarrow d_{\mathrm{C}}(X_m, X_n) < 2^{-(k+2)}.$$

Set

$$c_k := X_{N_k}, \qquad r_k := 2^{-(k+1)}, \qquad I_k := [c_k - r_k, c_k + r_k].$$

Because $N_{k+1} \geq N_k$,

$$|c_{k+1} - c_k| < 2^{-(k+2)} = r_{k+1}.$$

Hence every point of I_{k+1} lies within

$$r_{k+1} + |c_{k+1} - c_k| < 2r_{k+1} = r_k$$

of c_k, so

$$I_{k+1} \subseteq I_k.$$

The widths are

$$2r_k = 2^{-k} \longrightarrow_{\mathrm{C}} 0.$$

Finally, the center subsequence (c_k) is equivalent to X, because a Cauchy family and any cofinal tail subsequence have zero Cauchy distance. Therefore the nested intervals retain $[X]$. □

This theorem shows that Cauchy-family completion and nested-rational-interval completion report the same carrier once their equivalence has been proved.

56.6 Addition and additive orientation on completed reports

For Cauchy families $X = (X_n)$ and $Y = (Y_n)$, define stagewise families

$$X + Y := (X_n + Y_n), \qquad -X := (-X_n).$$

Lemma 56.15 (Stagewise sums and negatives are Cauchy)**.** *If X and Y are rational Cauchy families, then $X + Y$ and $-X$ are rational Cauchy families.*

Proof. For requested resolution ρ, place each input tail within $\rho/2$. Then

$$|(X_m + Y_m) - (X_n + Y_n)| \leq |X_m - X_n| + |Y_m - Y_n| < \rho.$$

Negation preserves residual magnitude exactly. □

Theorem 56.16 (Addition and negation descend to completion)**.** *The definitions*

$$[X] + [Y] := [X + Y], \qquad -[X] := [-X]$$

are independent of the chosen representatives.

Proof. Suppose $X \sim_{\mathrm{Cau}} X'$ and $Y \sim_{\mathrm{Cau}} Y'$. Then

$$|(X_n + Y_n) - (X'_n + Y'_n)| \leq |X_n - X'_n| + |Y_n - Y'_n| \longrightarrow_{\mathrm{C}} 0.$$

Thus $X + Y \sim_{\mathrm{Cau}} X' + Y'$. Negation preserves the same residual magnitude, so $-X \sim_{\mathrm{Cau}} -X'$. □

The additive zero and subtraction are then

$$0_{\mathbb{R}_{\mathrm{C}}} := [\underline{0}], \qquad [X] - [Y] := [X] + (-[Y]).$$

56.7 Boundedness of rational Cauchy families

Multiplication requires a bound because a small factor error can be amplified by an unbounded companion.

Lemma 56.17 (Cauchy boundedness)**.** *Every rational Cauchy family $X = (X_n)$ has a finite rational bound $M_X > 0$ such that*

$$|X_n| \le M_X$$

for every finite n.

Proof. Choose a tail index N such that

$$m, n \ge N \Longrightarrow |X_m - X_n| < 1.$$

Then every tail term satisfies

$$|X_m| \le |X_N| + 1.$$

The finite initial list

$$|X_0|, \ldots, |X_{N-1}|$$

has a largest retained rational magnitude. Choose M_X larger than that finite maximum and larger than $|X_N| + 1$. □

The bound is not unique and is not part of the completed scalar unless explicitly retained. Its role is to certify preservation of multiplication under the completion quotient.

56.8 Multiplication on completed reports

For Cauchy families X and Y, define the stagewise product

$$XY := (X_n Y_n).$$

Lemma 56.18 (Stagewise products are Cauchy)**.** *If X and Y are rational Cauchy families, then XY is rational Cauchy.*

Proof. Choose finite bounds $M_X, M_Y \ge 1$. For m, n,

$$|X_m Y_m - X_n Y_n| \le |X_m|\,|Y_m - Y_n| + |Y_n|\,|X_m - X_n|.$$

Given positive resolution ρ, choose a common tail on which

$$|Y_m - Y_n| < \frac{\rho}{2M_X} \quad \text{and} \quad |X_m - X_n| < \frac{\rho}{2M_Y}.$$

The displayed residual is then smaller than ρ. □

Theorem 56.19 (Multiplication descends to completion)**.** *The definition*

$$[X] \cdot [Y] := [XY]$$

is independent of the chosen representatives.

Proof. Suppose $X \sim_{\mathrm{Cau}} X'$ and $Y \sim_{\mathrm{Cau}} Y'$. Choose rational bounds M_X for X and $M_{Y'}$ for Y'. Then

$$|X_nY_n - X'_nY'_n| \le |X_n|\,|Y_n - Y'_n| + |Y'_n|\,|X_n - X'_n|.$$

Both right-hand residual families tend to zero, and the finite bounds prevent amplification from defeating any requested resolution. Hence

$$XY \sim_{\mathrm{Cau}} X'Y'.$$

□

Corollary 56.20 (The rational embedding preserves arithmetic)**.** *For $p, q \in \mathbb{Q}_{\mathrm{C}}$,*

$$\iota(p+q) = \iota(p) + \iota(q), \qquad \iota(pq) = \iota(p)\iota(q), \qquad \iota(-p) = -\iota(p).$$

At this point $\mathbb{R}_{\mathrm{C}}$ has completed addition, subtraction, and multiplication. Reciprocal construction for completed nonzero reports will require the signed separation and order developed in Section 58.

56.9 Completed zero versus finite exact zero

Definition 56.21 (Completed zero class)**.** *The completed zero is*

$$0_{\mathbb{R}_{\mathrm{C}}} := [\underline{0}] = \{X \in \mathrm{Cau}(\mathbb{Q}_{\mathrm{C}}) : X_n \to_{\mathrm{C}} 0\}.$$

A member of the zero class need not contain a finite exact-zero stage.

Example 56.22 (Positive stages in the completed-zero fiber)**.** *The family*

$$X_n = \frac{1}{n+1}$$

satisfies

$$X_n > 0$$

at every finite stage, yet

$$[X] = 0_{\mathbb{R}_{\mathrm{C}}}.$$

Example 56.23 (Exact zero at every stage)**.** *The constant family*

$$Z_n = 0$$

has a typed finite exact-zero report at every stage and also satisfies

$$[Z] = 0_{\mathbb{R}_{\mathrm{C}}}.$$

The two source families are not constructively identical:

$$X \neq_{\mathrm{C}} Z.$$

They become equal only after the completion quotient discards the distinction between their vanishing residual histories.

Theorem 56.24 (No nonzero rational enters the completed-zero class)**.** *For $q \in \mathbb{Q}_{\mathrm{C}}$,*

$$\iota(q) = 0_{\mathbb{R}_{\mathrm{C}}} \quad \Longleftrightarrow \quad q = 0.$$

Proof. This is injectivity of the rational embedding applied to q and 0. □

Principle 56.25 (Typed completed zero)**.** *A completed zero reports zero Cauchy distance from the constant-zero family. It does not assert that every representative has a finite exact zero, that every stage lacks positive occurrences, or that every zero representative shares one construction history.*

56.10 Information discarded by real completion

The quotient map κ identifies all Cauchy-equivalent approximation families. Consequently, a completed scalar generally does not retain:

(i) the source rule that generated the approximations;

(ii) the chosen rational representatives;

(iii) the base or digit grammar of a positional name;

(iv) the side or orientation of approach;

(v) monotonicity, alternation, or finite overshoot;

(vi) the rate at which resolutions are met;

(vii) the witness map $\rho \mapsto N_\rho$;

(viii) whether any finite stage equals the completed value;

(ix) whether the family eventually stabilizes;

(x) the nested interval, recurrence, geometric, or algebraic certificate used to prove Cauchy coherence;

(xi) or the source-level zeros and cancellations occurring along the way.

Example 56.26 (One completed rational, different approximation histories)**.** *The families*

$$A_n = 1, \qquad B_n = 1 - 10^{-(n+1)}, \qquad C_n = 1 + (-1)^n 2^{-(n+1)}$$

all represent the same completed real:

$$[A] = [B] = [C] = \iota(1).$$

The first is stationary, the second approaches from below, and the third alternates around the target. Completion preserves their common scalar and discards those stagewise distinctions unless they are carried in a richer report.

Theorem 56.27 (Completion-fiber preservation)**.** *A property P of rational Cauchy families descends to a property of completed real reports exactly when*

$$X \sim_{\mathrm{Cau}} Y \Longrightarrow P(X) = P(Y).$$

An operation descends exactly when it respects $\sim_{\mathrm{Cau}}$ in every input lane.

Proof. This is the projection-fiber factorization theorem of Section 54 applied to the quotient map κ. □

56.11 Rational completion theorem

Theorem 56.28 (Constructive rational completion)**.** *The carrier $\mathbb{R}_{\mathrm{C}}$ satisfies the following completion obligations:*

(i) every rational Cauchy family determines one completed report;

(ii) two families determine the same report exactly when their synchronized residual tends to zero;

(iii) the finite rational carrier embeds injectively by constant families;

(iv) rational convergence agrees with the embedding;

(v) nested rational intervals of vanishing width determine the same completed reports as Cauchy families;

(vi) addition, negation, subtraction, and multiplication are well defined on completion fibers;

(vii) and the completed-zero class is distinct from the requirement of finite exact zero.

The completion is therefore not an unexplained continuum inserted beneath finite arithmetic. It is a declared quotient of coherent finite rational reports, together with explicit preservation proofs for the operations admitted on the quotient.

57 Irrational reports and completed roots

The completion carrier contains reports not equal to any embedded rational. These reports must be distinguished from their construction rules, positional names, and algebraic predicates.

57.1 Rational and irrational completed reports

Definition 57.1 (Rational completed real)**.** *A completed real $\alpha \in \mathbb{R}_{\mathrm{C}}$ is* rational *when there exists $q \in \mathbb{Q}_{\mathrm{C}}$ such that*

$$\alpha = \iota(q).$$

Definition 57.2 (Irrational completed real)**.** *A completed real α is* irrational *when no finite rational report embeds to it:*

$$\forall q \in \mathbb{Q}_{\mathrm{C}}, \qquad \alpha \neq \iota(q).$$

Irrationality is therefore a statement about the completion fiber. It does not mean that the report lacks a finite rule, a finite symbol, a root predicate, or finite rational approximations.

Principle 57.3 (Finite description does not imply rational value)**.** *A finite construction rule may generate a Cauchy family whose completed report is irrational. Finiteness of the rule and rationality of the completed scalar are different properties.*

57.2 Root interval construction

Let $A \in \mathbb{Q}_{\mathrm{C}}$ be positive and let $k \geq 2$ be a finite natural exponent. Section 51 constructed exact rational roots and finite residual root shells. Completion now assigns a target when rational shells contract without exact rational closure.

Choose rational bounds

$$0 \leq L_0 < U_0$$

satisfying

$$L_0^k \le A \le U_0^k.$$

For example, one may take

$$L_0 = 0, \qquad U_0 = \max(1, A).$$

Given $[L_n, U_n]$, form the rational midpoint

$$M_n := \frac{L_n + U_n}{2}.$$

Define the next interval by the exact finite comparison

$$[L_{n+1}, U_{n+1}] = \begin{cases} [L_n, M_n], & A \le M_n^k, \\ [M_n, U_n], & M_n^k < A. \end{cases}$$

If $M_n^k = A$, either branch may be replaced by the constant exact interval $[M_n, M_n]$.

Theorem 57.4 (Root intervals are nested and contractive)**.** *The root interval rule satisfies*

$$L_n \le L_{n+1} \le U_{n+1} \le U_n,$$

$$L_n^k \le A \le U_n^k,$$

and

$$U_n - L_n = 2^{-n}(U_0 - L_0).$$

Hence its widths converge to zero.

Proof. Each step retains one branch of a balanced two-branch refinement of the preceding coordinate interval. The primitive report retains the two whole branch roles and the selected enclosure; the later coordinate compiler may describe their equal widths using half notation. Nesting and the width formula then follow by finite induction. The branch is chosen precisely to preserve the power enclosure. Since $2^{-n} \to 0$, the widths converge to zero. □

Definition 57.5 (Completed nonnegative root report)**.** *The completed kth-root report generated by the interval rule is*

$$\sqrt[k]{A}_{\mathrm{C}} := [(L_n)] = [(U_n)] \in \mathbb{R}_{\mathrm{C}}.$$

The notation $\sqrt[k]{A}_{\mathrm{C}}$ abbreviates the completed interval fiber. It is not a claim that a radical symbol existed before the interval construction.

57.3 Closure of the root relation

Theorem 57.6 (Completed root relation)**.** *For the completed root report $\alpha = \sqrt[k]{A}_{\mathrm{C}}$,*

$$\alpha^k = \iota(A).$$

Proof. Let M be a positive rational bound for all endpoints; one may take $M = U_0 + 1$. For nonnegative $L_n \le U_n$,

$$U_n^k - L_n^k = (U_n - L_n) \sum_{j=0}^{k-1} U_n^{k-1-j} L_n^j.$$

Every term in the sum is at most M^{k-1}, so

$$0 \le U_n^k - L_n^k \le kM^{k-1}(U_n - L_n) \longrightarrow_{\mathrm{C}} 0.$$

Because

$$L_n^k \le A \le U_n^k,$$

we have

$$|L_n^k - A| \le U_n^k - L_n^k \longrightarrow_{\mathrm{C}} 0.$$

Therefore the product class $[(L_n^k)]$ equals the constant class $\iota(A)$, which is exactly

$$\alpha^k = \iota(A).$$

□

At every finite stage, the lower endpoint may retain a nonzero root residual

$$A - L_n^k > 0.$$

The completed root relation is exact because that residual family belongs to the completed-zero class, not because a positive finite residual becomes a finite exact zero.

57.4 Exact rational roots versus irrational completed roots

Theorem 57.7 (Exact-root collapse)**.** *If A has an exact nonnegative rational kth root q, so that*

$$q^k = A,$$

then every contracting root interval specification retaining q represents

$$\sqrt[k]{A}_{\mathrm{C}} = \iota(q).$$

Proof. For every stage,

$$L_n \le q \le U_n.$$

Interior choice invariance from Section 56 identifies the endpoint class with the constant family q. □

Corollary 57.8 (Non-rational root reports)**.** *If no $q \in \mathbb{Q}_{\mathrm{C}}$ satisfies $q^k = A$, then the completed root report generated above is irrational.*

Proof. If the completed root were $\iota(q)$ for some rational q, the well-defined power operation would give

$$\iota(q^k) = \iota(A).$$

Injectivity of ι would imply $q^k = A$, contrary to hypothesis. □

57.5 The completed report represented by $\sqrt{2}$

Begin with

$$I_0 = [1, 2].$$

The first midpoint is $3/2$, whose square is $9/4 > 2$, so

$$I_1 = \left[1, \frac{3}{2}\right].$$

The next midpoint is $5/4$, whose square is $25/16 < 2$, so

$$I_2 = \left[\frac{5}{4}, \frac{3}{2}\right].$$

The next midpoint is $11/8$, whose square is $121/64 < 2$, so

$$I_3 = \left[\frac{11}{8}, \frac{3}{2}\right].$$

The rule continues by exact rational squaring and comparison. The interval widths halve at every stage.

Definition 57.9 (Constructive square-root-of-two report)**.** *Let*

$$\sqrt{2}_{\mathrm{C}} := [(L_n)] = [(U_n)]$$

for the preceding bisection interval family.

The completed root theorem gives

$$(\sqrt{2}_{\mathrm{C}})^2 = \iota(2).$$

Theorem 57.10 (Irrationality of the completed square root of two)**.** *The report $\sqrt{2}_{\mathrm{C}}$ is irrational.*

Proof. Assume

$$\sqrt{2}_{\mathrm{C}} = \iota(p/q)$$

for positive integers p, q with their common replication factors cancelled. Squaring and using injectivity of the rational embedding gives

$$p^2 = 2q^2.$$

Thus p^2 is even. A finite parity decomposition shows that an odd integer has an odd square, so p must be even. Write $p = 2r$. Substitution gives

$$4r^2 = 2q^2, \qquad q^2 = 2r^2.$$

Hence q is also even. Both numerator and denominator therefore share a factor of two, contradicting the declared reduced construction of p/q. No rational report embeds to $\sqrt{2}_{\mathrm{C}}$. □

The contradiction concerns finite numerator-denominator structure. It does not depend on reading an infinite decimal expansion.

57.6 Four distinct reports associated with an irrational root

For a completed root such as $\sqrt{2}_{\mathrm{C}}$, at least four report layers must be distinguished.

(i) **Construction rule.** The finite bisection rule that maps one rational interval to the next.

(ii) **Positional expansion.** A base-dependent sequence of digit-prefix reports whose cylinder intervals retain the completed value.

(iii) **Root-defining relation.** The completed algebraic predicate

$$x^2 = 2.$$

(iv) **Completed scalar report.** The Cauchy equivalence class

$$\sqrt{2}_{\mathrm{C}} \in \mathbb{R}_{\mathrm{C}}.$$

These layers are related but not identical. The bisection rule is not a decimal string. A decimal string is not the equation $x^2 = 2$. The equation may have multiple roots after signed orientation is admitted. The completed scalar suppresses many source rules capable of constructing the same root.

Principle 57.11 (Root-predicate preservation)**.** *The predicate $P(X) := [X]^k = \iota(A)$ descends to completion because multiplication is well defined on Cauchy fibers. A stagewise predicate such as "the nth approximation was produced by bisection" does not descend unless separately retained.*

57.7 Positional names for completed reals

Fix a finite base $b \geq 2$. A length-m radix prefix determines a rational cylinder interval of width b^{-m} consisting of all completed reports sharing that prefix under the declared digit convention.

Definition 57.12 (Completed positional name)**.** *A* base-b positional name *for a completed real α is a coherent family of finite digit prefixes*

$$D_1 \prec D_2 \prec D_3 \prec \cdots$$

whose associated rational cylinder intervals are nested, retain α, and have widths b^{-m}.

The positional name is therefore another nested interval specification. It reports the same completed scalar only after the interval-to-Cauchy equivalence of Section 56 is applied.

Theorem 57.13 (Eventually periodic positional names are rational)**.** *If a base-b positional name is eventually periodic, then its completed value is rational.*

Proof. Separate the finite nonperiodic prefix from one repeating block. The repeating tail is a finite geometric ratio whose common factor is b^{-r}, where r is the block length. Finite ratio arithmetic gives a rational closed form. Adding the finite prefix preserves rationality. □

Theorem 57.14 (Rational positional names eventually recur)**.** *Every rational completed real has an eventually terminating or eventually recurring base-b positional name.*

Proof. This is the finite remainder recurrence theorem of Section 49 applied to a finite rational representative. Completion identifies the generated prefix family with the embedded rational value. □

Corollary 57.15 (Irrational positional nonperiodicity)**.** *No base-b positional name of an irrational completed real is eventually periodic.*

This criterion concerns completed positional names, not the finiteness of the rule generating them. A finite algorithm may produce a nonperiodic digit family.

57.8 Dual positional reports and normalization

Some rational boundary values admit two positional names, such as a terminating name and a name with an indefinitely repeated maximal digit. Their finite prefixes differ, but their cylinder intervals determine the same completion class.

Example 57.16 (Two positional names, one completed scalar)**.** *Under the limit-value interpretation,*

$$1.000\ldots_{10} =_{\mathbb{R}_{\mathrm{C}}} 0.999\ldots_{10}.$$

The left rule stabilizes after the radix boundary. The right rule retains a positive residual at every finite prefix. They are distinct rule reports and equal completed scalars.

A canonical positional convention may prohibit an indefinitely repeated maximal digit in order to choose one representative. Such normalization selects a preferred name; it does not prove that the discarded name never existed.

57.9 Completed residual roots

Let L_n be the lower endpoint family for a completed kth root of A. Define the finite root residual

$$R_n := A - L_n^k \geq 0.$$

Theorem 57.17 (Root residual completion)**.** *The residual family satisfies*

$$[R] = 0_{\mathbb{R}_{\mathrm{C}}}.$$

If no finite endpoint is an exact root, then

$$R_n > 0$$

for every finite n even though its completed report is zero.

Proof. The completed root theorem proves $L_n^k \sim_{\mathrm{Cau}} A$, hence $R_n \to 0$. The second statement is the assumed failure of exact finite closure. □

This is the root analogue of the decimal-nine residual. Completion closes a vanishing residual family without retroactively replacing each finite residual by exact zero.

57.10 Completed-root preservation theorem

Theorem 57.18 (Completed-root report preservation)**.** *A completed root report retains:*

(i) its completed scalar class;

(ii) every algebraic predicate proved invariant under Cauchy equivalence, including the completed power relation;

(iii) and any separately attached source certificate.

It does not automatically retain:

(iv) the bisection, Newton, positional, geometric, or other approximation rule;

(v) the finite side of approach;

(vi) the exact residual at each stage;

(vii) the base of a positional name;

(viii) or the finite stage at which a requested precision was reached.

The distinction among rule, name, predicate, and scalar will be essential later when analytic functions receive the same scalar output from different source constructions.

58 Real order, bounds, and completeness

Completion creates scalar fibers. It does not by itself justify comparing those fibers. Signed order must be shown independent of representatives, and the different meanings of completeness must be stated rather than conflated.

58.1 Approximate non-strict order

Let $X = (X_n)$ and $Y = (Y_n)$ be rational Cauchy representatives.

Definition 58.1 (Completed non-strict order)**.** *Define*

$$[X] \leq_{\mathbb{R}_C} [Y]$$

when, for every positive rational resolution ρ, there exists a finite N_ρ such that

$$n \geq N_\rho \Longrightarrow X_n \leq Y_n + \rho.$$

This definition allows finite approximation noise. It does not require the chosen representatives to satisfy $X_n \leq Y_n$ at every stage.

Theorem 58.2 (Order is representative-independent)**.** *If $X \sim_{\mathrm{Cau}} X'$ and $Y \sim_{\mathrm{Cau}} Y'$, then*

$$[X] \leq [Y] \quad \Longleftrightarrow \quad [X'] \leq [Y'].$$

Proof. Assume $[X] \leq [Y]$ and request positive resolution ρ. On a common tail, require

$$|X_n - X'_n| < \rho/3, \qquad |Y_n - Y'_n| < \rho/3, \qquad X_n \leq Y_n + \rho/3.$$

Then

$$X'_n \leq X_n + \rho/3 \leq Y_n + 2\rho/3 \leq Y'_n + \rho.$$

The reverse implication is symmetric. □

58.2 Strict separation order

Definition 58.3 (Completed strict order)**.** *Define*

$$[X] <_{\mathbb{R}_C} [Y]$$

when there exist a positive rational separation δ and a finite index N such that

$$n \geq N \Longrightarrow X_n + \delta \leq Y_n.$$

Strict order therefore requires an eventually retained positive gap. A finite crossing or occasional equality does not destroy strict order when a later uniform separation exists.

Theorem 58.4 (Strict order is representative-independent)**.** *The relation $<_{\mathbb{R}_C}$ is constant on Cauchy equivalence classes.*

Proof. If $X_n + \delta \le Y_n$ eventually, choose a common tail on which

$$|X_n - X'_n| < \delta/3, \qquad |Y_n - Y'_n| < \delta/3.$$

Then

$$X'_n + \delta/3 \le X_n + 2\delta/3 \le Y_n - \delta/3 \le Y'_n.$$

Thus the replacement representatives retain the positive gap $\delta/3$. □

Definition 58.5 (Positive and negative completed reports)**.** *A completed real A is positive when*

$$0_{\mathbb{R}_C} < A,$$

and negative when

$$A < 0_{\mathbb{R}_C}.$$

This extends signed orientation from finite integer and rational reports to completed fibers.

58.3 Order laws

Theorem 58.6 (Completed order is a partial order)**.** *The relation $\le_{\mathbb{R}_C}$ is reflexive, transitive, and antisymmetric.*

Proof. Reflexivity is immediate. For transitivity, split an arbitrary requested resolution into two positive parts and combine the eventual inequalities. For antisymmetry, suppose $[X] \le [Y]$ and $[Y] \le [X]$. Given positive ρ, on a common tail one has

$$X_n \le Y_n + \rho \quad \text{and} \quad Y_n \le X_n + \rho.$$

Hence

$$|X_n - Y_n| \le \rho.$$

Using $\rho/2$ in the hypotheses gives strict containment below any requested ρ, so $X \sim_{\mathrm{Cau}} Y$ and $[X] = [Y]$. □

Theorem 58.7 (Compatibility with addition and negation)**.** *For completed reals A, B, C:*

$$A \le B \Longrightarrow A + C \le B + C,$$

$$A < B \Longrightarrow A + C < B + C,$$

and

$$A \le B \Longrightarrow -B \le -A.$$

Proof. All three statements follow from the corresponding finite rational inequalities applied on the eventual tails. Addition preserves the same residual gap, while negation reverses its orientation. □

Theorem 58.8 (Positive-product compatibility)**.** *If*

$$0 \le A, \qquad 0 \le B,$$

then

$$0 \le AB.$$

If in addition $0 < A$ *and* $0 < B$*, then* $0 < AB$*.*

Proof. For non-strict positivity, any finite negative excursions of chosen representatives are dominated by arbitrarily small positive resolutions; replacing them by equivalent nonnegative tail representatives, or applying the resolution definition directly, gives $0 \le AB$. For strict positivity, choose eventual positive gaps α and β so that representatives satisfy $A_n \ge \alpha$ and $B_n \ge \beta$. Then

$$A_n B_n \ge \alpha\beta > 0$$

on that tail. □

58.4 Linearity and its logical dependency

For arbitrary completed reports, a proof of

$$A \le B \quad \text{or} \quad B \le A$$

requires a decision about the completed sign of $B - A$. That decision is not contained in any fixed finite prefix.

Principle 58.9 (Located comparison principle)**.** *A completed pair* (A, B) *is* located *when, for every positive rational* ρ*, one can construct at least one of the reports*

$$A < B + \rho \qquad \text{or} \qquad B < A + \rho.$$

A family is located when such comparisons can be produced uniformly for its relevant members and rational bounds.

Principle 58.10 (Declared linear-order principle)**.** *If the logic admits the completed sign decision needed for every pair, then* $\le_{\mathbb{R}_C}$ *is a linear order. Under a stricter constructive reading, the preceding partial order and strict-separation relation remain valid, while trichotomy is asserted only for located or otherwise decidable pairs.*

This dependency must be visible. A geometric drawing of a line cannot supply a comparison decision that the report grammar has not constructed.

58.5 Completed intervals

Definition 58.11 (Closed completed interval)**.** *For* $A \le B$*, define*

$$[A, B]_{\mathbb{R}_C} := \{X \in \mathbb{R}_C : A \le X \le B\}.$$

Definition 58.12 (Open completed interval)**.** *For* $A < B$*, define*

$$(A, B)_{\mathbb{R}_C} := \{X \in \mathbb{R}_C : A < X < B\}.$$

The interval is a relational report between completed endpoints. It is not yet a physical segment or a claim that uncountably many objects have been physically placed between them.

Theorem 58.13 (Rational density)**.** *If $A < B$, then there exists $q \in \mathbb{Q}_{\mathrm{C}}$ such that*

$$A < \iota(q) < B.$$

Proof. Choose representatives and a positive rational gap δ with

$$A_n + \delta \leq B_n$$

eventually. Select one sufficiently late stage and choose the rational midpoint

$$q := \frac{A_n + B_n}{2}.$$

After making the representative errors smaller than $\delta/4$, the constant class $\iota(q)$ remains separated from both completed endpoints by a positive rational gap. □

Theorem 58.14 (Archimedean bound)**.** *For every $A \in \mathbb{R}_{\mathrm{C}}$, there exists a natural integer report N such that*

$$A < \iota(N).$$

Within a declared signed coordinate projection, a sufficiently large whole-count offset may likewise be displayed on the lower side of A; the opposing sign is frame syntax, not negative primitive multiplicity.

Proof. Choose a rational Cauchy representative of A. Cauchy boundedness supplies a rational magnitude bound M. Choose a natural integer $N > M + 1$. Then the constant report N is eventually separated above every representative term. Exchanging the two declared residual roles in the signed coordinate compiler gives the lower displayed bound; the primitive report remains the role-exchange construction. □

58.6 Upper and lower bounds

Definition 58.15 (Upper bound)**.** *Let S be a report family of completed reals. A completed real U is an* upper bound *of S when*

$$\forall X \in S, \qquad X \leq U.$$

Definition 58.16 (Lower bound)**.** *A completed real L is a* lower bound *of S when*

$$\forall X \in S, \qquad L \leq X.$$

Definition 58.17 (Least upper bound)**.** *A completed real S^* is the* least upper bound *or* supremum *of S when:*

(i) S^* is an upper bound of S;

(ii) every upper bound U of S satisfies $S^* \leq U$.

Write

$$S^* = \sup S.$$

Definition 58.18 (Greatest lower bound)**.** *A completed real S_* is the* greatest lower bound *or* infimum *of S when:*

(i) S_* *is a lower bound of* S*;*

(ii) every lower bound L *of* S *satisfies* $L \leq S_*$*.*

Write

$$S_* = \inf S.$$

Supremum and infimum are order reports. They do not require that the bound itself be a member of the family.

58.7 Nested-interval completeness

Theorem 58.19 (Nested completed interval theorem)**.** *Let*

$$I_n = [A_n, B_n]_{\mathbb{R}_{\mathrm{C}}}$$

be nonempty closed completed intervals satisfying

$$I_{n+1} \subseteq I_n$$

and

$$B_n - A_n \longrightarrow 0.$$

Then there exists exactly one completed real L *lying in every* I_n*.*

Proof. Choose rational approximations a_n, b_n to A_n, B_n at resolution smaller than $2^{-(n+3)}$, while preserving a nested rational enclosure by widening each endpoint by that resolution and intersecting with the preceding enclosure. This produces a nested rational interval specification of width tending to zero. Section 56 assigns it a completed real L. The approximation widths show $A_n \leq L \leq B_n$ for every fixed n. If L' also lies in every interval, then

$$|L - L'| \leq B_n - A_n$$

for every n, so the residual is below every positive resolution and $L = L'$. □

The theorem is the completed form of nested whole bounds. It supplies one scalar fiber, not a finite stage at which the interval literally has zero width.

58.8 Sequential Cauchy completeness

Definition 58.20 (Cauchy family of completed reals)**.** *A family* (A_n) *in* $\mathbb{R}_{\mathrm{C}}$ *is Cauchy when, for every positive rational resolution* ρ*, there exists* N_ρ *such that*

$$m, n \geq N_\rho \Longrightarrow |A_m - A_n| < \rho.$$

Here $|A_m - A_n| < \rho$ abbreviates strict separation of the residual magnitude below the embedded positive rational ρ.

Theorem 58.21 (Sequential completeness)**.** *Every Cauchy family of completed reals converges to a member of* $\mathbb{R}_{\mathrm{C}}$*.*

Proof. Choose an increasing index sequence N_k such that

$$m, n \geq N_k \Longrightarrow |A_m - A_n| < 2^{-(k+2)}.$$

For each k, choose a rational representative q_k of A_{N_k} within $2^{-(k+2)}$ of that completed report. Such a rational stage exists because every completed real is represented by a rational Cauchy family.

For $j \geq k$,

$$|q_j - q_k| \leq |q_j - A_{N_j}| + |A_{N_j} - A_{N_k}| + |A_{N_k} - q_k| < 2^{-k}.$$

Thus (q_k) is a rational Cauchy family and defines

$$L := [(q_k)] \in \mathbb{R}_{\mathrm{C}}.$$

The same estimates show $A_n \to L$. □

Sequential completeness is built into the Cauchy completion. It is not an additional picture of a filled coordinate line.

58.9 Located least-upper-bound construction

Full least-upper-bound claims require enough information to decide how a bounded family sits relative to rational cuts.

Definition 58.22 (Supremum-located family)**.** *An inhabited family $S \subseteq \mathbb{R}_{\mathrm{C}}$ is* supremum-located *when, for every rational pair $a < b$, one can construct at least one of the following reports:*

(i) b is an upper bound of S;

(ii) there exists $X \in S$ with $a < X$.

This form is deliberately weaker than deciding whether a single rational is exactly an upper bound. The positive gap $a < b$ leaves room for finite approximation.

Theorem 58.23 (Least upper bound for located bounded families)**.** *Every inhabited, supremum-located family of completed reals that has an upper bound has a least upper bound in $\mathbb{R}_{\mathrm{C}}$.*

Proof. By the Archimedean theorem, choose rational bounds $L_0 < U_0$ such that U_0 is an upper bound and some member of S lies above L_0.

At stage n, let

$$M_n := \frac{L_n + U_n}{2}.$$

Apply locatedness with a narrow rational buffer around M_n. If the upper test is returned, choose the upper branch endpoint so that the new U_{n+1} remains an upper bound. If the witness test is returned, choose L_{n+1} below the witnessed member, certifying that L_{n+1} is not an upper bound. The buffer can be chosen so that the interval width is at most a fixed fraction smaller than the preceding width.

This produces nested rational intervals

$$[L_n, U_n]$$

of vanishing width, with every U_n an upper bound and every L_n certified not to be an upper bound. Let

$$\alpha := \mathrm{Real}([L_n, U_n]).$$

To show that α is an upper bound, suppose some $X \in S$ satisfied $\alpha < X$. Rational separation would eventually place X above one of the upper endpoints U_n, contradicting that U_n is an upper bound.

To show leastness, let $Y < \alpha$. Rational separation and vanishing interval width eventually give $Y < L_n$. Since L_n is not an upper bound, some $X \in S$ satisfies $L_n < X$, and hence $Y < X$. Therefore Y is not an upper bound. Every upper bound must satisfy $\alpha \leq Y$. Thus $\alpha = \sup S$. □

Corollary 58.24 (Greatest lower bound for located bounded families)**.** *If S is inhabited, suitably infimum-located, and bounded below, then*

$$\inf S = -\sup(-S),$$

where

$$-S := \{-X : X \in S\}.$$

58.10 Completeness is a typed property

The word *complete* now has several nonidentical meanings.

Definition 58.25 (Sequential completeness)**.** *A carrier is sequentially complete when every Cauchy sequence in the carrier converges to a member of the carrier.*

Definition 58.26 (Nested-interval completeness)**.** *A carrier is nested-interval complete when every nested family of nonempty closed intervals whose widths tend to zero has exactly one common member.*

Definition 58.27 (Located order completeness)**.** *A carrier is located-order complete when every inhabited located family that is bounded above has a supremum, and dually for infima.*

Definition 58.28 (Full order completeness)**.** *A carrier is fully order complete when every inhabited family that is bounded above has a supremum, without an additional locatedness or decision certificate.*

Theorem 58.29 (Established completeness of the constructed carrier)**.** *The carrier $\mathbb{R}_{\mathrm{C}}$ constructed in Section 56 is:*

(i) sequentially complete;

(ii) nested-interval complete;

(iii) and order complete for inhabited located bounded families.

Principle 58.30 (Full order-completeness declaration)**.** *Full order completeness for arbitrary bounded report families requires an additional comprehension, choice, or sign-location principle strong enough to perform the needed cut decisions. If that principle is adopted, then every nonempty upper-bounded family has a supremum and every nonempty lower-bounded family has an infimum. If it is not adopted, no proof may silently use the full least-upper-bound property outside the located domain established above.*

This is not a defect hidden behind terminology. It is the explicit dependency separating Cauchy completion from unrestricted order completion.

58.11 Completed reciprocal for separated nonzero reports

The order now permits the missing reciprocal construction.

Definition 58.31 (Separated nonzero completed real)**.** *A completed real $A = [(A_n)]$ is* separated from zero *when there exists a positive rational δ and a finite N such that*

$$n \geq N \Longrightarrow |A_n| \geq \delta.$$

Every strictly positive or strictly negative completed real is separated from zero.

Theorem 58.32 (Completed reciprocal)**.** *If A is separated from zero, then the tail family*

$$A_n^{-1}$$

is Cauchy, its completion class is independent of representatives and finite initial choices, and

$$A \cdot A^{-1} = 1.$$

Proof. For tail indices,

$$\left|\frac{1}{A_m} - \frac{1}{A_n}\right| = \frac{|A_m - A_n|}{|A_m A_n|} \leq \frac{|A_m - A_n|}{\delta^2}.$$

Cauchy coherence of A therefore gives Cauchy coherence of the reciprocal family. Equivalent representatives remain close enough to preserve a smaller positive lower bound, and the same estimate proves representative independence. Finally, every tail stage satisfies $A_n A_n^{-1} = 1$, so the product class is the embedded unit. □

Principle 58.33 (Nonzero versus apart from zero)**.** *Under a fully linear classical order, $A \neq 0$ implies either $A > 0$ or $A < 0$, hence separation from zero. Under a stricter constructive reading, mere negated equality need not supply a positive separation witness. Reciprocal construction therefore requires apartness from zero, not merely the absence of a proof of equality.*

58.12 Part XIII completion summary

The construction developed in Sections 55–58 is

$$\begin{aligned}\text{finite signed ratios} &\longrightarrow \text{finite residual resolutions} \longrightarrow \text{Cauchy tail coherence}\\ &\longrightarrow \mathrm{Cau}(\mathbb{Q}_{\mathrm{C}})/\!\sim_{\mathrm{Cau}} \longrightarrow \mathbb{R}_{\mathrm{C}}\\ &\longrightarrow \text{completed arithmetic, roots, order, bounds, and typed completeness.}\end{aligned}$$

The essential conclusions are:

(1) a real report is a Cauchy-equivalence class of finite rational approximation families;

(2) finite rationals embed by constant families and retain their distinctions;

(3) nested rational intervals and Cauchy families yield the same completed carrier;

(4) completed equality forgets rule, rate, side, base, and finite-stage history unless separately retained;

(5) completed zero need not contain a finite exact-zero stage;

(6) irrational reports are completed values outside the rational embedding, not objects without finite descriptions;

(7) a completed root closes a vanishing residual family without turning any positive finite residual into exact zero;

(8) positional expansion, root predicate, construction rule, and completed scalar are different report layers;

(9) signed order is defined by eventual finite-resolution comparison and must be proved independent of representatives;

(10) the Cauchy completion is sequentially and nested-interval complete;

(11) least upper bounds are constructed for located bounded families;

(12) unrestricted order completeness is an additional declared logical commitment rather than a hidden consequence of the word "real";

(13) and completed reciprocals require a positive separation from zero.

The next part reconstructs the relational content that conventional geometry normally compresses. It begins from primitive constraint reports, their equivalence classes, jointly sufficient closure, and constraint rank; only afterward may a coordinate compiler introduce line, distance, angle, area, volume, or Euclidean dimension notation.

Part XIV
Constraint Rank and Conditional Geometry

59 Primitive constraints and equivalence ranking

The preceding parts constructed whole report occurrences, comparison, matching, quotient–remainder, ratio, finite continuation, and projection tests. None requires a Euclidean background. This part therefore begins with constraint reports and treats conventional geometry as a later compiler.

59.1 Property-relative difference

Definition 59.1 (Constraint report)**.** *Let A and B be comparable under property grammar P. If the comparison realizes a distinction, the retained result*

$$D_P(A, B)$$

is a constraint report*. It records that the two source reports are not equivalent under the declared comparison, together with the criterion, source roles, outcome, and lineage.*

A constraint report is whole. It is not initially a signed number, length, interval, vector, or coordinate displacement.

Definition 59.2 (Constraint zero)**.** *If the same comparison closes with no admitted distinction, its typed zero is the equivalence certificate*

$$0_{P;A,B} := Z_P(A, B).$$

The zero is a report of comparison closure, not an empty space between the sources.

59.2 Equivalence ranking of constraints

Definition 59.3 (Constraint-equivalence relation)**.** *Let D_1 and D_2 be two retained constraint reports admitted to a common comparison grammar $\mathcal{D}$. Write*

$$D_1 \equiv_{\mathcal{D}} D_2$$

when comparison of the two constraints closes with no admitted distinction under $\mathcal{D}$.

Definition 59.4 (Finite constraint class)**.** *A finite constraint class is a presently grouped family*

$$\mathfrak{C} = [D]_{\mathcal{D}} = (D_1, \ldots, D_n; \mathcal{D}, \Gamma)$$

of source-distinct whole constraints equivalent under $\mathcal{D}$.

Principle 59.5 (Equivalence before magnitude)**.** *The statement that two differences are equivalent is available before any scalar magnitude is assigned to either one. A coordinate length is therefore a possible later encoding of a constraint-equivalence class, not the primitive source of the equivalence.*

59.3 Count and ratio from constraint recurrence

If n source-distinct constraints belong to one equivalence class, their multiplicity is the whole count n. If two classes $\mathfrak{C}_A$ and $\mathfrak{C}_B$ are comparable by repeated matching, construct whole replication carriers M, N and retain the exact extraction

$$M\mathfrak{C}_A = Q\,(N\mathfrak{C}_B) + R.$$

The quotient Q and residual R are the primitive comparison data. Any later scalar ratio is a projection of this retained ledger.

Principle 59.6 (Measurement by ratio, not by hidden continuum)**.** *PCL does not require a continuum of positions between two reports in order to compare their constraints. Repetition, matching, quotient, and residual are sufficient for finite ratio measurement whenever the active grammar supplies comparable constraint classes.*

59.4 Occurrence exclusivity and refinement

Theorem 59.7 (Occurrence exclusivity)**.** *If two retained reports differ under any property declared relevant to complete identity, then they require different complete occurrence identities.*

Proof. If the same complete index denoted both while the distinction remained relevant, the index would erase the very distinction by which the reports were two. A projection may later identify them only after proving that the forgotten property is irrelevant to the target predicate. □

Definition 59.8 (Constructive refinement)**.** *A proposed refinement $X \rightsquigarrow X'$ is constructively new only when the active grammar retains a new distinction between the two stages or adds a new distinguishable internal occurrence whose role was absent before.*

Theorem 59.9 (No refinement from instruction alone)**.** *A continuation rule that says "refine again" does not prove that another distinguishable refinement exists.*

Proof. If the next proposed stage is equivalent to the previous stage under every declared criterion, no new report occurrence or constraint has been constructed. If it is distinguishable, the distinguishing report is the certificate that a new stage exists. Thus realizable distinction, not an unbounded instruction, licenses refinement. □

60 Constraint rank and whole closure

60.1 Joint sufficiency

Definition 60.1 (Constraint packet)**.** *A finite constraint packet for a report state X is*

$$\mathcal{C}_X = (C_1, \dots, C_r; \Gamma),$$

where each C_j is a retained comparison or dependency role relevant to the identity of X.

Definition 60.2 (Jointly sufficient packet)**.** *The packet $\mathcal{C}_X$ is jointly sufficient under grammar $\mathcal{M}$ when agreement on every C_j is sufficient to certify the declared equivalence of two candidate states:*

$$(A \equiv_{C_1} B, \dots, A \equiv_{C_r} B) \Longrightarrow A \equiv_{\mathcal{M}} B.$$

Definition 60.3 (Irredundant constraint)**.** *A constraint C_j in a jointly sufficient packet is irredundant when deleting it destroys sufficiency: there exist candidate reports A, B that agree under every retained C_i with $i \neq j$ but remain distinguishable under the complete grammar.*

60.2 Primitive constraint rank

Definition 60.4 (Constraint rank)**.** *The primitive constraint rank of X under grammar $\mathcal{M}$ is the least whole count r for which a jointly sufficient packet of r individually irredundant constraint roles has been constructed. Write*

$$\operatorname{rank}_{\mathcal{M}}(X) = r.$$

This is a whole-count report of necessary constraint roles, not a Euclidean dimension coordinate.

Theorem 60.5 (Minimum semantic closure rank is three)**.** *For the complete primitive measurement report*

$$\mathfrak{T} = (P, M, R; \Gamma),$$

the three semantic roles property, realized measurement, and retained report form an irredundant rank-three packet with respect to the meaning of a retained measured distinction. No packet with fewer than these three dependency roles retains that same complete meaning.

Proof. This is the triadic semantic irreducibility theorem of Section 1 restated in rank language. Removing any one role changes the complete semantic report. No claim about physical spatial dimension follows from this rank count. □

60.3 No rank creation by stacking or sweeping

Theorem 60.6 (Repetition does not create independent constraint rank)**.** *Repeating, stacking, sequencing, or sweeping copies of a report does not by itself increase the independent constraint rank of each copy.*

Proof. Repetition introduces source and repetition indices. Those are additional indexing roles in the composite ledger, but they do not manufacture a new property criterion internal to each repeated report. A higher constraint rank exists only when an additional irredundant constraint role is actually constructed and shown necessary for closure. □

Principle 60.7 (Space is not derived by accumulating lower-rank extent)**.** *PCL does not claim that one-dimensional repetition generates a two-dimensional object or that two-dimensional sweeping generates a three-dimensional object. If a placement grammar requires several independent irredundant constraints, those constraints must already be present in the placement report or be separately constructed. A coordinate embedding may display them; it cannot retroactively create their independence.*

60.4 Whole closure rather than continuum filling

Definition 60.8 (Whole constraint closure)**.** *A whole closure is a report*

$$\mathcal{W}_X = \mathrm{Close}(C_1, \ldots, C_r; \Gamma)$$

certifying that the jointly sufficient constraint packet for X has been satisfied under the declared grammar.

The word *whole* means that the declared grouping is complete relative to its own constraint grammar. It does not mean that an uncountable continuum of unreported positions has been inserted into the object.

Theorem 60.9 (Boundary data do not manufacture filling)**.** *An abstract finite cycle or boundary ledger does not by itself imply an additional filled continuum object. Conversely, when a whole closure is primitively given by its jointly sufficient constraint packet, no separate continuum-filling operation is required to make that closure a whole report.*

Proof. Boundary projection can forget whether an additional closure role was declared, so filling is not constant on the boundary fiber. A whole constraint closure, however, is identified by the complete packet that constructed it. Its wholeness is therefore a report property of that closure packet, not the result of adding infinitely many hidden points after the fact. □

60.5 Data-index rank and constraint rank

Principle 60.10 (Product indices are not geometric dimensions)**.** *An ordered product*

$$I_1 \times \cdots \times I_d$$

has d displayed index roles. This is data-index arity. It equals primitive constraint rank only after a theorem proves that those index roles correspond to jointly sufficient and individually irredundant constraints for the property under study.

Principle 60.11 (Constraint rank is typed)**.** *The same report may have different ranks under different grammars because different properties may require different minimal sufficient packets. Rank must therefore retain the grammar and target identity predicate; a bare dimension number is an information-losing projection.*

61 Conditional coordinate geometry

61.1 Geometry as a provenance-preserving compiler

Definition 61.1 (Coordinate geometry compiler)**.** *A coordinate geometry compiler is a declared map*

$$\kappa_{\mathcal{F}} : \mathcal{X} \longrightarrow \mathcal{V}_{\mathcal{F}}$$

from previously constructed PCL reports to a conventional coordinate grammar $\mathcal{V}_{\mathcal{F}}$. The frame record $\mathcal{F}$ retains every additional assumption used by the compiler, including any origin, axis roles, orientation, unit relations, order, metric, orthogonality, continuity, or completion convention.

Principle 61.2 (Coordinates do not prove their own prerequisites)**.** *A successful coordinate representation does not prove that its origin, axes, signed directions, exact orthogonality, continuum, or metric existed primitively. Those are either separately constructed constraint relations or declared structure internal to the coordinate compiler.*

61.2 Line, distance, area, and volume are compiled predicates

Definition 61.3 (Compiled line relation)**.** *Inside a declared geometry compiler, a* line *is whatever coordinate or incidence predicate the compiler declares to represent one-dimensional alignment. Its source ledger must retain the primitive constraint reports whose coordinate image is being called a line.*

Definition 61.4 (Compiled distance)**.** *A distance formula such as*

$$d_{\mathcal{F}}(p, q) = \sqrt{(\Delta x_1)^2 + \cdots + (\Delta x_d)^2}$$

is a coordinate projection available only after the frame has supplied signed coordinate differences, multiplication, addition, root evaluation, and the relevant metric grammar. It is not the primitive definition of difference.

Principle 61.5 (No exact orthogonality from finite cycling)**.** *A finite four-state cycle proves exact cyclic return after four transitions if that return is constructed. It does not by itself prove a Euclidean right angle, perpendicular axes, or a quarter turn. Orthogonality is a separate coordinate-frame predicate unless independently constructed by another constraint grammar.*

Principle 61.6 (Area and volume are coordinate or closure measures)**.** *A scalar area or volume is never inferred merely from two or three index roles. Such a scalar is admitted only after a declared measurement compiler specifies which whole closures are being compared, which unit closure is used, how overlap is handled, and how the scalar projection preserves the relevant constraint ledger.*

61.3 Finite cyclic orientation before angle

Definition 61.7 (Finite orientation cycle)**.** *A finite orientation cycle is a whole transition ledger*

$$\mathcal{O}_n = (q_0, \ldots, q_{n-1}; h_0, \ldots, h_{n-1}; \Gamma)$$

with a declared successor transition from each q_j to the next and an exact retained return from q_{n-1} to q_0.

For a whole transition count K, quotient–remainder gives

$$K = Wn + J, \qquad 0 \le J < n.$$

The quotient W counts complete returns and the remainder J identifies the terminal state in the declared cycle. No radian, circle, right angle, or completed real angle is needed for this finite orientation report.

Principle 61.8 (Angle is a later projection)**.** *A conventional angle value is a coordinate projection of finite orientation and winding data through an independently declared Euclidean or other angular grammar. Equal projected terminal angles need not identify equal winding quotients, hinge order, source cycle, or transition history.*

61.4 Conditional Euclidean circle and pi compiler

If a later Euclidean compiler independently declares a metric plane, radial normalization, exact perpendicular/tangent predicates, root evaluators, and a real-coordinate refinement grammar, then the usual nested regular-polygon construction may be run entirely inside that compiler. Its finite inner and outer perimeter reports can be refined at each requested resolution, and their common coordinate ratio may be denoted $\pi_{\mathcal{F}}$.

Principle 61.9 (Pi is not a primitive circle constructor)**.** *The symbol $\pi_{\mathcal{F}}$ denotes a coordinate invariant internal to the declared Euclidean compiler. Its construction does not retroactively prove primitive circles, exact orthogonality, a completed real continuum, or infinitely divisible lengths. Any use of $\pi_{\mathcal{F}}$ downstream retains the frame and refinement provenance that licensed the symbol.*

61.5 Conditional geometric preservation

Theorem 61.10 (Coordinate-fiber obligation)**.** *Let P be any geometric predicate asserted from a coordinate report $\kappa_{\mathcal{F}}(X)$. The predicate descends to the primitive source report independently of coordinate representation only if P is constant on every relevant fiber of $\kappa_{\mathcal{F}}$ and on every admissible frame-change fiber claimed to preserve it.*

Proof. If two primitive reports have the same coordinate projection but different values of P, then the coordinate report is insufficient to determine P. This is the projection-fiber factorization theorem of Part XII applied to the geometry compiler. □

61.6 Part XIV summary

The primitive dependency carried forward is now

$$\begin{aligned}\text{property sensitivity} &\longrightarrow \text{realized distinction} \longrightarrow \text{constraint report} \longrightarrow \text{constraint equivalence}\\ &\longrightarrow \text{whole count and ratio} \longrightarrow \text{jointly sufficient closure} \longrightarrow \text{constraint rank}.\end{aligned}$$

Conventional geometry enters only afterward through a provenance-preserving compiler:

$$\begin{aligned}\text{constraint closure} &\longrightarrow \text{declared coordinate frame}\\ &\longrightarrow \text{line, distance, angle, area, and volume projections}.\end{aligned}$$

No geometric projection is permitted to retroactively justify the primitive constraint relations on which its own definition depends.

Part XV
Algebraic Carriers and Conditional Coordinate Calculi

Principle 61.11 (Algebraic carrier status)**.** *An algebraic carrier may be a finite native PCL report family or a conventional projection grammar. Statements that $\mathbb{Z}_{\mathrm{C}}$, $\mathbb{R}_{\mathrm{C}}$, or later $\mathbb{C}_{\mathrm{C}}$ satisfy familiar algebraic laws mean that the declared coordinate compiler preserves those laws on its admissible inputs. They do not assert primitive existence of completed signed, real, or complex domains.*

62 Identity, inverse, and algebraic closure

The arithmetic, mapping, equality, finite-resolution, constraint-rank, and conditional coordinate constructions already established can now be compiled into algebraic carriers. The order of dependence matters. A group, ring, or field is not introduced as a primitive universe containing unexplained operations. It is a report system whose carrier, operations, identities, inverses, closure rules, and equality relation have each already been constructed or explicitly declared.

62.1 Algebraic carriers are typed report systems

Definition 62.1 (Algebraic carrier specification)**.** *An* algebraic carrier specification *is a report*

$$\mathcal{A} = (A, \mathcal{O}, \mathcal{E}, \mathcal{D}, \Gamma),$$

where:

(i) A *is a declared carrier of report objects;*

(ii) $\mathcal{O}$ *is a family of total or partial operations on those reports;*

(iii) $\mathcal{E}$ *is the equality or congruence grammar under which outputs are compared;*

(iv) $\mathcal{D}$ *records the admissible domains of partial operations;*

(v) Γ *records any source, orientation, quotient, boundary, completion, or construction data retained by the system.*

The notation

$$(A, +, \cdot)$$

is therefore a compressed report. It is meaningful only after the meanings of A, $+$, $\cdot$, and equality on A have been fixed.

Principle 62.2 (No primitive algebraic universe)**.** *An algebraic structure may summarize previously constructed relations, but it may not be used retroactively to define the objects or operations from which it was compiled.*

Remark 62.3 (Relation to the earlier PCA manuscript)**.** *The earlier* Primitive Constructive Algebra *manuscript introduced construction ledgers, boundary lanes, graded equality, and the projection-collapse principle. In the present manuscript those ideas are no longer primitive assumptions. They are recompiled from the prior PCL constructions of reports, occurrence identity, signed orientation, quotient–remainder states, mappings, equality fibers, completion, boundary, and measure. A PCA-style ledger is consequently one admissible choice of* Γ*, not the foundation of the carrier itself.*

62.2 Closure as an operation obligation

Definition 62.4 (Operational closure)**.** *Let $\star$ be a k-ary operation with declared admissible domain $D_\star \subseteq A^k$. The carrier A is* closed under $\star$ *when*

$$(a_1, \ldots, a_k) \in D_\star \quad \Longrightarrow \quad \star(a_1, \ldots, a_k) \in A.$$

If $D_\star = A^k$, the operation is total on A. Otherwise it is partial.

Closure is not a claim that every attempted construction succeeds. It states that every *admissible* operation returns a report in the declared carrier.

Definition 62.5 (Typed failure and typed residual)**.** *For a partial constructive operation, an inadmissible input may return a typed failure marker $\bot_\star$, while an incomplete but meaningful construction may return a typed residual report $R_\star$. Neither is silently identified with the carrier identity unless a theorem proves that identification.*

Principle 62.6 (Closure is relative)**.** *The same operation may be closed on one carrier and not on another. For example, natural-number subtraction is not closed on $\mathbb{N}_{\mathrm{C}}$, signed subtraction is closed on $\mathbb{Z}_{\mathrm{C}}$, and reciprocal formation is closed only on the declared nonzero or apart subcarrier.*

62.3 Identity reports

Definition 62.7 (Left and right identity)**.** *Let $\star$ be a binary operation on A. A report $e_L \in A$ is a* left identity *when*

$$e_L \star a =_{\mathcal{E}} a$$

for every admissible $a \in A$. A report $e_R \in A$ is a right identity *when*

$$a \star e_R =_{\mathcal{E}} a$$

for every admissible $a \in A$. A two-sided identity satisfies both conditions.

The equality symbol is part of the definition. An object may be an identity under scalar equality while retaining a nontrivial source ledger under constructive equality.

Theorem 62.8 (Uniqueness of a two-sided identity)**.** *If e and e' are two-sided identities for the same operation and equality grammar, then*

$$e =_{\mathcal{E}} e'.$$

Proof. Using e as a left identity and e' as a right identity,

$$e \star e' =_{\mathcal{E}} e' \qquad \text{and} \qquad e \star e' =_{\mathcal{E}} e.$$

Transitivity gives $e =_{\mathcal{E}} e'$. □

Definition 62.9 (Additive and multiplicative identities)**.** *The additive identity is the typed zero report 0_A satisfying*

$$0_A + a =_A a = a + 0_A.$$

The multiplicative identity is the typed unit report 1_A satisfying

$$1_A a =_A a = a 1_A.$$

The zero and unit are not imported from an external number system. In each carrier they are the identities compiled from the earlier no-contribution closure and one-copy constructions.

62.4 Inverse reports

Definition 62.10 (Left and right inverse)**.** *Let e be the identity for $\star$. A report b is a left inverse of a when*

$$b \star a =_{\mathcal{E}} e,$$

and a right inverse when

$$a \star b =_{\mathcal{E}} e.$$

A two-sided inverse satisfies both.

Theorem 62.11 (Uniqueness of a two-sided inverse)**.** *In an associative carrier with identity e, a two-sided inverse of a is unique under the declared equality.*

Proof. If b and c are both two-sided inverses of a, then

$$b =_{\mathcal{E}} b \star e =_{\mathcal{E}} b \star (a \star c) =_{\mathcal{E}} (b \star a) \star c =_{\mathcal{E}} e \star c =_{\mathcal{E}} c.$$

□

Definition 62.12 (Additive inverse in a signed projection grammar)**.** *When a declared signed coordinate compiler is in force, the glyph $-a$ denotes the coordinate shadow obtained by exchanging the two unmatched residual roles of the underlying comparison report. The coordinate identity*

$$a + (-a) =_A 0_A$$

is valid inside that compiler when the rematching ledger closes. It does not construct a primitive negative object.

Definition 62.13 (Multiplicative inverse)**.** *A multiplicative inverse a^{-1} is a report satisfying*

$$aa^{-1} =_A 1_A = a^{-1}a.$$

Its construction is admissible only on the subcarrier for which inverse existence has been proved.

Principle 62.14 (Inverse existence is not symbolic permission)**.** *Writing a^{-1} does not construct an inverse. It references a previously established inverse report and its admissibility conditions.*

62.5 Semigroups, monoids, and groups

Definition 62.15 (Semigroup)**.** *A* constructive semigroup *is a carrier with one closed associative binary operation.*

Definition 62.16 (Monoid)**.** *A* constructive monoid *is a constructive semigroup with a two-sided identity.*

Definition 62.17 (Group)**.** *A* constructive group *is a constructive monoid in which every carrier report has a two-sided inverse.*

Definition 62.18 (Abelian group)**.** *A constructive group is* abelian *when*

$$a \star b =_A b \star a$$

for every $a, b \in A$.

The equality here may be scalar or quotient equality even when the transposed source ledgers differ. Commutativity is therefore a theorem about the selected carrier report, not a claim that the two evaluation histories are identical.

Theorem 62.19 (Compiled additive coordinate grammars)**.** *Under their declared coordinate equalities, the conventional projections customarily written as integer, rational, and real addition satisfy the corresponding abelian-group laws whenever the source-preserving join, residual-role exchange, ratio compiler, and refinement compiler are all admissible. The natural-count grammar remains a commutative monoid without requiring signed projection.*

Proof. The scalar group identities are shadows of source-preserving joins followed by declared role exchange and rematching. Ratio and real-coordinate compilers preserve those identities only after their fiber-preservation obligations are proved. The theorem is therefore internal to the compiled coordinate grammar and does not erase the underlying matching or refinement reports. □

62.6 Multiplicative carriers and units

Definition 62.20 (Unit of a multiplicative carrier)**.** *A report $u \in A$ is a* unit *when it has a multiplicative inverse in A. The unit carrier is*

$$A^{\times} := \{u \in A : u^{-1} \text{ exists in } A\}.$$

Theorem 62.21 (Units form a group)**.** *If multiplication on A is associative with identity 1_A, then $(A^{\times}, \cdot, 1_A)$ is a group.*

Proof. The identity is a unit. If u and v are units, then

$$(uv)^{-1} = v^{-1}u^{-1},$$

so the product is a unit. Associativity is inherited from A, and inverse uniqueness was proved above. □

For commutative multiplication, the unit group is abelian.

62.7 Semirings, rings, and integral domains

Definition 62.22 (Constructive semiring)**.** *A* constructive semiring *is a carrier A with operations $+$ and $\cdot$ such that:*

(i) $(A, +, 0_A)$ *is a commutative monoid;*

(ii) $(A, \cdot, 1_A)$ *is a monoid;*

(iii) multiplication distributes over addition;

(iv) $0_A a = a 0_A = 0_A$.

Definition 62.23 (Constructive ring)**.** *A* constructive ring *is a constructive semiring whose additive monoid is an abelian group.*

Definition 62.24 (Commutative ring)**.** *A constructive ring is commutative when multiplication is commutative under the carrier equality.*

Definition 62.25 (Zero divisor)**.** *A nonzero report a is a* zero divisor *when there exists a nonzero report b such that*

$$ab = 0_A.$$

Definition 62.26 (Constructive integral domain)**.** *A* constructive integral domain *is a commutative ring with $1_A \neq 0_A$ and no nonzero zero divisors.*

Theorem 62.27 (Conditional arithmetic structures)**.** *The conventional signed-integer, rational-coordinate, and real-coordinate compilers inherit the familiar ring or field identities exactly to the extent that their declared projections preserve the underlying finite operations and equality fibers. These are algebraic facts about the compilers, not proofs that completed domains of negative, real, or complex objects exist primitively.*

The theorem compiles earlier operation laws while retaining the matching, ratio, and refinement ledgers that made those coordinate identities meaningful.

62.8 Fields and apartness

Mere negation of equality is not always enough to execute a reciprocal construction in a constructive carrier. The inverse algorithm requires positive separation from zero.

Definition 62.28 (Apartness from zero)**.** *For an ordered completed carrier, write*

$$a \# 0_A$$

when there exists a positive rational resolution ρ such that

$$|a| \geq \rho.$$

This is a positive witness that a is separated from zero.

Definition 62.29 (Constructive field)**.** *A* constructive field *is a commutative ring with $1 \neq 0$ together with an apartness relation $\#$ such that every $a\#0$ has a multiplicative inverse.*

Theorem 62.30 (Rational field)**.** *Every nonzero finite constructive rational is apart from zero and has a finite reciprocal. Hence $\mathbb{Q}_{\mathrm{C}}$ is a constructive field.*

Proof. A nonzero rational has a reduced finite numerator with positive whole magnitude and a positive denominator. Its magnitude is bounded below by the reciprocal of that denominator. The reciprocal construction of Sections 36–38 therefore applies. □

Theorem 62.31 (Completed-real field under apartness)**.** *Every completed real report $x\#0$ has a completed reciprocal. Thus $\mathbb{R}_{\mathrm{C}}$ is a constructive field relative to the apartness relation.*

Proof. Section 58 constructed the reciprocal of every completed report positively separated from zero. Representative independence follows from the Cauchy-fiber preservation theorem. □

Remark 62.32 (Inequality and invertibility)**.** *In a classical logic one often replaces $x\#0$ by $x \neq 0$. PCL retains the distinction. A negative statement that equality has not been proved is not automatically the positive finite information needed to build an inverse.*

62.9 Congruences and quotient carriers

Definition 62.33 (Algebraic congruence)**.** *An equivalence relation $\sim$ on an algebraic carrier is a* congruence *for a k-ary operation $\star$ when*

$$a_j \sim b_j \text{ for every } j \quad \Longrightarrow \quad \star(a_1, \dots, a_k) \sim \star(b_1, \dots, b_k).$$

Theorem 62.34 (Operation descent to a quotient)**.** *An operation descends to a well-defined operation on the quotient carrier $A/\sim$ exactly when $\sim$ is a congruence for that operation.*

Proof. A quotient operation must be independent of the chosen representative. This is exactly the stated congruence condition. It is the operation form of the projection-fiber factorization theorem from Section 54. □

Example 62.35 (QR remainder projection as a quotient report)**.** *Let $n > 0$ be a finite whole carrier. For presented integers a and b, retain the complete quotient–remainder ledgers*

$$a = q_a n + r_a, \qquad b = q_b n + r_b, \qquad 0 \le r_a, r_b < n.$$

Declare $a \sim_n b$ exactly when $r_a = r_b$. Addition and multiplication respect this remainder-equivalence because their output quotient and remainder can be reconstructed by ordinary quotient–remainder extraction from the exact integer producers. A remainder-only quotient grammar may therefore inherit those projected operations, while the full PCL ledger retains q_a, q_b whenever future reasoning depends on the quotient lanes.

Principle 62.36 (Quotient carrier discipline)**.** *A quotient carrier is valid for exactly those operations and predicates proved constant on its fibers. It does not make the erased source distinctions unreal.*

62.10 Homomorphisms and isomorphisms

Definition 62.37 (Homomorphism)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be carriers with corresponding operations. A mapping*

$$F : A \to B$$

is a homomorphism *when it preserves every declared operation and identity, for example*

$$F(a + a') = F(a) + F(a'), \qquad F(aa') = F(a)F(a'), \qquad F(0_A) = 0_B, \qquad F(1_A) = 1_B.$$

Definition 62.38 (Isomorphism)**.** *A bijective homomorphism whose inverse is also a homomorphism is an* isomorphism*. It certifies structural equivalence under the declared algebraic grammar, not necessarily source-ledger identity.*

Definition 62.39 (Kernel and image)**.** *For an additive homomorphism $F : A \to B$, define*

$$\ker F := \{a \in A : F(a) = 0_B\}, \qquad \operatorname{im} F := \{F(a) : a \in A\}.$$

The kernel is a zero-output fiber in the input carrier. It is not the same object as a complete evaluation ledger zero, which will be constructed in Section 64.

62.11 Operational closure versus algebraic closure

The phrase *algebraic closure* is used in two different senses and must be separated.

Definition 62.40 (Operationally closed carrier)**.** *A carrier is operationally closed under an operation when every admissible use of that operation remains in the carrier.*

Definition 62.41 (Polynomial report)**.** *A finite polynomial report over a commutative ring A is a finite coefficient ledger*

$$p(T) = a_0 + a_1T + \cdots + a_nT^n$$

with each $a_j \in A$ and finite operational depth.

Definition 62.42 (Algebraic element)**.** *Let B extend A. A report $\alpha \in B$ is* algebraic over A *when some nonzero finite polynomial report $p \in A[T]$ satisfies*

$$p(\alpha) = 0_B.$$

Definition 62.43 (Algebraically closed field)**.** *A field carrier F is* algebraically closed *when every nonconstant finite polynomial report in $F[T]$ has at least one root in F.*

This is a later completeness property of a field with respect to polynomial root obligations. It is not the same as closure under addition or multiplication.

Definition 62.44 (Adjoining a root)**.** *If $p(T) \in A[T]$ has no root in A, an extension may be constructed by adjoining a formal report α and imposing the smallest operation congruence containing*

$$p(\alpha) \sim 0.$$

The resulting quotient is admissible only after the congruence and noncollapse conditions have been proved.

The next section applies this compiler to the polynomial

$$T^2 + 1,$$

not by importing a mysterious scalar square root of -1, but by constructing an oriented two-lane report whose transverse unit squares to orientation reversal.

62.12 Section 62 preservation theorem

Theorem 62.45 (Algebraic-carrier preservation)**.** *A compressed algebraic carrier report is sufficient for a property P only when:*

(i) every operation used by P is closed on its declared admissible domain;

(ii) the carrier equality is a congruence for those operations;

(iii) identities and inverses are interpreted relative to that equality;

(iv) quotient projections preserve P on every fiber;

(v) and every partial operation retains its admissibility witness.

Without these conditions, equal algebraic outputs need not certify equal constructions.

63 Conditional complex-coordinate compiler

PCL does not claim here that a complex plane, exact transverse axis, exact orthogonality, or completed set of complex numbers has been constructively demonstrated to exist. This section records only how conventional complex algebra may be entered as a provenance-preserving coordinate compiler once a user declares the required signed two-role frame.

63.1 Primitive source retained beneath the compiler

The native PCL source is a finite relational report: finite refinement evaluators for the two displayed magnitude roles, together with a finite transition/orientation ledger and all matching residuals. No pair of perpendicular axes is inferred from those ingredients.

Principle 63.1 (No quarter-turn from cycle count alone)**.** *A four-state cyclic ledger proves return after four declared transitions. It does not prove that one transition is a Euclidean right angle or that the two displayed roles are orthogonal. Any interpretation as a quarter-turn belongs to an additional Euclidean coordinate frame.*

63.2 Declared complex pair grammar

Suppose a conventional coordinate compiler $\mathcal{F}_{\mathbb{C}}$ is declared with two formal coordinate roles $\mathbf{e}$ and $\mathbf{j}$, signed-coordinate arithmetic on their coefficients, and the algebraic reduction rule

$$\mathbf{j}^2 = -\mathbf{e}.$$

This equation is part of the declared compiler. It is not used as evidence that a primitive object $\mathbf{i}$, an orthogonal plane, or negative existence has been independently constructed.

Definition 63.2 (Complex-coordinate report)**.** *A complex-coordinate report is*

$$\mathcal{C} = (\mathcal{S}; a, b; \mathcal{F}_{\mathbb{C}}),$$

where $\mathcal{S}$ is the retained primitive source/refinement/transition ledger and

$$(a, b)_{\mathcal{F}_{\mathbb{C}}}$$

is its conventional two-coordinate shadow. The familiar notation

$$a + \mathbf{i}b$$

is a further abbreviation of that shadow.

Definition 63.3 (Conventional complex grammar)**.** *The symbol $\mathbb{C}_{\mathrm{C}}$ denotes this declared coordinate compiler and the operations proved to be preserved by it. It does not denote a primitively completed set of all complex numbers.*

63.3 Internal pair algebra

Inside the declared compiler define

$$(a, b) + (c, d) := (a + c, b + d)$$

and

$$(a, b)(c, d) := (ac - bd, ad + bc).$$

Here every displayed minus sign is a signed-coordinate operation whose complete PCL state still carries the underlying residual-role exchange.

The conventional additive identity is the pair shadow $(0, 0)$ and the conventional coordinate inverse of (a, b) is $(-a, -b)$. These statements are algebraic identities of the compiler. They are not assertions of primitive negative or orthogonal existence.

Theorem 63.4 (Internal complex ring laws)**.** *If the coefficient compiler supplies the required commutative ring identities and the declared pair operations preserve its equality fibers, then the pair grammar satisfies the usual commutative ring laws.*

63.4 Formal complex unit

The glyph

$$\mathbf{i} := (0, 1)$$

is the conventional name of the second coordinate basis token. Under the declared reduction rule,

$$\mathbf{i}^2 = -1.$$

This is an internal coordinate identity, not a foundational existence theorem and not a proof of Euclidean orthogonality.

63.5 Conjugation, norm, and division as compiled operations

Within the same conventional grammar one may define

$$\overline{(a, b)} = (a, -b), \qquad N(a, b) = a^2 + b^2,$$

and, when the norm coordinate is constructively separated from its zero report, the usual quotient formulas. Each operation is admissible only with the retained coefficient/refinement and nonzero-separation certificates required to execute it.

Principle 63.5 (Complex projection cannot ground its prerequisites)**.** *No theorem proved solely in the pair grammar may be used retroactively to establish the primitive existence of signed numbers, exact real coordinates, a perpendicular axis, a right angle, or the complex domain itself. Those are prerequisites or declarations of the compiler, not consequences of it.*

Theorem 63.6 (Complex-coordinate preservation)**.** *Any property inferred from a conventional complex-coordinate value is a property of the richer PCL source report only after that property is proved constant on the projection fiber of $\mathcal{F}_{\mathbb{C}}$. In particular, equality with the complex scalar zero does not automatically imply equality of source, refinement, transition, or closure ledgers.*

64 Complex zero and complete operator zero

The symbol 0 now occurs in several constructed carriers. Their scalar reports may agree while their types and obligations differ. This section separates those zero predicates before any analytic function, functional equation, or zeta zero is discussed.

64.1 Zero is typed by carrier and obligation

Definition 64.1 (Scalar real zero)**.** *The completed-real zero is*

$$0_{\mathbb{R}} := [\underline{0}],$$

the Cauchy-equivalence class of the constant rational-zero family.

Definition 64.2 (Signed zero)**.** *A signed state is zero under integer or real signed equality when its positive and negative lanes balance. Its normalized report is* $0_{\mathbb{Z}}$ *or* $0_{\mathbb{R}}$*, while its cancellation ledger may remain nontrivial.*

Definition 64.3 (Pair zero)**.** *The oriented-pair zero is*

$$0_{\text{pair}} := (0_{\mathbb{R}}, 0_{\mathbb{R}}).$$

Definition 64.4 (Complex zero)**.** *The complex zero is the additive identity*

$$0_{\mathbb{C}} := 0_{\text{pair}}.$$

It is one admitted member of the complex carrier and not literal absence.

Pair zero and complex zero are the same carrier element under the complex pair grammar. They are named separately to distinguish the representational statement "both lanes report real zero" from the algebraic statement "this report is the complex additive identity."

Remark 64.5 (Many source reports, one complex zero)**.** *The following may all project to* $0_{\mathbb{C}}$*:*

$$(0,0), \qquad z + (-z), \qquad (a,b) + (-a,-b),$$

and any completed pair whose two component families converge to real zero. They need not have equal source or evaluation ledgers.

64.2 Evaluation ledgers

Definition 64.6 (Complex evaluation ledger)**.** *Let*

$$f : D \to \mathbb{C}_{\mathrm{C}}$$

be a complex-valued mapping. A declared evaluation ledger at $s \in D$ *is a report*

$$\Lambda_f(s) = (s, f, \mathcal{T}_s, \mathcal{S}_s, \mathcal{R}_s, \mathcal{B}_s, \mathcal{O}_s, \gamma_s, f(s)),$$

where, as relevant:

(i) $\mathcal{T}_s$ *is the finite or completed operation tree;*

(ii) $\mathcal{S}_s$ *is the family of source contributions and branch labels;*

(iii) $\mathcal{R}_s$ *is the residual or unresolved-obligation ledger;*

(iv) $\mathcal{B}_s$ *is the boundary or quotient ledger;*

(v) $\mathcal{O}_s$ *is the orientation, conjugation, reflection, or winding ledger;*

(vi) γ_s *is the construction or continuation history;*

(vii) and $f(s)$ *is the final complex output report.*

Not every application requires every coordinate. The signature must state which coordinates are relevant and operational.

Definition 64.7 (Complex output projection)**.** *Define*

$$\pi_{\mathbb{C}} : \Lambda_f(D) \to \mathbb{C}_{\mathrm{C}}, \qquad \pi_{\mathbb{C}}(\Lambda_f(s)) := f(s).$$

This projection forgets every ledger coordinate not encoded in the final oriented pair.

64.3 Residual and source closure

Definition 64.8 (Residual closure zero)**.** *A ledger has* residual closure zero*, written*

$$Z_{\mathrm{res}}(\Lambda) = 1,$$

when every residual obligation declared by the operation grammar has been discharged and the residual channel reports its typed zero

$$\mathcal{R} = 0_{\mathrm{res}}.$$

Definition 64.9 (Boundary closure zero)**.** *A ledger has* boundary closure zero *when every declared quotient, carry, exposed-boundary, or interface obligation is either exactly matched or retained in the typed closure state prescribed by the operation.*

Definition 64.10 (Source closure zero)**.** *A ledger has* source closure zero*, written*

$$Z_{\mathrm{src}}(\Lambda) = 1,$$

when every admitted source contribution is accounted for by the declared matching, cancellation, transport, or completion rule, with no unmatched source lane remaining.

Source closure does not mean the source ledger is empty. It means that its obligations close under the declared relation.

Example 64.11 (Scalar cancellation without ledger identity)**.** *Let* $z \neq 0_{\mathbb{C}}$*. The evaluation*

$$z + (-z) = 0_{\mathbb{C}}$$

has scalar output zero and may have exact source closure through an explicit opposing pairing. A primitive constant-zero evaluation also outputs $0_{\mathbb{C}}$ *but contains no such pair. The two ledgers share an output and differ in source history.*

Example 64.12 (Output zero with an unresolved auxiliary ledger)**.** *An operation may produce two complex branches that cancel in the final pair while retaining a nonzero declared remainder, boundary crossing, or unmet certificate obligation. Then*

$$\pi_{\mathbb{C}}(\Lambda) = 0_{\mathbb{C}}$$

but

$$Z_{\mathrm{res}}(\Lambda) = 0.$$

The scalar cancellation does not erase the unresolved ledger coordinate.

64.4 Root, kernel, and zero operator

Definition 64.13 (Root of a complex-valued mapping)**.** *A report $s \in D$ is a* complex root *of f when*

$$f(s) = 0_{\mathbb{C}}.$$

The root set is

$$Z(f) := \{s \in D : f(s) = 0_{\mathbb{C}}\}.$$

Definition 64.14 (Kernel)**.** *When f is a homomorphism, its kernel is*

$$\ker f := Z(f).$$

The word kernel *additionally records that the zero fiber is compatible with the algebraic operations preserved by f.*

Definition 64.15 (Zero operator)**.** *The* zero operator *from A to B is the mapping*

$$\mathbf{0}_{A\to B} : A \to B$$

satisfying

$$\mathbf{0}_{A\to B}(a) = 0_B$$

for every $a \in A$.

Theorem 64.16 (A root is not a zero operator)**.** *The statement*

$$f(s) = 0_{\mathbb{C}}$$

at one input does not imply

$$f = \mathbf{0}_{D\to\mathbb{C}_{\mathrm{C}}}.$$

Proof. The first statement concerns one input fiber. The second quantifies over every admissible input. A single root provides no such universal report. □

Definition 64.17 (Operator equality)**.** *Two operators $F, G : A \to B$ are extensionally equal when*

$$F(a) =_B G(a)$$

for every $a \in A$. They are constructively operator-equal only when the additional rule, domain, evaluation, and source structures declared relevant are also preserved.

The zero operator is therefore a typed mapping object, not merely the scalar zero of its codomain.

64.5 Complete ledger zero

Definition 64.18 (Zero signature)**.** *A* zero signature *is a declared finite family of zero obligations*

$$\Sigma = \{Z_{\mathbb{C}}, Z_{\mathrm{res}}, Z_{\mathrm{src}}, Z_{\mathrm{bdry}}, Z_{\mathrm{orient}}, \ldots\}$$

relevant to a class of evaluation ledgers.

Definition 64.19 (Complete ledger zero)**.** *A ledger* Λ *has* complete zero relative to Σ*, written*

$$Z_\Sigma(\Lambda) = 1,$$

when every zero obligation in Σ *is satisfied. In particular, when* $Z_{\mathbb{C}} \in \Sigma$*,*

$$\pi_{\mathbb{C}}(\Lambda) = 0_{\mathbb{C}}$$

is required but may not be sufficient.

There is no untyped universal ledger zero independent of a declared operation grammar. The signature states what must close.

Definition 64.20 (Complete operator zero)**.** *An operator* $F : A \to B$ *is a* complete operator zero relative to Σ *when, for every admissible input* $a \in A$*,*

$$Z_\Sigma(\Lambda_F(a)) = 1.$$

This is stronger than extensional equality with the scalar zero operator whenever Σ includes source, residual, boundary, orientation, or rule obligations not determined by the final codomain value.

64.6 The zero hierarchy

The principal zero reports now form the following typed hierarchy:

literal absence	:	no report exists,
zero token	:	one report object denotes no declared contribution,
signed zero	:	opposing magnitude lanes balance,
real zero	:	a completed-real equivalence class,
pair zero	:	$(0_{\mathbb{R}}, 0_{\mathbb{R}})$,
complex zero	:	the additive identity of $\mathbb{C}_{\mathrm{C}}$,
residual zero	:	all declared residual obligations close,
source zero	:	all declared source obligations are accounted for,
zero operator	:	every input receives codomain zero,
complete ledger zero	:	every obligation in Σ closes.

No upward implication is automatic merely because the displayed scalar symbol is 0.

64.7 Scalar zero is not automatically complete zero

Theorem 64.21 (No automatic zero upgrade)**.** *Let* $\pi_{\mathbb{C}} : \mathcal{L} \to \mathbb{C}_{\mathrm{C}}$ *be the complex output projection on a ledger carrier* $\mathcal{L}$*, and let* Z_Σ *be a stronger zero predicate. From*

$$\pi_{\mathbb{C}}(\Lambda) = 0_{\mathbb{C}}$$

it follows that $Z_\Sigma(\Lambda) = 1$ *only if* Z_Σ *is constant on the complex-output fiber over* $0_{\mathbb{C}}$*.*

Proof. If the predicate is not constant on the zero fiber, there exist ledgers Λ_1 and Λ_2 with

$$\pi_{\mathbb{C}}(\Lambda_1) = \pi_{\mathbb{C}}(\Lambda_2) = 0_{\mathbb{C}}$$

but

$$Z_\Sigma(\Lambda_1) \neq Z_\Sigma(\Lambda_2).$$

The common scalar output then cannot determine the stronger predicate. □

Theorem 64.22 (Zero-predicate factorization criterion)**.** *The following are equivalent:*

(i) there exists a predicate

$$\widehat{Z}_\Sigma : \mathbb{C}_{\mathrm{C}} \to \{0, 1\}$$

such that

$$Z_\Sigma = \widehat{Z}_\Sigma \circ \pi_{\mathbb{C}};$$

(ii) whenever two ledgers have the same complex output, they have the same Σ-zero status:

$$\pi_{\mathbb{C}}(\Lambda_1) = \pi_{\mathbb{C}}(\Lambda_2) \Longrightarrow Z_\Sigma(\Lambda_1) = Z_\Sigma(\Lambda_2).$$

Proof. This is the predicate specialization of the projection-fiber preservation theorem from Section 54. □

Corollary 64.23 (When complex zero is sufficient)**.** *If Z_Σ factors through $\pi_{\mathbb{C}}$ and*

$$\widehat{Z}_\Sigma(0_{\mathbb{C}}) = 1,$$

then

$$f(s) = 0_{\mathbb{C}} \Longrightarrow Z_\Sigma(\Lambda_f(s)) = 1.$$

Without the factorization theorem, the implication is unsupported.

64.8 Complete zero and construction equivalence

Theorem 64.24 (Complete zero need not imply ledger identity)**.** *Even when two ledgers both satisfy $Z_\Sigma = 1$, they need not be constructively equal. Complete zero classifies closure status relative to Σ; it does not necessarily identify source histories, operation trees, or certificates.*

Proof. Two distinct source pairings may each close every obligation in Σ. Unless Σ includes full source and rule identity, both are complete zero ledgers but remain distinct constructions. □

Thus complete zero is itself a projection unless its signature contains the entire declared ledger.

64.9 Zero-preserving maps

Definition 64.25 (Zero-preserving map)**.** *A mapping $F : A \to B$ is scalar-zero-preserving when*

$$a = 0_A \Longrightarrow F(a) = 0_B.$$

It is Σ-zero-preserving when complete zero of an input ledger implies complete zero of the output ledger under the translated signature.

Theorem 64.26 (Homomorphisms preserve scalar identity zero)**.** *Every additive homomorphism satisfies*

$$F(0_A) = 0_B.$$

Proof. Since $0_A + 0_A = 0_A$,

$$F(0_A) + F(0_A) = F(0_A).$$

Add the inverse of $F(0_A)$ to both sides to obtain $F(0_A) = 0_B$. □

This theorem concerns the carrier identity. Preservation of source, residual, or operator zero requires additional ledger theorems.

64.10 Application gate for complex analysis

Any later use of a statement

$$f(s) = 0_{\mathbb{C}}$$

must declare which of the following is intended:

(1) only the final longitudinal and transverse reports vanish;

(2) every finite or completed residual channel closes;

(3) reflected, conjugate, quotient, and source lanes are all accounted for;

(4) the evaluation is zero under a declared operator predicate;

(5) or the complete ledger satisfies a specified zero signature Σ.

Principle 64.27 (Complex-zero audit)**.** *A scalar complex root may be promoted to a stronger constructive zero only by one of two routes:*

(i) directly verify the additional zero obligations on the evaluation ledger; or

(ii) prove that the stronger predicate factors through the complex output projection.

No appeal to the written symbol 0 *can replace this obligation.*

64.11 Part XV summary

Sections 62–64 compile the prior manuscript into the following dependency chain:

$$\begin{gathered}\text{finite and completed operations} \longrightarrow \text{typed carriers and closure} \\ \longrightarrow \text{identities and inverses} \longrightarrow \text{groups, rings, and fields} \\ \longrightarrow \text{congruences and quotient carriers} \\ \longrightarrow \text{oriented pairs and } \mathbf{i}^2 = -1 \\ \longrightarrow \text{complex field operations} \\ \longrightarrow \text{typed zero ledgers and fiber preservation.}\end{gathered}$$

The essential conclusions are:

(1) algebraic carriers are compiled report systems, not primitive universes;

(2) closure is relative to an operation, admissible domain, carrier, and equality;

(3) identities and inverses are typed and unique only under the declared equality;

(4) $\mathbb{N}_{\mathrm{C}}$, $\mathbb{Z}_{\mathrm{C}}$, $\mathbb{Q}_{\mathrm{C}}$, and $\mathbb{R}_{\mathrm{C}}$ support the expected additive and multiplicative structures through previously proved construction laws;

(5) constructive field inversion requires positive apartness from zero;

(6) quotient operations are valid exactly when the quotient relation is a congruence;

(7) operational closure and algebraic closure are different obligations;

(8) when the conventional complex compiler is declared, its unit token is a formal coordinate operator and not a primitive existence claim;

(9) $\mathbb{C}_{\mathrm{C}}$ is the oriented-pair field and equivalently the quadratic quotient $\mathbb{R}_{\mathrm{C}}[T]/(T^2+1)$;

(10) conjugation is transverse reflection, norm is $z\overline{z}$, magnitude is its completed nonnegative root, and division is conjugate-norm repair;

(11) complex scalar zero, residual zero, source zero, zero operator, and complete ledger zero are different typed predicates;

(12) $f(s) = 0_{\mathbb{C}}$ certifies only complex output zero unless every stronger zero obligation is directly checked or proved constant on the complex-output fiber;

(13) and even complete zero relative to a signature need not erase distinct construction histories.

The next part should construct finite sequences, partial sums, series rules, and completed aggregation before continuity, differentiation, integration, exponential, logarithmic, gamma, and analytic continuation are introduced.

Part XVI
Analysis

65 Series and completed aggregation

Analysis begins only after finite arithmetic, signed orientation, completion, mappings, and equality preservation have already been constructed. A series is not primitive addition over an already completed infinity. It is first an indexed rule that produces finite partial-sum ledgers; a completed aggregate appears only after a convergence or summability criterion has been declared.

65.1 Finite partial sums

Definition 65.1 (Series term family)**.** *Let I be an indexed construction carrier and let A be an additive carrier. A **term family** is a mapping*

$$a : I \to A.$$

When I is a natural-number prefix carrier, write its values as

$$a_0, a_1, \ldots, a_n.$$

The family includes both the term reports and the rule or table by which the terms are generated.

Definition 65.2 (Finite partial-sum ledger)**.** *For a finite prefix $[0,n]$, the* ***finite partial-sum ledger*** *is*

$$S_n = (a_0, a_1, \ldots, a_n; +, \gamma_n)$$

whose scalar additive report is

$$s_n = \sum_{k=0}^{n} a_k.$$

The symbol $\sum$ abbreviates repeated source-preserving addition over the retained prefix. It does not erase the term indices, grouping rule, or addition order unless a later theorem proves that those data are irrelevant.

Principle 65.3 (Finite-first aggregation)**.** *Every series construction begins with finite partial sums. No completed aggregate may be inferred from the notation*

$$\sum_{k=0}^{\infty} a_k$$

without a declared completion rule for the partial-sum family.

Definition 65.4 (Partial-sum mapping)**.** *The partial-sum mapping of a term family is*

$$\Sigma_a : \mathbb{N}_{\mathrm{C}} \to A, \qquad \Sigma_a(n) = \sum_{k=0}^{n} a_k.$$

Thus a series rule is the composed construction

$$a_k \;\rightsquigarrow\; S_n \;\rightsquigarrow\; \Sigma_a(n).$$

65.2 Resolution-certified convergence of partial-sum ledgers

Let S_N be the finite source-preserving aggregation through stage N. For a later finite stage $M > N$, write the exact finite decomposition

$$S_M = S_N \sqcup T_{N,M},$$

where $T_{N,M}$ is the retained finite tail ledger.

Definition 65.5 (Resolution-certified series convergence)**.** *A series rule is resolution-certified convergent under observable signature Λ when, for every presented positive finite resolution ρ, one can construct a finite witness N_ρ and a finite tail-bound rule $\mathcal{B}_\rho$ such that every lawful finite $M > N_\rho$ satisfies*

$$\mathrm{Mag}_\Lambda\big(\mathrm{Norm}(T_{N_\rho,M})\big) < \rho.$$

No target drawn from an already-completed continuum is required by this definition. When an independently constructed target evaluator L is also available and every finite stage beyond N_ρ lies within ρ of L, the conventional limit notation may be used as a later compression.

Definition 65.6 (Series convergence certificate)**.** *A series convergence certificate records the term rule, finite witness N_ρ, tail-bound rule, observable signature, aggregation/cancellation grammar, and all residual obligations required to justify the requested resolution.*

Theorem 65.7 (Finite-tail criterion)**.** *A finite aggregation rule is resolution-certified convergent exactly when it admits such a finite witness and tail-bound certificate for every requested positive finite resolution.*

Proof. This is the constructive content of convergence: every actual request is answered by a finite prefix together with a proof that every later finite extension changes the declared observable by less than the requested amount. Conversely, such a certificate is precisely what the definition requires. □

65.3 Absolute convergence

Definition 65.8 (Absolute series)**.** *For terms in a signed or complex completed carrier, define the magnitude term family*

$$b_k = |a_k|.$$

The original series is ***absolutely convergent*** *when*

$$\sum_{k=0}^{\infty} b_k$$

converges.

Theorem 65.9 (Absolute convergence implies Cauchy convergence)**.** *If the positive magnitude tails admit a resolution certificate, then the oriented series tails admit the same resolution certificate.*

Proof. For $m > n$, the signed tail satisfies

$$\left| \sum_{k=n+1}^{m} a_k \right| \leq \sum_{k=n+1}^{m} |a_k|.$$

If the magnitude series converges, its tails are eventually smaller than every positive resolution. Hence every finite oriented tail is bounded by the certified positive tail, which supplies resolution-certified convergence without requiring a pre-existing completed target. □

Principle 65.10 (Magnitude domination is a ledger theorem)**.** *Absolute convergence is not an aesthetic preference for positive terms. It is a constructive domination theorem: every signed or oriented tail is bounded by a positive tail whose completed disappearance has been proved.*

65.4 Conditional convergence

Definition 65.11 (Conditional convergence)**.** *A series is* ***conditionally convergent*** *when*

$$\sum a_k$$

converges but

$$\sum |a_k|$$

does not converge. Equivalently, signed cancellation is necessary for the completed aggregate.

Remark 65.12 (Why conditional convergence preserves more history)**.** *For an absolutely convergent series, large signed rearrangements are dominated by a convergent positive magnitude ledger. For a conditionally convergent series, cancellation order may contribute to the completed value. Therefore order and grouping cannot be casually quotiented away.*

65.5 Rearrangement

Definition 65.13 (Rearranged term family)**.** *Let $\rho : \mathbb{N}_C \to \mathbb{N}_C$ be a bijective reindexing. The rearranged family is*

$$a_k^\rho = a_{\rho(k)}.$$

Its partial sums are

$$s_n^\rho = \sum_{k=0}^{n} a_{\rho(k)}.$$

The rearrangement ledger retains ρ.

Theorem 65.14 (Absolute rearrangement preservation)**.** *If $\sum a_k$ is absolutely convergent, every bijective rearrangement has the same completed sum.*

Proof. Given a positive resolution, choose a finite prefix after which the absolute tail is smaller than the resolution. Every rearranged finite prefix eventually contains all terms from the original large finite core; the remaining discrepancy is contained in the absolutely small tail. Thus rearranged and original partial sums become arbitrarily close to the same limit. □

Principle 65.15 (Rearrangement audit)**.** *A rearranged infinite aggregation is valid only after the property under discussion is proved constant over the rearrangement fiber. Absolute convergence supplies such a theorem for the completed scalar sum. Conditional convergence does not supply it automatically.*

65.6 Source preservation

Definition 65.16 (Source-preserving series ledger)**.** *A source-preserving series ledger is*

$$\Lambda_{\sum a} = (a, \Sigma_a, (s_n), \gamma, +, \mathcal{C}),$$

where a is the term rule, Σ_a is the partial-sum mapping, (s_n) is the partial-sum family, γ records grouping and order, $+$ is the additive carrier operation, and $\mathcal{C}$ is the convergence certificate.

Theorem 65.17 (Completed scalar sum does not recover the series ledger)**.** *Two series may have the same completed sum while having different term rules, partial-sum paths, convergence rates, cancellation histories, or rearrangement ledgers.*

Proof. Let one series be $1 + (-1) + 0 + 0 + \cdots$ and another be the constant zero series. Both may project to completed sum 0, but the first has a nontrivial finite cancellation ledger and the second does not. Equality of completed scalar sums therefore does not imply equality of series ledgers. □

66 Continuity

Continuity is a property of a completed mapping between completed carriers. It is not inherited from a visual curve, a coordinate drawing, or an assumed continuum. It must be stated through input and output relations.

66.1 Input-output completion

Definition 66.1 (Completed mapping)**.** *Let X and Y be completed carriers and let*

$$f : X \to Y$$

be a mapping. An evaluation of f at x has an evaluation ledger

$$\Lambda_f(x) = (x, f, x \mapsto f(x), \gamma_x).$$

Continuity concerns the behavior of these output reports as input reports vary inside the completed carrier.

Definition 66.2 (Continuity at a point)**.** *Let X and Y carry completed distance or resolution relations. The mapping $f : X \to Y$ is **continuous at** x_0 when for every positive output resolution ε there exists a positive input resolution δ such that*

$$d_X(x, x_0) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x_0)) < \varepsilon.$$

This is a relation between completed input differences and completed output differences.

Definition 66.3 (Sequential continuity)**.** *The mapping f is sequentially continuous at x_0 when every convergent input family*

$$x_n \to x_0$$

produces a convergent output family

$$f(x_n) \to f(x_0).$$

Theorem 66.4 (Metric equivalence of the two continuity reports)**.** *For mappings between completed metric carriers, epsilon–delta continuity at x_0 is equivalent to sequential continuity at x_0.*

Proof. If epsilon–delta continuity holds and $x_n \to x_0$, choose the corresponding δ for a given ε. Eventually $d_X(x_n, x_0) < \delta$, so eventually $d_Y(f(x_n), f(x_0)) < \varepsilon$.

Conversely, if epsilon–delta continuity fails, then some $\varepsilon_0 > 0$ has the property that for every positive δ there is an x with $d_X(x, x_0) < \delta$ but $d_Y(f(x), f(x_0)) \geq \varepsilon_0$. Choosing such an x_n for $\delta = 1/n$ constructs a sequence converging to x_0 whose output fails to converge to $f(x_0)$. □

66.2 Continuity and projection fibers

Principle 66.5 (Continuity is not source preservation)**.** *Continuity of the scalar or coordinate output of f does not imply preservation of every evaluation ledger coordinate. To promote continuity to a stronger source statement, the stronger predicate must factor through the completed output relation.*

Definition 66.6 (Ledger-continuity)**.** *Let $\Lambda_f(x)$ be a selected evaluation ledger. The map f is **ledger-continuous** for a ledger coordinate P when*

$$x_n \to x_0 \quad \Longrightarrow \quad P(\Lambda_f(x_n)) \to P(\Lambda_f(x_0))$$

inside the declared carrier for that ledger coordinate.

Remark 66.7. *Ordinary continuity usually concerns final values. Primitive Constructive Logic permits that convention, but it does not allow a final-value continuity theorem to be silently used as a source-ledger continuity theorem.*

66.3 Uniform continuity

Definition 66.8 (Uniform continuity)**.** *A mapping $f : X \to Y$ is **uniformly continuous** on a domain $D \subseteq X$ when for every positive output resolution ε there exists one positive input resolution δ such that for all $x, y \in D$,*

$$d_X(x, y) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(y)) < \varepsilon.$$

The same δ works throughout the declared domain.

Theorem 66.9 (Continuous image of a convergent family)**.** *If f is continuous at x_0 and $x_n \to x_0$, then $f(x_n) \to f(x_0)$.*

Proof. This is the forward implication in the metric equivalence theorem. It is recorded separately because it is the continuity rule most often used in later analysis. □

66.4 Composition

Theorem 66.10 (Composition of continuous mappings)**.** *If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.*

Proof. Given an output resolution for Z, continuity of g supplies a resolution in Y. Continuity of f supplies a resolution in X that forces the Y-output inside that resolution. The composed implication is exactly continuity of $g \circ f$. □

67 Differentiation

Differentiation is the completed report of signed ratio states produced by changing an input and comparing the resulting output. No derivative is primitive. Every finite difference quotient remains a finite ratio object; the derivative appears only after a limiting operation is applied to the family of such quotients.

67.1 Finite difference quotient

Definition 67.1 (Input increment)**.** *Let $f : D \subseteq \mathbb{R}_{\mathrm{C}} \to \mathbb{R}_{\mathrm{C}}$ and let x be a point of D. A nonzero signed input increment is a report h such that $h \neq 0$ and $x + h \in D$.*

Definition 67.2 (Finite difference quotient)**.** *The **finite difference quotient** of f at x with increment h is the signed ratio state*

$$Q_f(x; h) = \frac{f(x + h) - f(x)}{h}.$$

Its complete ledger retains

$$(x, h, x + h, f(x), f(x + h), f(x + h) - f(x), h, \textit{division rule}).$$

Principle 67.3 (Finite quotients are not derivatives)**.** *For $h \neq 0$, $Q_f(x; h)$ is a finite completed ratio. It is not yet the derivative. The derivative is a later limit report of the family $h \mapsto Q_f(x; h)$ as h approaches zero through nonzero increments.*

67.2 Derivative as a limit report

Definition 67.4 (Derivative)**.** *The function f is **differentiable at** x with derivative L when*

$$\lim_{h\to 0,\ h\neq 0} Q_f(x;h) = L.$$

Write

$$f'(x) = L.$$

The derivative ledger retains the difference-quotient family and the convergence certificate.

Definition 67.5 (Residual form of differentiability)**.** *Equivalently, f is differentiable at x with derivative L when there exists a residual family $\eta(h)$ such that*

$$f(x+h) = f(x) + Lh + h\eta(h)$$

with

$$\eta(h) \to 0.$$

This separates the linear carrier Lh from the vanishing residual report.

Theorem 67.6 (Differentiability implies continuity)**.** *If f is differentiable at x, then f is continuous at x.*

Proof. By the residual form,

$$f(x+h) - f(x) = Lh + h\eta(h).$$

As $h \to 0$, both Lh and $h\eta(h)$ converge to zero, since L is fixed and $\eta(h)$ remains eventually bounded while tending to zero. Hence $f(x+h) \to f(x)$. □

67.3 Basic derivative laws

Theorem 67.7 (Linearity of derivative)**.** *If f and g are differentiable at x, then $f+g$ is differentiable at x and*

$$(f+g)'(x) = f'(x) + g'(x).$$

For a scalar report c,

$$(cf)'(x) = cf'(x).$$

Proof. The finite difference quotient of $f+g$ is the sum of the finite difference quotients of f and g. Taking limits and using algebraic preservation of limits gives the result. Scalar multiplication is identical. □

Theorem 67.8 (Product rule)**.** *If f and g are differentiable at x, then fg is differentiable at x and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Proof. For nonzero h,

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \frac{f(x+h)-f(x)}{h}g(x+h) + f(x)\frac{g(x+h)-g(x)}{h}.$$

The difference quotients converge to $f'(x)$ and $g'(x)$, and differentiability implies $g(x+h) \to g(x)$. Limit preservation gives the stated formula. □

Theorem 67.9 (Chain rule)**.** *If f is differentiable at x and g is differentiable at $f(x)$, then $g \circ f$ is differentiable at x and*

$$(g \circ f)'(x) = g'(f(x))f'(x).$$

Proof. The input change h produces an intermediate change

$$k(h) = f(x+h) - f(x).$$

Differentiability of f gives $k(h) = f'(x)h + h\eta(h)$, so $k(h) \to 0$. Differentiability of g at $f(x)$ gives

$$g(f(x)+k) - g(f(x)) = g'(f(x))k + k\theta(k)$$

with $\theta(k) \to 0$. Substitute $k = k(h)$ and divide by h. The residual terms vanish, leaving $g'(f(x))f'(x)$. □

67.4 Complex differentiability

Definition 67.10 (Complex difference quotient)**.** *For $f : D \subseteq \mathbb{C}_{\mathrm{C}} \to \mathbb{C}_{\mathrm{C}}$, the finite complex difference quotient is*

$$Q_f(z;h) = \frac{f(z+h) - f(z)}{h}, \qquad h \in \mathbb{C}_{\mathrm{C}},\ h \neq 0.$$

The denominator may approach zero through many orientation lanes.

Definition 67.11 (Complex derivative)**.** *The complex derivative $f'(z)$ exists when $Q_f(z;h)$ has the same completed limit for every nonzero complex approach $h \to 0$ admitted by the domain.*

Principle 67.12 (Complex differentiability is stronger than two real directional derivatives)**.** *A complex derivative requires approach-lane invariance. Agreement along only the longitudinal and transverse axes is evidence, not the full theorem.*

68 Integration

Integration is a conditional accumulation compiler over finitely partitioned coordinate domains. It must retain finite partitions, cell-width reports, refinement, lower and upper accumulation, and the declared convergence certificate. No primitive geometric area is inferred merely from the notation of a partition cell.

68.1 Finite partitions and cells

Definition 68.1 (Interval partition)**.** *For a completed real interval $[a,b]$, a finite partition is an ordered report*

$$P = (x_0, x_1, \ldots, x_n)$$

with

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Each subinterval $[x_{j-1}, x_j]$ is a whole cell in the partition ledger.

Definition 68.2 (Mesh)**.** *The mesh of P is*

$$\|P\| = \max_{1 \le j \le n} (x_j - x_{j-1}).$$

The mesh reports the largest retained cell width.

Definition 68.3 (Tagged Riemann ledger)**.** *A tagged partition includes tags $t_j \in [x_{j-1}, x_j]$. Its finite accumulation report is*

$$R(f, P, t) = \sum_{j=1}^{n} f(t_j)(x_j - x_{j-1}).$$

The ledger retains every cell, tag, width, height report, and product cell.

68.2 Lower and upper accumulation

Definition 68.4 (Lower and upper cell reports)**.** *For a bounded function $f : [a, b] \to \mathbb{R}_{\mathrm{C}}$ and subinterval $I_j = [x_{j-1}, x_j]$, define*

$$m_j = \inf_{x \in I_j} f(x), \qquad M_j = \sup_{x \in I_j} f(x),$$

when these infimum and supremum reports exist.

Definition 68.5 (Lower and upper sums)**.** *The lower and upper sums are*

$$L(f, P) = \sum_{j=1}^{n} m_j(x_j - x_{j-1}),$$

$$U(f, P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}).$$

They are finite accumulation ledgers whose cell values come from lower and upper bounds.

Definition 68.6 (Riemann integrability)**.** *A bounded function f is Riemann integrable on $[a, b]$ when the lower and upper accumulations close to the same completed report:*

$$\sup_P L(f, P) = \inf_P U(f, P).$$

The common completed value is

$$\int_a^b f(x)\, dx.$$

Principle 68.7 (Integral as completed squeeze)**.** *An integral is not a primitive area value. Within the declared coordinate-analysis compiler it is the resolution-certified agreement of lower and upper finite accumulation ledgers under refinement.*

68.3 Area-bearing interpretation

Remark 68.8 (Why integration waits for measure)**.** *The product $f(t_j)(x_j - x_{j-1})$ is a coordinate accumulation report whose width factor is licensed by the active ratio/coordinate grammar. Interpreting that product as literal geometric area is optional visualization inside a declared geometry compiler, not a foundational prerequisite for finite accumulation.*

68.4 Refinement

Definition 68.9 (Partition refinement)**.** *A partition Q refines P when every partition point of P also occurs in Q. Refinement subdivides cells into smaller whole cells; it does not create fractional indices.*

Theorem 68.10 (Refinement monotonicity)**.** *If Q refines P, then*

$$L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P).$$

Proof. On each original cell, subdivision can only raise the sum of local infimum-width products and can only lower the sum of local supremum-width products. Summing across all original cells gives the inequalities. □

68.5 Fundamental theorem as a preservation result

Theorem 68.11 (Integral of a derivative)**.** *If F is continuously differentiable on $[a,b]$, then*

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

Proof sketch. For each partition cell, the mean-value relation or a uniform continuity argument compares

$$F(x_j) - F(x_{j-1})$$

with

$$F'(t_j)(x_j - x_{j-1}).$$

The finite telescoping sum of the exact differences is $F(b) - F(a)$. Uniform continuity of F' forces the total error between the derivative accumulation ledger and the telescoping ledger to vanish under refinement. The completed reports are therefore equal. □

Principle 68.12 (Integration preserves declared accumulation, not every source)**.** *Equality of completed integrals does not by itself identify integrands, partitions, geometric cells, or physical accumulation histories. Any stronger identification requires a fiber-preservation theorem.*

69 Exponential, logarithmic, and gamma reports

The zeta compiler needs logarithmic scale, exponentiation, orientation transport, and gamma-type accumulation. None of these may be defined by a completed infinite object that presupposes the very continuum being constructed.

69.1 Finite quotient–remainder logarithm

Let $N > 0$ and let $b > 1$ be finite whole source reports. Repeatedly apply quotient–remainder division

$$N = bQ_0 + R_0, \qquad Q_0 = bQ_1 + R_1, \qquad \dots$$

until the quotient no longer contains a complete b-carrier. The finite depth determines the whole multiplicative shell h satisfying the compiled power comparison

$$b^h \le N < b^{h+1}.$$

The powers here are finite repeated-product reports already constructed in Part IX.

For a refinement radix $r \geq 2$, retain whole counts A_m, D_m with

$$\frac{A_m}{D_m} \leq \log_b N < \frac{A_m + 1}{D_m}, \qquad (A_0, D_0) = (h, 1).$$

To refine, set $D_{m+1} = rD_m$. For each $d \in \{0, \ldots, r-1\}$ compare only whole finite powers

$$N^{D_{m+1}} \quad \text{and} \quad b^{rA_m + d}$$

by quotient–remainder extraction. Choose the greatest d for which the second is completely contained in the first, and set

$$A_{m+1} = rA_m + d.$$

No fractional power is used in the refinement.

Definition 69.1 (Finite logarithm evaluator)**.** *The report*

$$\mathcal{L}_b(N; \rho)$$

is the finite QR/power-comparison ledger obtained by refining until

$$\frac{1}{D_m} < \rho.$$

It returns the bracket $[A_m/D_m, (A_m + 1)/D_m]$ *together with every retained quotient, remainder, and comparison witness.*

A finer request invokes another finite construction. It does not reveal digits from an already completed infinite real object.

69.2 Finite rational exponential compiler

For a rational exponent report A/D with $D > 0$, define

$$\mathrm{Exp}_b(A/D) := \mathrm{Root}_D(\mathrm{Pow}_A(B_b)),$$

where negative orientation of A is handled by first constructing the positive-depth report and then applying the already constructed reciprocal grammar. Exact rational roots close exactly; otherwise the root evaluator returns a finite shell and retained residual at the requested resolution.

Thus logarithm and exponential are not defined circularly. The logarithm uses QR and finite whole-power comparisons; rational exponentiation uses finite powers and inverse root reconstruction. Inverse identities are proved afterward when their closure hypotheses are satisfied.

69.3 Orientation exponential and complex coordinate projection

For a finite rational orientation demand C/D cycles, choose a finite orientation ledger with K whole transitions. Resolve

$$KC = QD + R, \qquad 0 \leq R < D,$$

and then

$$Q = WK + J, \qquad 0 \leq J < K.$$

The quotient W retains complete returns, J the terminal orientation role, and R the unresolved orientation demand at that finite resolution. Choosing K as a whole multiple of D makes $R = 0$ and represents every finite rational orientation demand exactly.

Only after the four-state coordinate compiler of Section 63 is chosen may such a finite orientation report be projected to the familiar exponential phase notation. The symbol $\exp(z)$ is therefore a compiled coordinate evaluator whose actual construction consists of finite magnitude and finite orientation reports.

69.4 Complex powers as transformed report packets

For a positive finite source n and an instruction report Σ carrying attenuation and orientation roles, construct first the logarithmic source report $\mathcal{L}_b(n;\rho)$, then the finite attenuation magnitude and finite orientation packet. Denote the resulting source-preserving transformed term by

$$\mathcal{T}_n[\Sigma;\rho].$$

When the complex coordinate compiler is applied and its preservation theorem holds, the scalar shadow may be written conventionally as

$$n^{-s} = \exp(-s\log n).$$

This formula is a theorem about the compiled projection. It is not the primitive definition of the finite power, logarithm, or orientation operations used to build the packet.

69.5 Gamma as a finite-window evaluator

For an input report s in a region where the endpoint estimates close, a gamma evaluation at requested resolution ρ consists of a finite window

$$[\varepsilon, R], \qquad 0 < \varepsilon < R,$$

a finite partition of that interval, finite evaluations of

$$t^{s-1}e^{-t},$$

and two explicit boundary certificates whose total contribution is below ρ. The reported value is the finite partition accumulation plus retained lower- and upper-boundary obligations.

The familiar notation

$$\Gamma(s)$$

therefore abbreviates this finite evaluator grammar. Integration by parts on a finite window gives the recurrence packet

$$\Gamma(s+1) = s\Gamma(s)$$

up to the retained finite boundary terms; when those terms are certified below every requested resolution, the recurrence is admitted as a report-preserving theorem. Stepwise use of this recurrence extends the evaluator only where the denominator report is constructively separated from zero.

70 Analytic continuation

Analytic continuation is not permission to replace one function with another merely because symbols match. It is a domain-extension construction with a preservation theorem.

70.1 Original report and extension target

Definition 70.1 (Analytic germ)**.** *An **analytic germ** at a point or connected domain consists of a local power-series rule, its convergence neighborhood, and its evaluation ledger. Write*

$$\mathcal{G} = (U, f, \Lambda_f).$$

Definition 70.2 (Continuation step)**.** *A continuation step from (U, f) to (V, g) is admissible when $U \cap V$ is nonempty and*

$$f(z) = g(z)$$

for every z in a nontrivial connected overlap where both sides are defined, with equality understood as completed scalar equality unless a stronger ledger equality is explicitly proved.

Definition 70.3 (Analytic continuation ledger)**.** *An analytic continuation ledger from an initial germ $\mathcal{G}_0$ to a target point z is a finite or coherently completed chain*

$$(U_0, f_0) \rightsquigarrow (U_1, f_1) \rightsquigarrow \cdots \rightsquigarrow (U_m, f_m)$$

with $z \in U_m$ and equality on every overlap. The ledger retains the domains, overlap witnesses, identity theorem used, and any singularities or branch choices crossed or avoided.

70.2 Identity theorem

Theorem 70.4 (Identity theorem)**.** *Let U be a connected complex domain. If analytic functions $f, g : U \to \mathbb{C}_{\mathrm{C}}$ agree on a set with an accumulation point in U, then they agree throughout U.*

Proof sketch. At an accumulation point, all coefficients of the local power series of $f - g$ vanish, so $f - g$ vanishes on a neighborhood. The set on which $f - g$ vanishes is both open and closed inside the connected continuation domain, hence it is all of U. □

Principle 70.5 (What analytic continuation preserves)**.** *Analytic continuation preserves the scalar analytic function specified by the identity theorem. It does not automatically preserve source formula, term-by-term convergence, product expansion, branch history, integration path, or operator-zero ledger unless those are separately carried and shown invariant.*

70.3 Branching and monodromy

Definition 70.6 (Branch ledger)**.** *A branch ledger records the path, cuts, winding counts, and singularity avoidances used to continue a multivalued analytic report such as a logarithm or power.*

Definition 70.7 (Monodromy report)**.** *If continuation along two paths with the same endpoints gives different branch values, the discrepancy is the monodromy report. It is not contradiction; it is retained orientation and winding history.*

Principle 70.8 (Single-valuedness obligation)**.** *A continued report may be treated as a single-valued function on a domain only after the continuation ledger proves path-independence on that domain or a branch selection is declared.*

70.4 Continuation versus original series

Definition 70.9 (Series domain)**.** *The series domain of a power series or Dirichlet series is the region where its defining partial sums converge to the reported value.*

Definition 70.10 (Continued domain)**.** *The continued domain is the region reached by analytic continuation of the original germ, minus declared singularities and branch exclusions.*

Principle 70.11 (No termwise inheritance outside convergence)**.** *Outside the series domain, the continued value is not the value of the original divergent termwise series. It is the value of the continued analytic report.*

70.5 Zeros under continuation

Definition 70.12 (Continued zero)**.** *A continued zero of an analytically continued report F at z is a point where*

$$F(z) = 0_{\mathbb{C}}.$$

The zero ledger includes the continuation path or domain certificate used to define $F(z)$.

Principle 70.13 (Continuation-zero audit)**.** *A zero obtained after analytic continuation certifies complex scalar zero of the continued report. It certifies complete source or operator zero only if the stronger zero predicate is proved invariant under the continuation chain and constant on the complex-output fiber.*

70.6 Analysis summary

Sections 65–70 construct the analysis layer required before any zeta application:

$$\begin{gathered}\text{finite term families} \longrightarrow \text{partial-sum ledgers} \longrightarrow \text{series convergence} \longrightarrow \text{completed aggregation},\\ \text{completed input-output relations} \longrightarrow \text{continuity},\\ \text{signed finite difference quotients} \longrightarrow \text{derivatives as completed quotient limits},\\ \text{finite partitions and lower/upper accumulation} \longrightarrow \text{completed integrals},\\ \text{series and integral compilers} \longrightarrow \exp, \log, \Gamma,\\ \text{local analytic germs and overlap identities} \longrightarrow \text{analytic continuation}.\end{gathered}$$

The essential conclusions are:

(1) a series is completed only through convergence of its partial-sum ledgers;

(2) absolute convergence supplies a rearrangement-preservation theorem, while conditional convergence retains order-sensitive cancellation history;

(3) continuity is a property of completed input-output relations, not a drawing of a curve;

(4) a derivative is a limit of signed ratio states, not a finite quotient;

(5) integration is completed lower/upper accumulation over finite area-bearing partition ledgers;

(6) exponentials, logarithms, complex powers, and gamma reports are compiled analytic objects with branch and boundary obligations;

(7) analytic continuation preserves the scalar analytic report supplied by the identity theorem, not every source construction by default;

(8) branch ledgers and monodromy record path-sensitive continuation history;

(9) and zeros of continued functions require the same complex-zero audit established in Section 64.

The next part can construct the zeta report itself: finite Dirichlet partial sums, the half-plane of convergence, Euler product where admissible, completed continuation, reflection, and the exact zero-predicate audit required before any one-to-two balance-coordinate theorem is stated.

Part XVII
Zeta Projection Audit, Continuation, and Reflection Diagnostics

71 Finite construction of the zeta source report

Principle 71.1 (Zeta-domain status)**.** *The ordinary statement $s \in \mathbb{C}$, the glyph n^{-s}, and later real-part or complex-zero notation belong to the conventional coordinate compiler, not to the primitive PCL layer. PCL first retains the finite source transformation and only then permits those scalar shadows conditionally. No claim in this part establishes a completed complex domain as a primitive totality.*

71.1 One transformed source term

Let n be a positive finite source-count report and let

$$\Sigma = (\mathcal{A}, \Omega)$$

carry a finite attenuation instruction $\mathcal{A}$ and a finite orientation-rate instruction Ω. At requested resolution ρ, construct

$$\mathcal{T}_n[\Sigma; \rho] = (n, \mathcal{L}_b(n; \rho), M_n, \Phi_n, \Gamma_n),$$

where M_n is the finite attenuation-magnitude report, Φ_n is the finite cyclic-orientation report, and Γ_n retains source and construction lineage.

The conventional expression n^{-s} is admitted only after the complex coordinate compiler has been proved to preserve these fields for the property under study.

71.2 Finite source-preserving zeta aggregation

For a positive finite stage N, define the retained term ledger

$$\mathfrak{D}_N[\Sigma; \rho] = [\mathcal{T}_1, \ldots, \mathcal{T}_N]$$

and its primitive aggregation

$$\mathfrak{J}_N[\Sigma; \rho] = \mathcal{T}_1 \sqcup \cdots \sqcup \mathcal{T}_N.$$

Joining occurs before cancellation or coordinate projection. Equal projected term values at different source indices remain different occurrences.

Let $\mathcal{K}$ be the declared finite orientation-closure grammar already constructed from opposing pairs, triangular closures, and lawful finite ring closures. Define

$$\mathfrak{C}_N = \mathrm{Norm}_{\mathcal{K}}(\mathfrak{J}_N),$$

retaining every closure matching and every unmatched residual.

When the complex coordinate compiler is applied, write

$$\Pi_\rho(\mathfrak{C}_N) = (X_N, Y_N).$$

Only then may the scalar shorthand

$$\zeta_{N,\rho}(s) = X_N + \mathbf{i}Y_N$$

be used.

71.3 Resolution-certified Dirichlet convergence for attenuation greater than one

Suppose the attenuation instruction contains a finite positive gap certificate

$$\mathcal{A} \geq 1 + \Delta, \qquad \Delta > 0.$$

Partition the source indices into finite dyadic blocks

$$B_j = [2^j, 2^{j+1} - 1].$$

Every block contains 2^j whole source indices, and every term in B_j has magnitude bounded by the finite power report

$$(2^j)^{-(1+\Delta)}.$$

Hence the positive magnitude of the whole block satisfies

$$\mathrm{Mag}(B_j) \leq 2^j (2^j)^{-(1+\Delta)} = 2^{-j\Delta}.$$

Set $c = 2^{-\Delta}$, so $0 < c < 1$. Any finite later block sequence obeys the finite geometric bound

$$c^J + c^{J+1} + \cdots + c^K = c^J \frac{1 - c^{K-J+1}}{1 - c} < \frac{c^J}{1 - c}.$$

Given a positive finite resolution ρ, the QR logarithm/power-shell compiler constructs a finite J with

$$\frac{c^J}{1 - c} < \rho.$$

Then

$$N_\rho = 2^J - 1$$

is a finite convergence witness for every later finite stage. This proves resolution-certified convergence without performing an infinite addition.

71.4 Constructive divergence boundary

The individual-term test immediately excludes attenuation $\mathcal{A} \le 0$, because the magnitude of the newly joined term does not become smaller than every positive resolution.

For $0 < \mathcal{A} \le 1$, fix a sufficiently narrow finite orientation sector with positive projection floor $\lambda > 0$. Choose finite whole counts $u, v > 0$ such that the multiplicative window ratio $1 + u/v$ keeps the orientation change inside that sector. For any sufficiently large finite N, divide

$$uN = vQ_N + R_N, \qquad 0 \le R_N < v,$$

and set $M_N = N + Q_N$. The finite block $N+1, \dots, M_N$ stays inside the selected orientation sector and contains at least a positive constant multiple of N terms. Its directed aggregate therefore has a finite lower bound

$$D_N \ge CN^{1-\mathcal{A}}$$

for a fixed positive report C determined by the sector and window.

If $0 < \mathcal{A} < 1$, this lower bound does not vanish and eventually grows; if $\mathcal{A} = 1$, it remains at least $C > 0$. Thus there is a positive finite resolution $\rho < C$ for which no proposed finite tail witness succeeds. The failure packet

$$\mathcal{D}_{\text{fail}} = (\mathcal{A}, \Omega, \rho, N, u, v, Q_N, R_N, D_N)$$

is retained as part of the Dirichlet-source report.

Theorem 71.2 (Dirichlet source boundary)**.** *Under the declared magnitude and orientation compiler, the original Dirichlet source construction is resolution-certified convergent for attenuation strictly greater than one and has explicit finite nonconvergence witnesses at and below one.*

72 A distinct paired continuation route

72.1 Finite eta-source pairing

Define the finite paired source packet by an explicit opposing-role instruction, whose later signed coordinate shadow is alternating subtraction:

$$\Delta_k(\Sigma) = \mathrm{PairOpp}(\mathcal{T}_{2k-1}[\Sigma], \mathcal{T}_{2k}[\Sigma])$$

and finite paired stage

$$\mathfrak{E}_K[\Sigma] = \Delta_1 \sqcup \cdots \sqcup \Delta_K.$$

For a fixed finite input instruction, the local finite logarithm/exponential compiler gives a finite response bound of the form

$$\mathrm{Mag}(\Delta_k) \le C_\Sigma k^{-(\mathcal{A}+1)}.$$

Therefore the paired source rule has a positive-tail convergence certificate whenever $\mathcal{A} > 0$.

72.2 Exact finite overlap with the Dirichlet stages

Let $D_N(s)$ denote the scalar coordinate shadow of the finite Dirichlet stage when that projection is lawful. The finite paired shadow satisfies the exact identity

$$E_K(s) = D_{2K}(s) - 2^{1-s} D_K(s).$$

This is a finite source reindexing identity: separate the odd and even source indices and compile the even terms by their common finite factor. No limiting argument is needed to establish the stage identity.

On attenuation greater than one, both Dirichlet stages have resolution certificates. Therefore the exact finite identity yields the overlap theorem

$$\eta(s) = (1 - 2^{1-s})\zeta_D(s)$$

as equality of projected evaluators on that shared domain.

For $0 < \mathcal{A} < 1$, the original Dirichlet report retains its nonconvergence certificate, while the paired report converges. Since the magnitude of 2^{1-s} differs constructively from the unit magnitude in this strip, the denominator report $1 - 2^{1-s}$ is separated from zero and the continued evaluator

$$\zeta_\eta(s) = \frac{\eta(s)}{1 - 2^{1-s}}$$

is defined by a distinct source route. It is not the value of the divergent Dirichlet termwise construction.

Principle 72.1 (No retroactive convergence)**.** *Agreement of the eta continuation with the Dirichlet evaluator on the overlap does not rewrite the divergent Dirichlet source report outside that overlap. The two routes remain distinguishable construction ledgers even where their scalar coordinate shadows agree.*

73 Finite reflection compiler before completed symmetry

73.1 Finite Gaussian source packets

For a positive finite scale report x, whole cutoff M, and finite exponential evaluator, define

$$g_x(n) = \mathrm{Exp}(-\pi x n^2)$$

and the finite symmetric packet

$$\Theta_M(x) = g_x(0) + \sum_{n=1}^{M} (g_x(n) + g_x(-n)).$$

The displayed negative index is an orientation/source label for the finite symmetric source packet; it does not require a completed set of all integers. Also define

$$\Psi_M(x) = \sum_{n=1}^{M} e^{-\pi n^2 x}$$

as the later scalar shorthand of the positive-side packet.

73.2 Finite Fourier–Gaussian bridge certificate

This subsection deliberately enters a conventional real/orientation coordinate compiler. Every negative coordinate and Fourier phase appearing below is a coordinate shadow of retained mirrored source roles and finite transition ledgers; it is not primitive signed or complex existence.

Use a finite balanced two-boundary cell with a retained center role. After entering the conventional signed ratio-coordinate compiler, this cell may be normalized and displayed as

$$-\frac{1}{2} \le u \le \frac{1}{2}.$$

The endpoint glyphs are coordinate shadows of the balanced whole-cell construction; no primitive half-occurrence or negative count is introduced. In that compiler construct

$$G_{M,x}(u) = \sum_{n=-M}^{M} e^{-\pi x(n+u)^2}.$$

The source ledger actually retains a center occurrence and M mirrored left/right source pairs; the signed labels are only the conventional coordinate names of those mirror roles. Gaussian evenness gives the exact finite boundary closure

$$G_{M,x}(-1/2) = G_{M,x}(1/2).$$

For each whole frequency label k in a finite range, form the finite coefficient

$$C_{M,k}(x) = \int_{-1/2}^{1/2} G_{M,x}(u)\, \mathcal{O}_{-k}(u)\, du,$$

where $\mathcal{O}_{-k}$ is a finite orientation compiler and the usual exponential phase is only its conventional coordinate shadow. A finite change of variable in each of the $2M+1$ source cells joins the pieces exactly into

$$C_{M,k}(x) = \int_{-L}^{L} e^{-\pi x t^2} \mathcal{O}_{-k}(t)\, dt, \qquad 2L = 2M+1.$$

The whole relation $2L = 2M+1$ is retained; the conventional coordinate compiler may abbreviate it by $L = M + 1/2$.

The finite Gaussian normalization and integration-by-parts ledger produce

$$C_{M,k}(x) = x^{-1/2} e^{-\pi k^2/x} + \varepsilon_{M,k}(x),$$

with an explicit finite endpoint bound. For every finite frequency cap K, the residual can be bounded uniformly using the Gaussian endpoint factor $e^{-\pi x L^2}$ and the retained finite integration certificate.

Reconstruct the center with the finite positive Fejér stage

$$F_{K,M,x}(0) = \sum_{k=-K}^{K} \left(1 - \frac{|k|}{K+1}\right) C_{M,k}(x)$$

and retain the reconstruction residual E_F. Releasing the finite Fejér weights on a smaller reflected cutoff $H < K$ and retaining both the weight defect and reflected Gaussian tail yields

$$\boxed{\Theta_M(x) = x^{-1/2}\Theta_H(1/x) + E_P, \qquad |E_P| < \eta.}$$

For every requested positive finite η, the cutoffs and finite partitions are chosen constructively until this inequality is certified. The usual theta reciprocal-scale identity is therefore a later zero-residual projection of a finite bridge rule, not an actually completed infinite sum.

73.3 Finite reflected Mellin packets

Choose a finite scale endpoint $X > 1$ and a finite partition

$$1 = x_0 < x_1 < \cdots < x_J = X.$$

At each tagged cell construct the paired exponent-ratio lanes. Primitive division of an exponent instruction by two is not used: if an exponent report has numerator instruction E against denominator carrier B_E, its one-to-two scaled exponent is retained as the same numerator instruction against the doubled denominator carrier $D_2 \times B_E$. The later conventional coordinate compiler may display the two lanes as

$$L_j(s) = t_j^{s/2}, \qquad R_j(s) = t_j^{(1-s)/2}.$$

Thus every written "/2" below is a coordinate abbreviation of denominator replication, not literal division of a primitive occurrence. Define the finite symmetric accumulation

$$I_{M,P,X}[s] = \sum_{j=1}^{J} \Psi_M(t_j)\big(L_j(s) + R_j(s)\big)\frac{\Delta x_j}{t_j}.$$

At every finite stage the reflection instruction

$$\mathcal{R} : s \longmapsto 1 - s$$

exchanges L_j and R_j. Therefore

$$\boxed{I_{M,P,X}[1-s] = I_{M,P,X}[s]}$$

by literal lane permutation before any completion or scalar analytic theorem is invoked.

73.4 Native finite horizontal balance comparison

After a conventional signed two-role coordinate compiler has been explicitly declared, a finite coordinate shadow may be written

$$s = \frac{A + \mathbf{j}T}{B}, \qquad B > 0.$$

The reflection compiler maps the retained numerator roles to their complementary roles. The primitive horizontal test is not the signed difference $2A - B$. Construct the doubled whole carrier $D_2 \times A$ and maximally match it against B:

$$D_2 \times A \cong C_H \sqcup R_A, \qquad B \cong C_H \sqcup R_B.$$

Reflection exchanges which complementary residual role can remain. The reflection-fixed horizontal balance condition is

$$\boxed{R_A = 0_A \quad \text{and} \quad R_B = 0_B,}$$

whose count shadow is $2A = B$.

Only inside the later signed coordinate compiler may this comparison be compressed to

$$H_B := 2A - B,$$

with reflection displayed as $H_B \mapsto -H_B$. Neither the minus sign nor the scalar imbalance is primitive.

For every retained source node $x > 1$, construct a finite root/ratio carrier using the one-to-$4B$ exponent relation. A conventional power-coordinate compiler may abbreviate this carrier by

$$V = x^{1/(4B)}.$$

When the signed coordinate shadow H_B is also admitted, the paired radial diagnostic may be displayed as

$$U_{x,B} = V^{H_B}, \qquad U^{\leftarrow}_{x,B} = V^{-H_B}.$$

The complete PCL report retains the two opposing radial roles; it does not contain a primitive negative power. Within the declared coordinate compiler their displayed difference vanishes exactly when the primitive whole-count balance comparison closes.

73.5 Finite boundary split and overlap theorem

Before naming any completed ξ report, define the finite Mellin window

$$\mathcal{M}_{M,X}[s] = \int_{1/X}^{X} \Psi_M(x) x^{s/2} \frac{dx}{x}$$

as a finite partition evaluator with retained integration residual. Because Ψ_M is a finite source sum, source and partition joins may be exchanged by finite distributivity. For each source n, the finite change of variable $t = \pi n^2 x$ gives

$$\int_{1/X}^{X} x^{s/2-1} e^{-\pi n^2 x}\, dx = \pi^{-s/2} n^{-s} \int_{\pi n^2/X}^{\pi n^2 X} t^{s/2-1} e^{-t}\, dt.$$

Thus on the already valid Dirichlet side the finite Mellin packet overlaps the finite Dirichlet/Gamma compiler up to explicitly retained boundary residuals.

Split the finite Mellin window at 1. In the lower finite interval use $x = 1/u$ and insert the finite Poisson bridge. The result is

$$\mathcal{M}_{M,X}[s] = I_{M,X}[s] + B_X(s) + E^{P}_{M,X}(s),$$

where the symmetric source packet is built from the paired lanes $u^{s/2}$ and $u^{(1-s)/2}$ and

$$B_X(s) = \frac{X^{(1-s)/2} - 1}{1-s} + \frac{X^{-s/2} - 1}{s}.$$

Avoid any primitive half-normalization by doubling the conventional boundary identity. The finite boundary packet satisfies

$$s(s-1) B_X(s) = 1 + \widehat{E}^{\mathrm{bd}}_X(s),$$

where, inside the declared coordinate compiler,

$$\widehat{E}^{\mathrm{bd}}_X(s) = -s X^{(1-s)/2} + (s-1) X^{-s/2}.$$

The unit on the right is a whole carrier. The traditional factor $1/2$ used in the conventional ξ normalization is therefore unnecessary for reflection or zero analysis and is retained only as an optional one-to-two ratio-coordinate projection.

Collect the Dirichlet, gamma-window, Gaussian, Poisson, partition, and boundary residuals into

$$E_{\mathrm{total}} = E_{\mathrm{Dir}} + E_{\Gamma} + E_{\mathrm{Gauss}} + E_{\mathrm{Poisson}} + E_{\mathrm{part}} + E_{\mathrm{bd}}.$$

For every requested positive finite ρ, choose finite cutoffs and partitions so that

$$|E_{\mathrm{total}}| < \rho.$$

This finite overlap certificate is what licenses the later scalar continuation identity.

74 Completed reflection notation as a projection theorem

74.1 The symmetric evaluator before the name xi

Define the unhalved finite symmetric stage

$$\widehat{\mathfrak{R}}_{M,P,X}[s] = 1 + s(s-1)I_{M,P,X}[s].$$

This form uses only the whole unit carrier and avoids introducing the one-to-two normalization into the operative reflection ledger. At each finite stage,

$$\widehat{\mathfrak{R}}_{M,P,X}[1-s] = \widehat{\mathfrak{R}}_{M,P,X}[s]$$

because the polynomial factor is invariant and the paired source lanes are exchanged.

Definition 74.1 (Resolution-certified reflection evaluator)**.** *The report $\widehat{\mathfrak{R}}[s;\rho]$ is the finite unhalved stage together with all source, Poisson, integration, boundary, and overlap certificates chosen so that the total unresolved effect is below the requested finite resolution ρ.*

Only after the overlap theorem has related this evaluator to the Dirichlet/Gamma construction on their common domain may a conventional normalization be introduced. The traditional ξ coordinate is a one-to-two ratio-coordinate projection of the unhalved evaluator; symbolically a conventional compiler may display

$$\xi(s) = \frac{\pi_{\mathrm{scalar}}(\widehat{\mathfrak{R}}[s])}{2}.$$

Primitive PCL retains instead the whole relation between the unhalved evaluator and the two-occurrence normalization carrier. Reflection requires no division by two:

$$\boxed{\pi_{\mathrm{scalar}}(\widehat{\mathfrak{R}}[s]) = \pi_{\mathrm{scalar}}(\widehat{\mathfrak{R}}[1-s]).}$$

This is the scalar coordinate shadow of the stagewise typed transport

$$\boxed{\mathcal{R} : \widehat{\mathfrak{R}}[s] \longrightarrow \widehat{\mathfrak{R}}[1-s].}$$

The transport retains source indices, finite partitions, reflection pairing, residual certificates, and genealogy.

74.2 Conjugation as a distinct orientation transport

Conjugation reverses the transverse orientation role while retaining the horizontal attenuation role. It is a different finite transport from reflection. Conventional fourfold scalar symmetry is admitted only after the two transports and their coordinate projections have been separately constructed.

75 Zero-predicate and reflection-diagnostic audit

75.1 Scalar zero is only a coordinate predicate

When the conventional complex-coordinate compiler is invoked, an evaluated reflection report may have a scalar shadow customarily written $\xi(s)$. Equality of that shadow with the pair-grammar zero means only that the selected coordinate value vanished. It does not, by itself, close source, continuation, reflection, refinement, or radial diagnostics.

Principle 75.1 (No promotion of scalar zero)**.** *A stronger report field may be required to vanish at a scalar zero only after an independent preservation theorem proves that field constant on the scalar-zero fiber. Attaching a field to the evaluator and then defining "complete zero" to set that field to zero does not establish the required implication.*

75.2 Reflection-fixed diagnostic

Inside the declared rational two-role coordinate compiler, retain first the primitive whole-count comparison

$$D_2 \times A \cong C_H \sqcup R_A, \qquad B \cong C_H \sqcup R_B.$$

The horizontal reflection-fixed predicate is

$$\boxed{R_A = 0_A \quad \text{and} \quad R_B = 0_B,}$$

with count shadow $2A = B$.

If a later signed coordinate frame is invoked, the same comparison may be displayed by

$$H_B = 2A - B.$$

At a retained positive source node $x > 1$, the conventional radial diagnostic may then be displayed as

$$\Delta_{x,B} = V^{H_B} - V^{-H_B}, \qquad V = x^{1/(4B)} > 1.$$

All minus signs and reciprocal exponents here belong to the conventional signed coordinate grammar; the complete PCL report retains paired source roles and their matching history.

Reflection exchanges the two radial roles. Within the declared coordinate compiler the displayed radial difference is zero exactly when the primitive whole-count balance comparison closes.

Theorem 75.2 (Reflection-fixed coordinate diagnostic)**.** *If a finite reflected coordinate packet is required, by an independently declared fixed-point predicate, to equal its own reflected packet without exchanging distinct radial roles, then the whole-count horizontal balance comparison closes:*

$$R_A = 0_A, \qquad R_B = 0_B,$$

and hence $2A = B$ at the count level. A later ratio-coordinate compiler may display the corresponding ratio class with the traditional half-coordinate glyph.

Remark 75.3 (This is not an ordinary zeta-zero theorem)**.** *Reflection transport is valid for every admissible input packet; reflection fixedness is an additional predicate. A scalar zero of the conventional zeta or xi coordinate does not currently imply reflection fixedness or radial-diagnostic zero. Consequently PCL does not identify the theorem above with the ordinary Riemann Hypothesis, whose usual statement also presupposes a completed complex domain and real-part coordinate that have not been established as primitive existences here.*

75.3 Structural counterexample to zero-promotion from symmetry alone

Reflection symmetry and positive paired source weights alone cannot force scalar zeros onto the reflection-fixed coordinate. Finite symmetric pair evaluators can be constructed whose scalar coordinate shadow vanishes away from the fixed coordinate while every radial diagnostic has one common nonzero orientation. Therefore any future route from conventional scalar zero to closure of the primitive balance comparison $2A = B$ requires a property intrinsic to the actual zeta source construction, not reflection symmetry by itself and not a radial field appended to the definition of zero.

76 Dependency summary for provenance-preserving zeta analysis

The repaired dependency chain is

distinction → retained report → whole occurrence → finite grouping,
→ matching and unmatched residuals → count and quotient–remainder,
→ finite powers/roots/refinement rules → optional signed coordinate compiler,
→ optional real-coordinate evaluator grammar → optional complex-pair compiler,
→ finite zeta source transformations → finite continuation certificates,
→ finite Gaussian/Mellin reflection transport → conditional scalar projection,
→ zero-predicate audit and reflection diagnostics.

The central rule is that domain entry never deletes provenance. Conventional negative, real, geometric, complex, and analytic expressions may be extremely useful compressed calculi, but PCL carries the prior matching, refinement, transition, source, residual, and closure reports alongside them. No later coordinate abstraction may be used retroactively to justify the primitives that its own construction required.

In particular, this manuscript does not claim that a completed real continuum, exact orthogonal frame, or completed complex domain has been established as primitive existence. It therefore does not present the ordinary Riemann Hypothesis as a presently grounded PCL proposition. The zeta work here is a provenance audit of finite source compilers and their conventional coordinate shadows, with unresolved foundational obligations stated explicitly.